

# Week 10: Low-Fidelity Prototypes

COMS 4170 • Spring 2026

Prof. Celeste Layne  
Tuesday March 31, 2026



# The biggest misconception about creativity

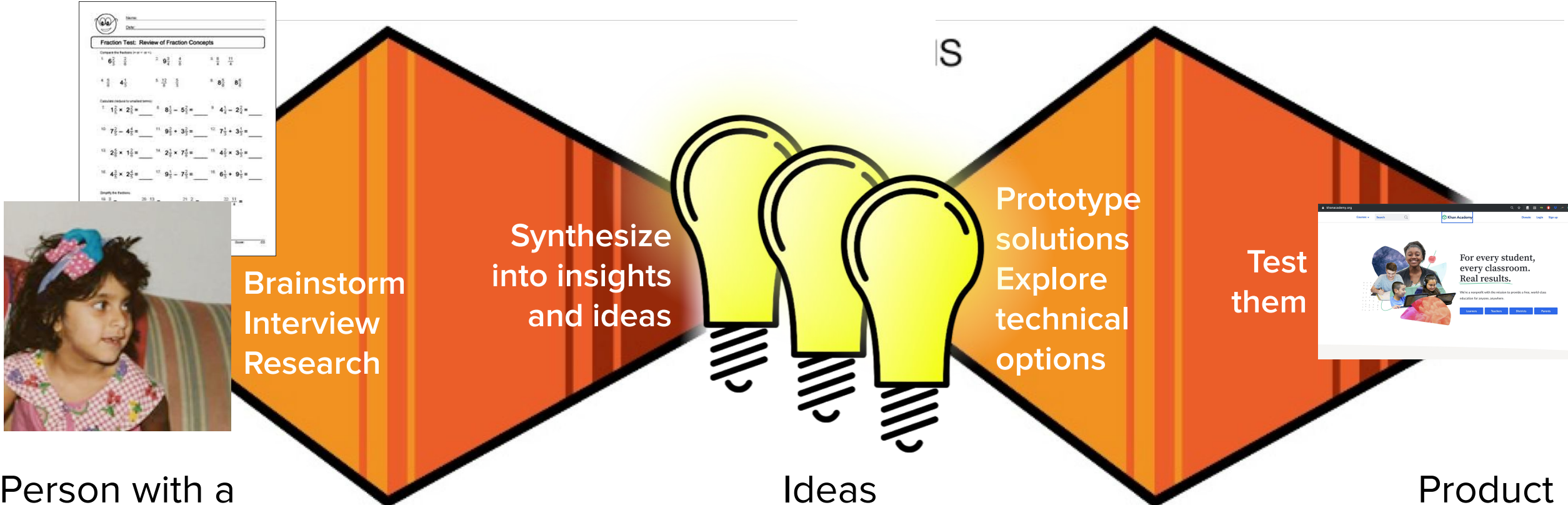


Idea



Product

# Creativity is a Process



**Understand the problem**

**Solve the problem**

# Design Challenge

- Design and build a **web** application
- To help a user learn an introductory topic **interactively**
- Within a **domain of your choosing**
- The interactive experience must center around **media**  
**(images, video, audio)**
- The interactive experience should help users assess their learning with a quiz
- Users should keep learning through feedback from the quiz
- It should take under 10 minutes total

# The USER is your TA

- ...and more generally, people in this class
- This way, you can test your designing on people in this class.
- Your TA must also want to learn this topic, and must be able to give you feedback on it.
- Your TA will also grade you, so listen to their feedback and take it

# Design Challenge

The TOPIC must be in a domain of your choosing; but it must be focused enough to teach in 10 minutes.

- Examples of broad domains:
  - chess, basketball, art history, music etc
- Examples of focused topics:
  - Chess: How and when to perform 3 different opening moves in chess for chess beginners
  - Basketball: How to run a pick and roll in basketball for casual NBA fans
  - Art History: How to tell Impressionist paintings from Post-Impressionist paintings for ArtHum students
  - Music: How to mix a drop swap for aspiring DJs

# Design Insight for Teaching:

People learn through interaction and feedback, not from reading long streams of text.



Learning to walk



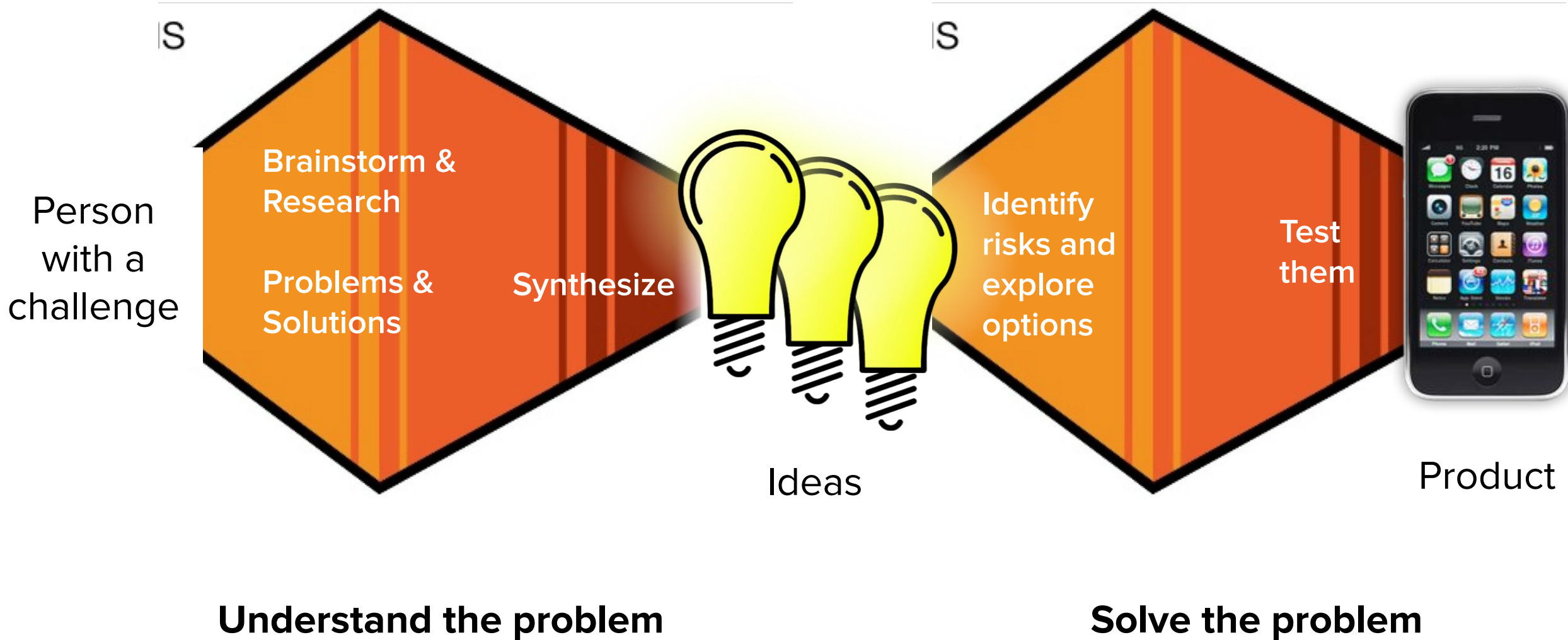
Playing an instrument

```
void loop() {  
  float humidity = 452;  
  for( int counter = 0; counter < 1000; ++counter ) {  
    digitalWrite(13, HIGH); // set the LED on  
    delay( 200 );           // wait for a second  
    digitalWrite(13, LOW);  // set the LED off  
    Serial.print( "Hello " );  
    Serial.println( counter );  
    delay(200);             // wait for a second  
    humidity += 0.5;  
  }  
}
```

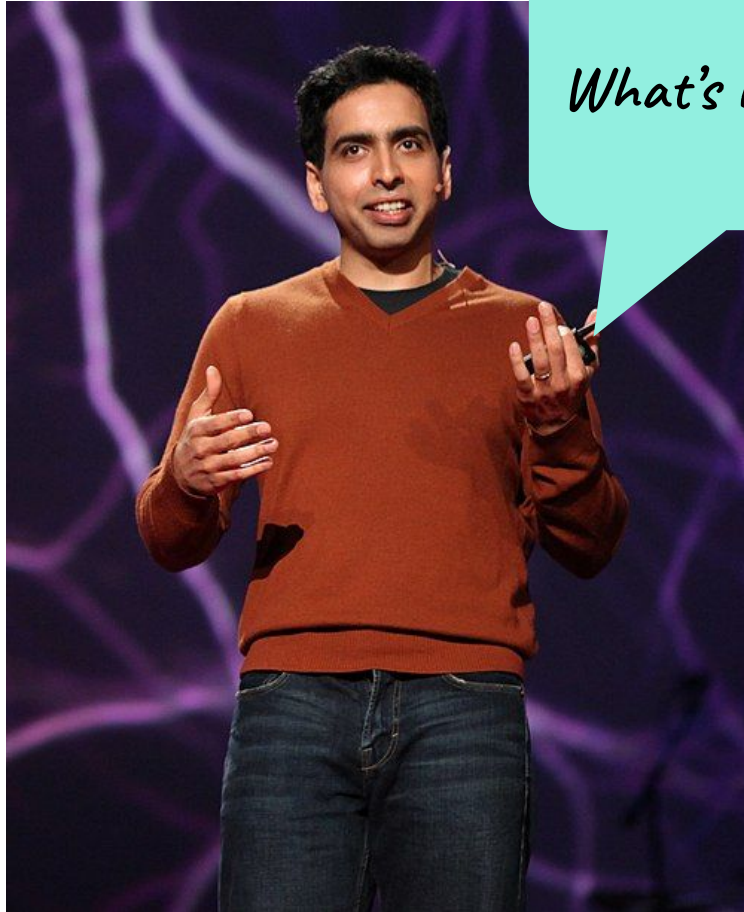


Learning to code

# The Double Diamond Flare and Focus



# Design is a process where you work with users to understand their problems



*What's wrong?*

Math Test: Review of Fraction Concepts

Calculate (reduce to smallest terms):

|                                          |                                          |                                          |
|------------------------------------------|------------------------------------------|------------------------------------------|
| 1. $1\frac{1}{2} \times 2\frac{2}{3} =$  | 8. $8\frac{5}{8} - 5\frac{3}{4} =$       | 15. $4\frac{1}{4} - 2\frac{3}{4} =$      |
| 10. $7\frac{2}{3} - 4\frac{4}{5} =$      | 11. $9\frac{5}{6} \div 3\frac{1}{2} =$   | 12. $7\frac{1}{3} + 3\frac{1}{3} =$      |
| 13. $2\frac{5}{6} \times 1\frac{2}{3} =$ | 14. $2\frac{1}{2} \times 7\frac{3}{4} =$ | 15. $4\frac{2}{3} \times 3\frac{1}{3} =$ |
| 16. $4\frac{3}{4} \times 2\frac{2}{3} =$ | 17. $9\frac{5}{6} - 7\frac{1}{3} =$      | 18. $6\frac{1}{3} \div 9\frac{1}{9} =$   |

Simplify the Fractions:

|                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 19. $\frac{10}{15} =$ | 20. $\frac{13}{18} =$ | 21. $\frac{12}{18} =$ | 22. $\frac{11}{14} =$ |
| 23. $\frac{14}{28} =$ | 24. $\frac{22}{33} =$ | 25. $\frac{15}{25} =$ |                       |

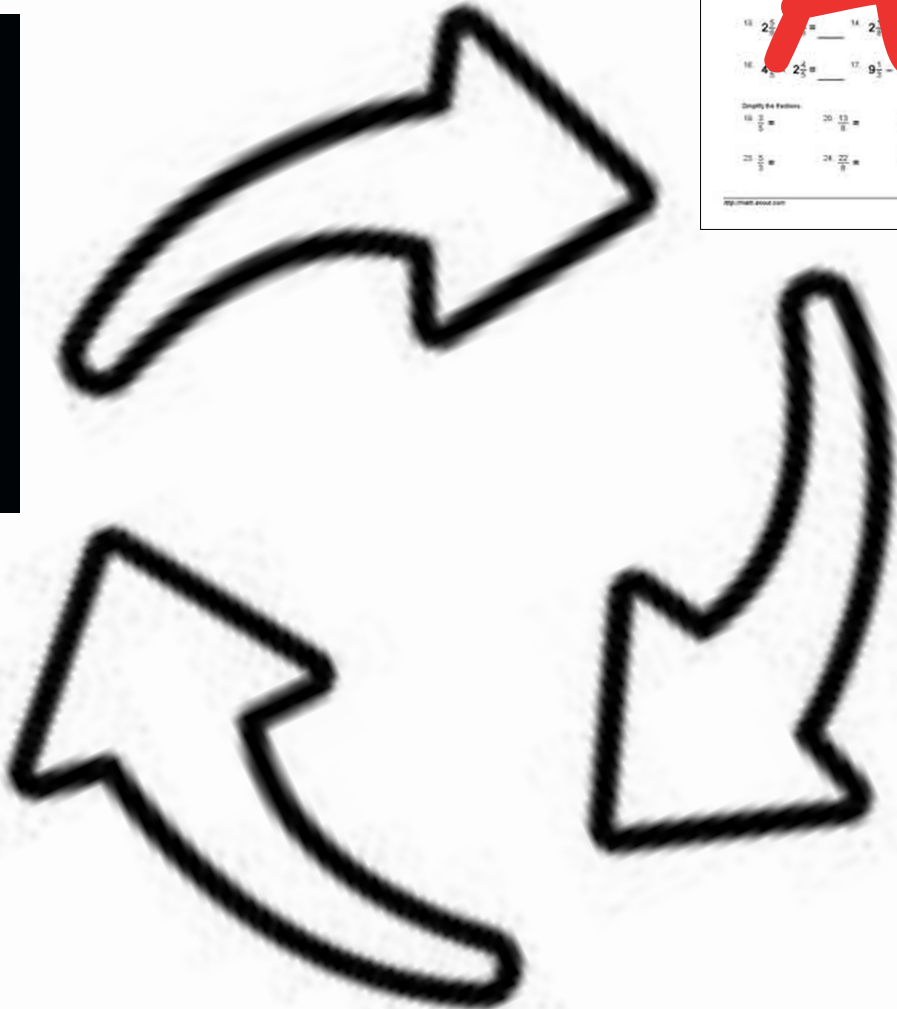
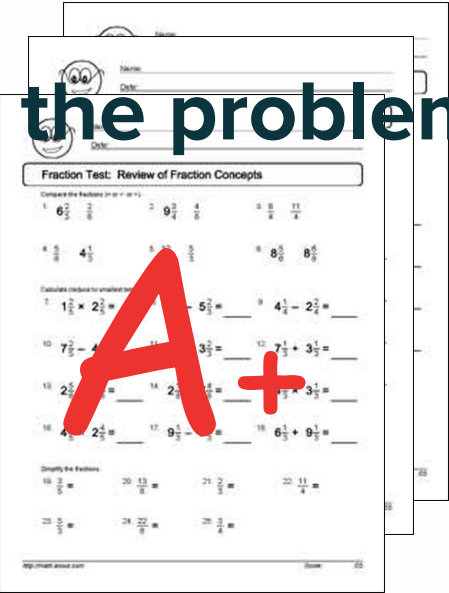
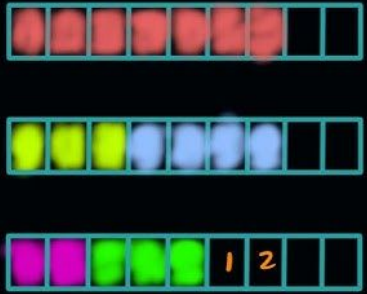
http://math.about.com

*I just don't know what to do.*

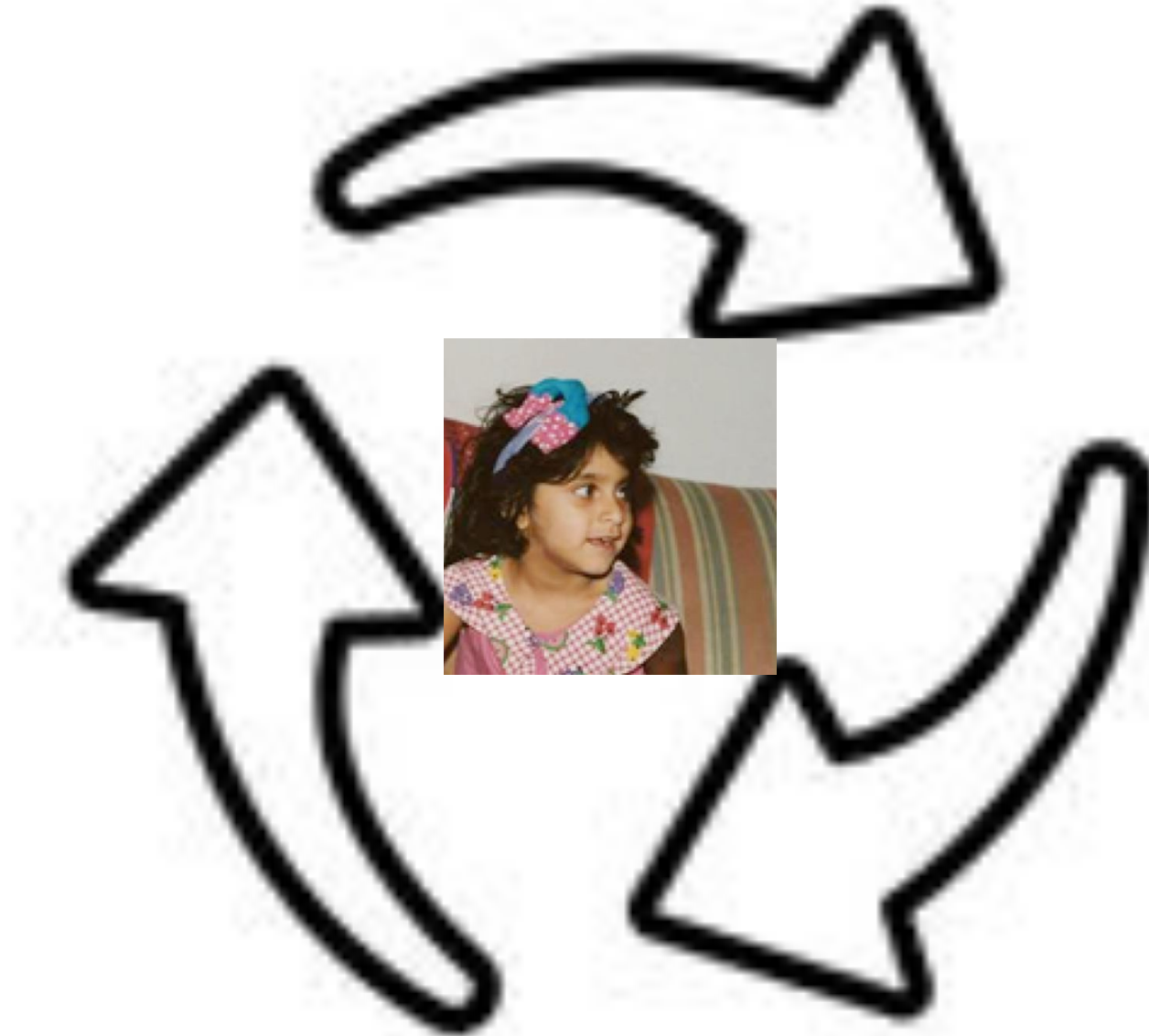


...and test solutions with users until it solves the problem

Multiply  $1\frac{3}{4} \times 1\frac{1}{5}$ . Simplify your answer and write it as a mixed fraction.

$$\frac{2}{9} + \frac{3}{9} + \frac{7}{9}$$


**The user is at the center of the process**

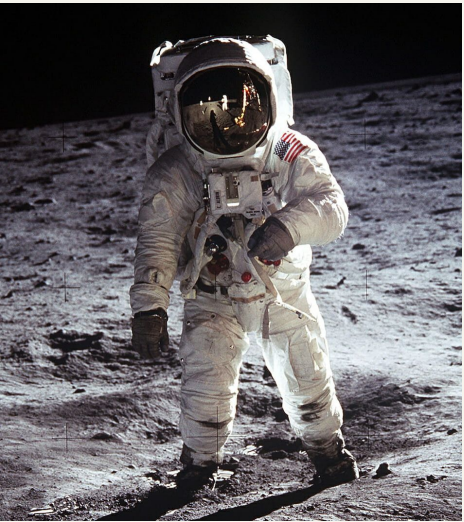


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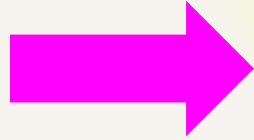
# The Design Process



# What I expect the design process to look like.



Challenge



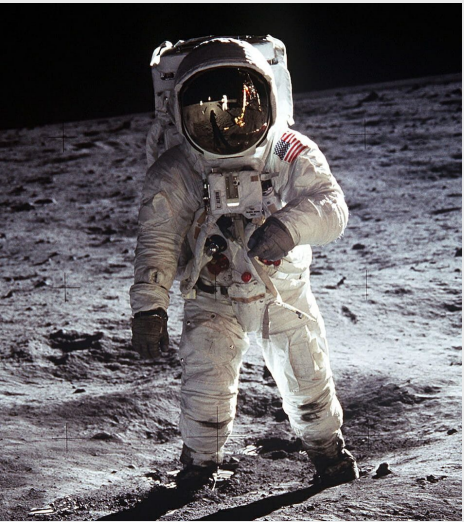
Idea



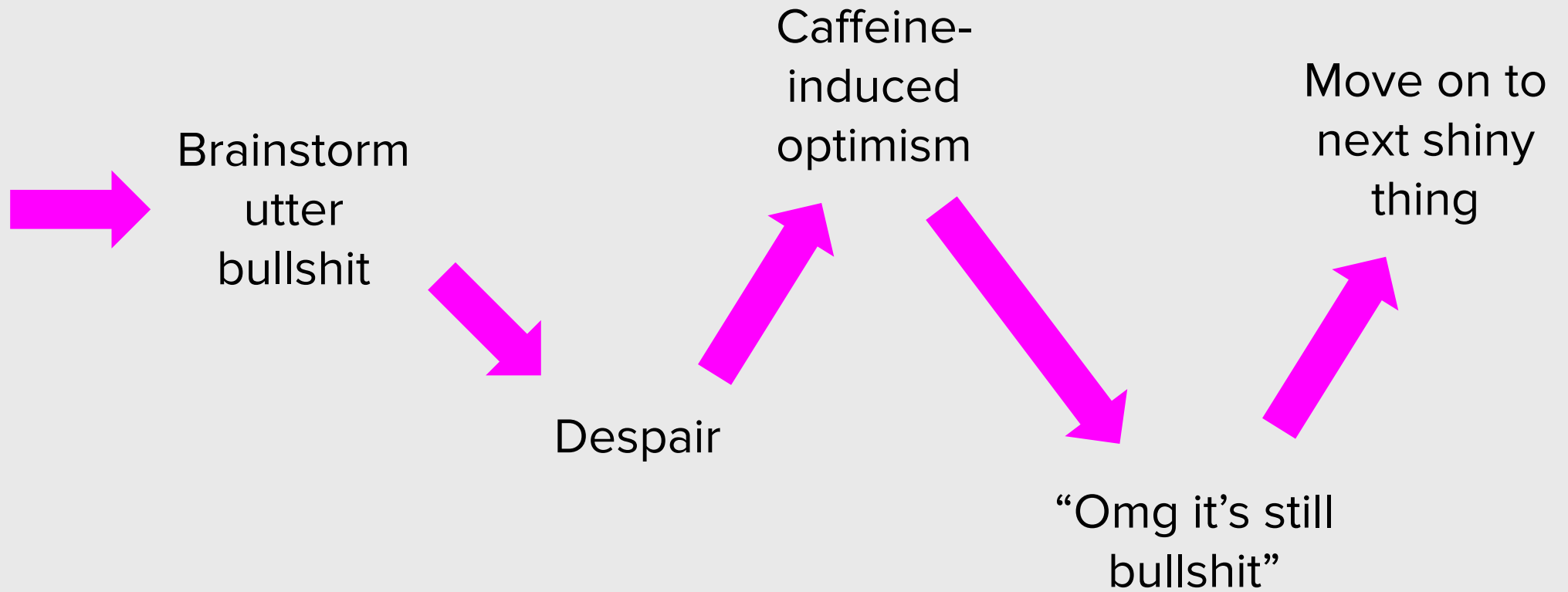
Product



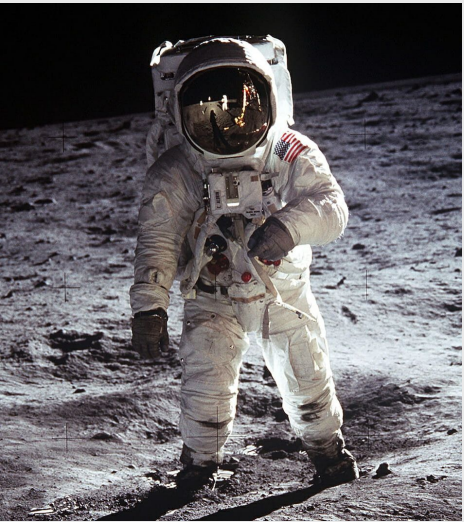
# What the design process actually looks like



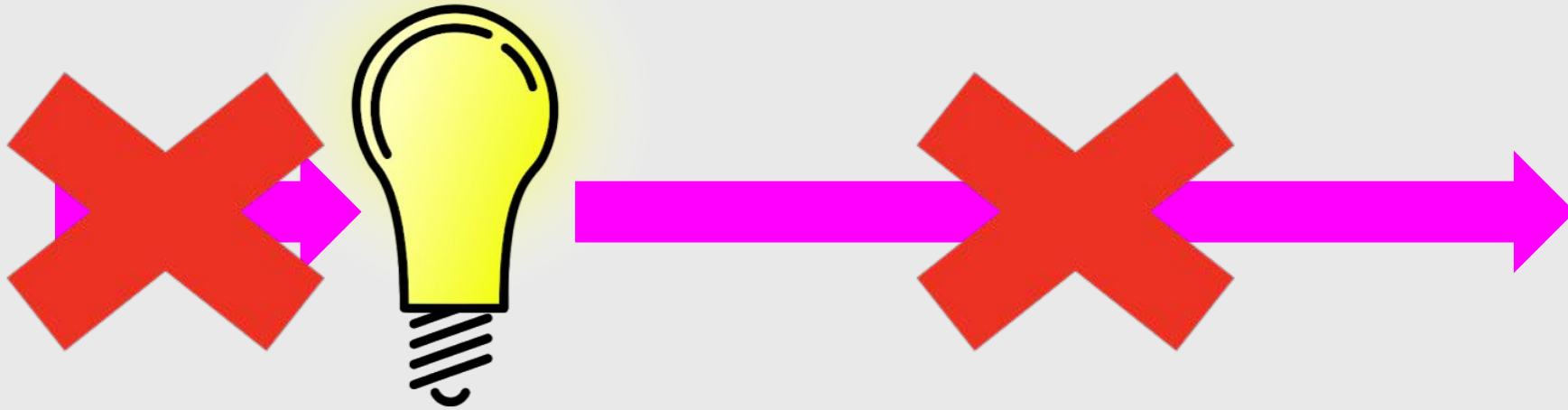
Challenge



**If you expect the design process to be straightforward, you will probably be frustrated**



Challenge

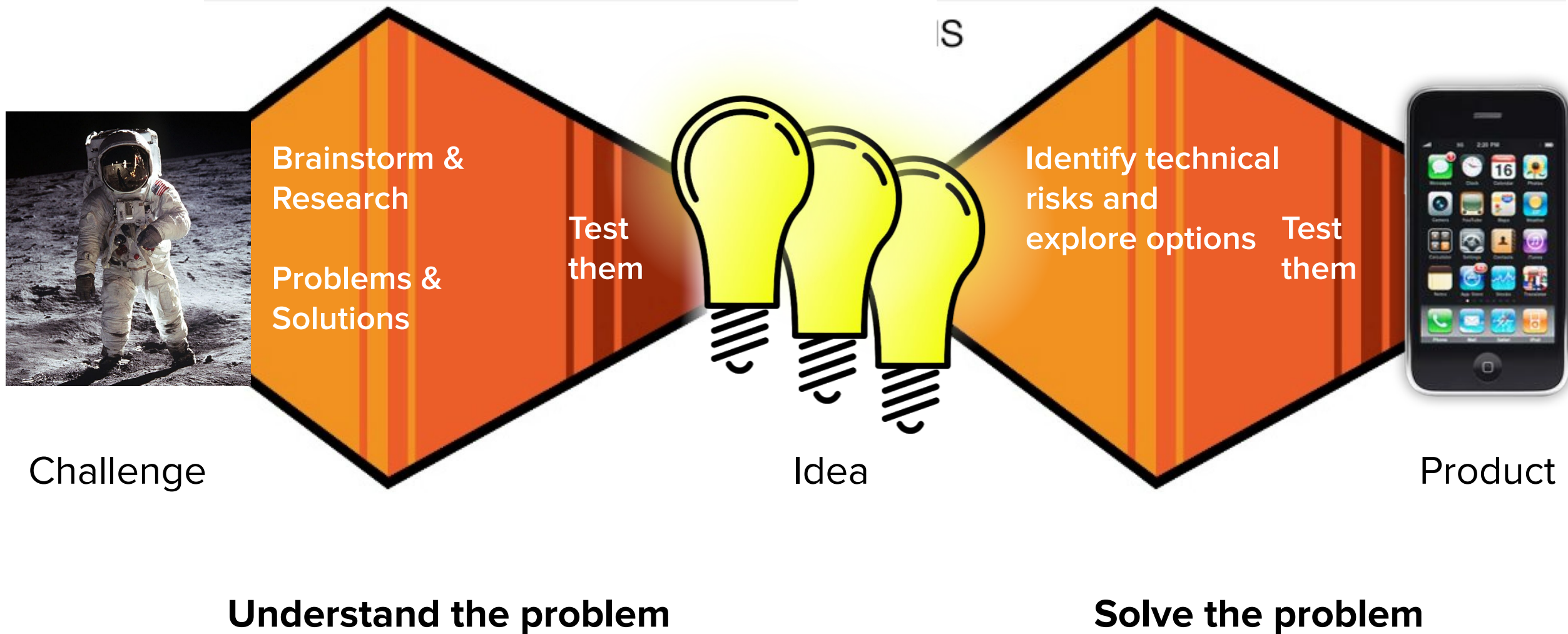


Idea



Product

# The Double Diamond Flare and Focus



# Phase 1: Understand the Problem



# How do we start to test ideas?

S

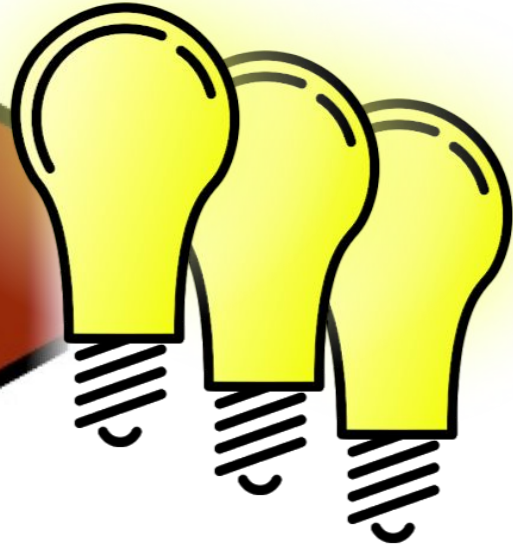


Challenge

Brainstorm &  
Research

Problems &  
Solutions

Test  
them



Idea

**Understand the problem**

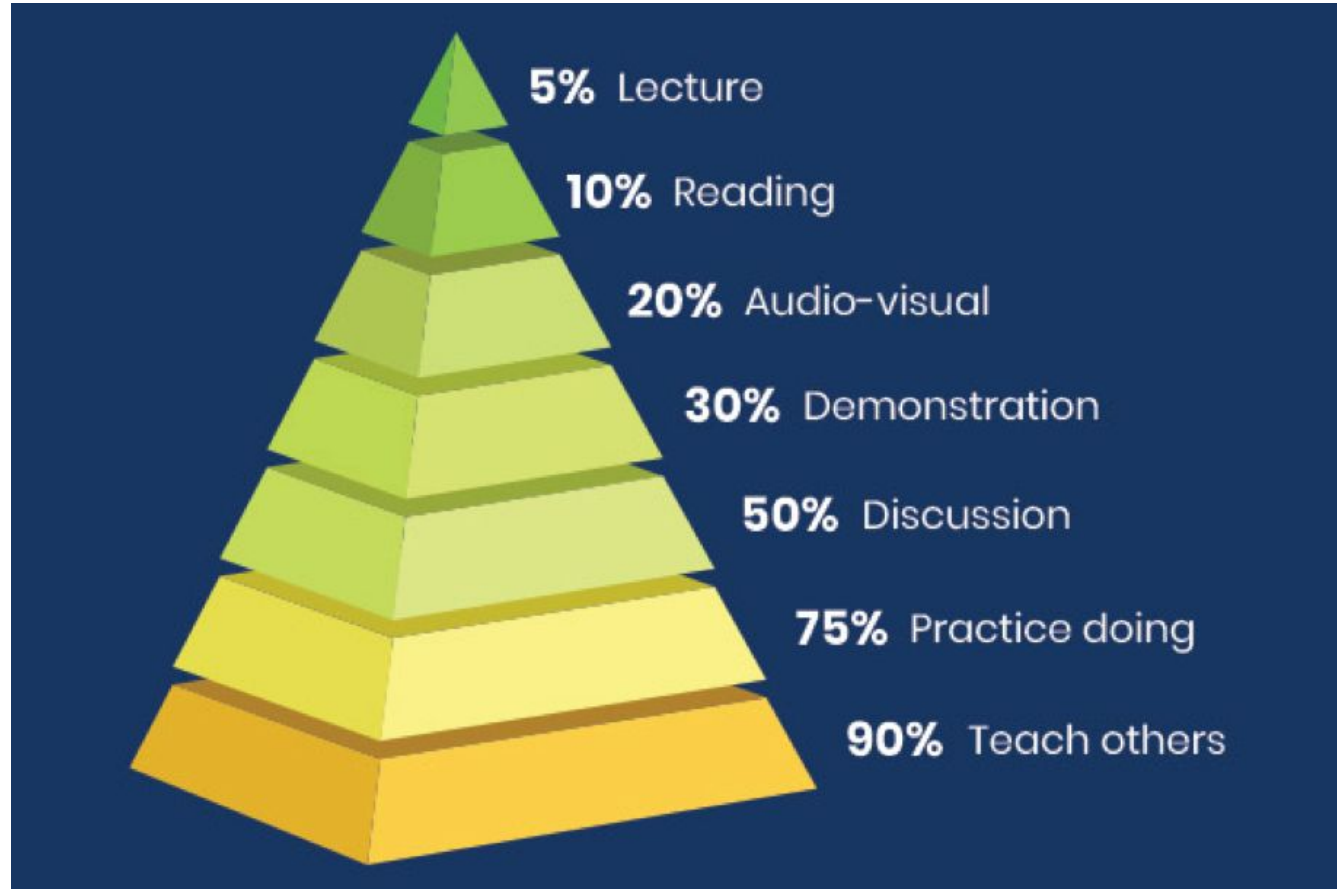
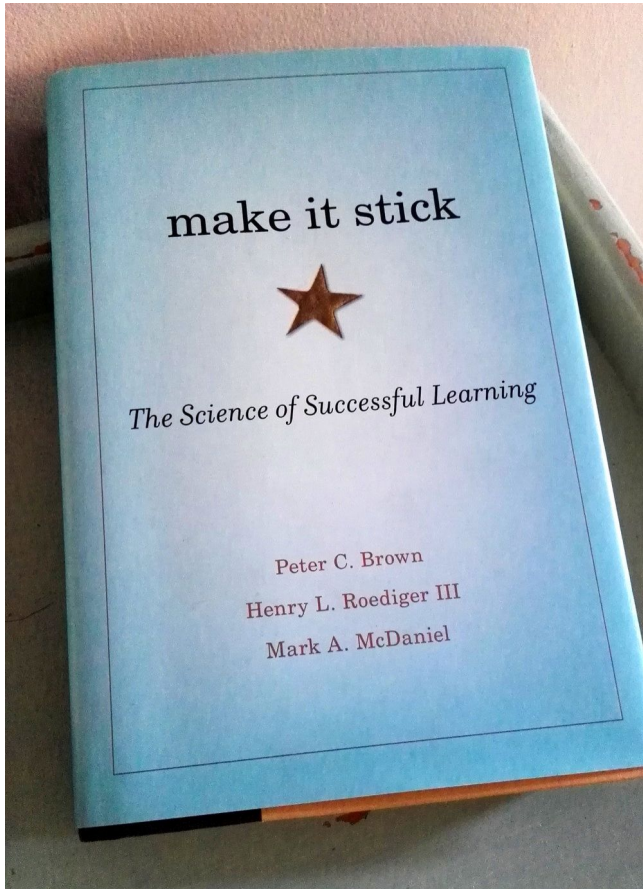
# Brainstorm ideas for domains and specific problems



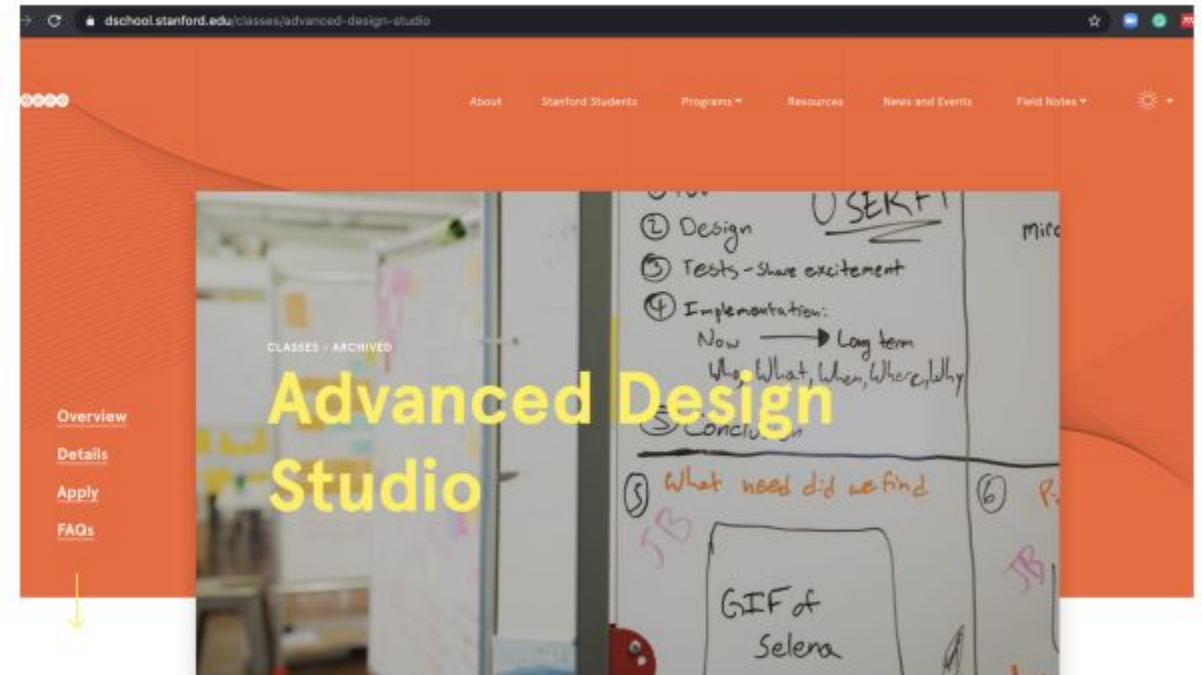
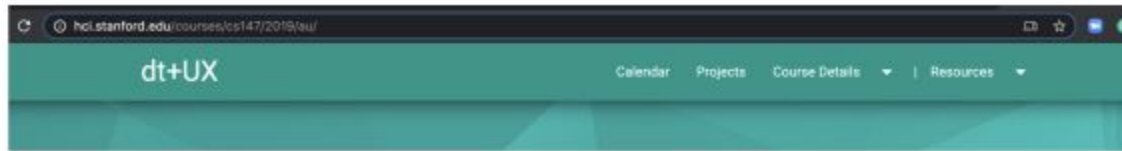
The best way to have a good idea  
is to have lots of ideas.

- Linus Pauling

# Read books, papers, theories, scientific evidence to get insights into the problem.



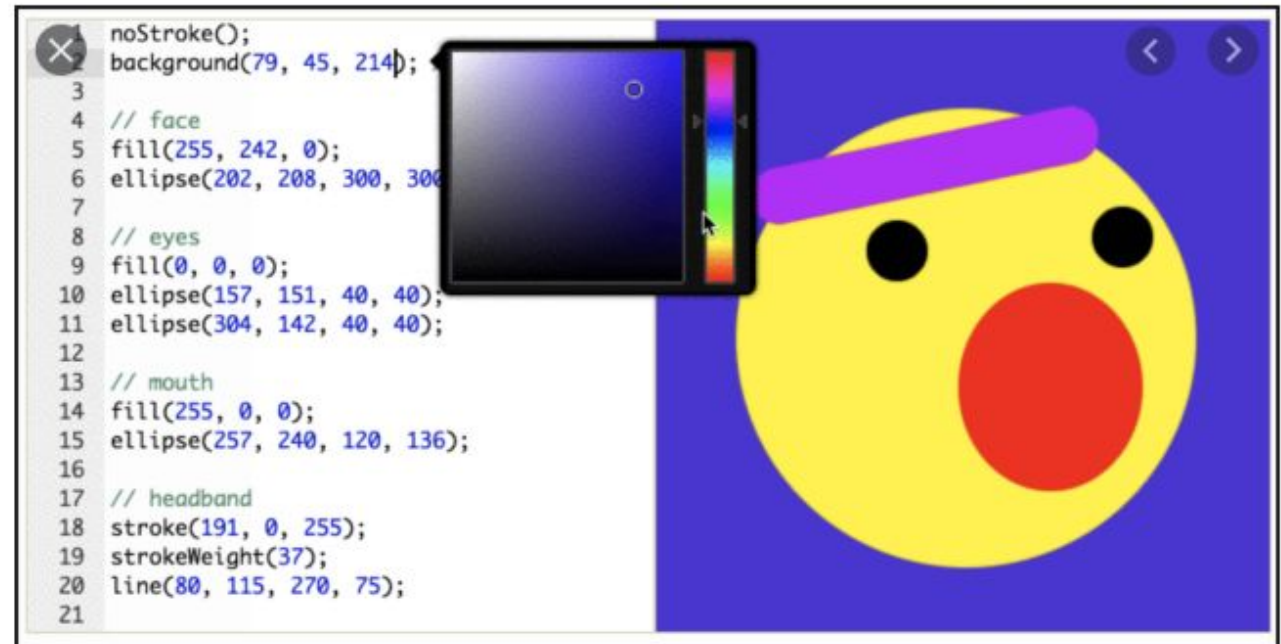
# Direct Competitors: Find other solutions to the problem



Learn from them.

# Indirect Competitor Research:

## Find solutions to other similar problem



**Steal good ideas.**

# Talk to a user: Learn about their experience



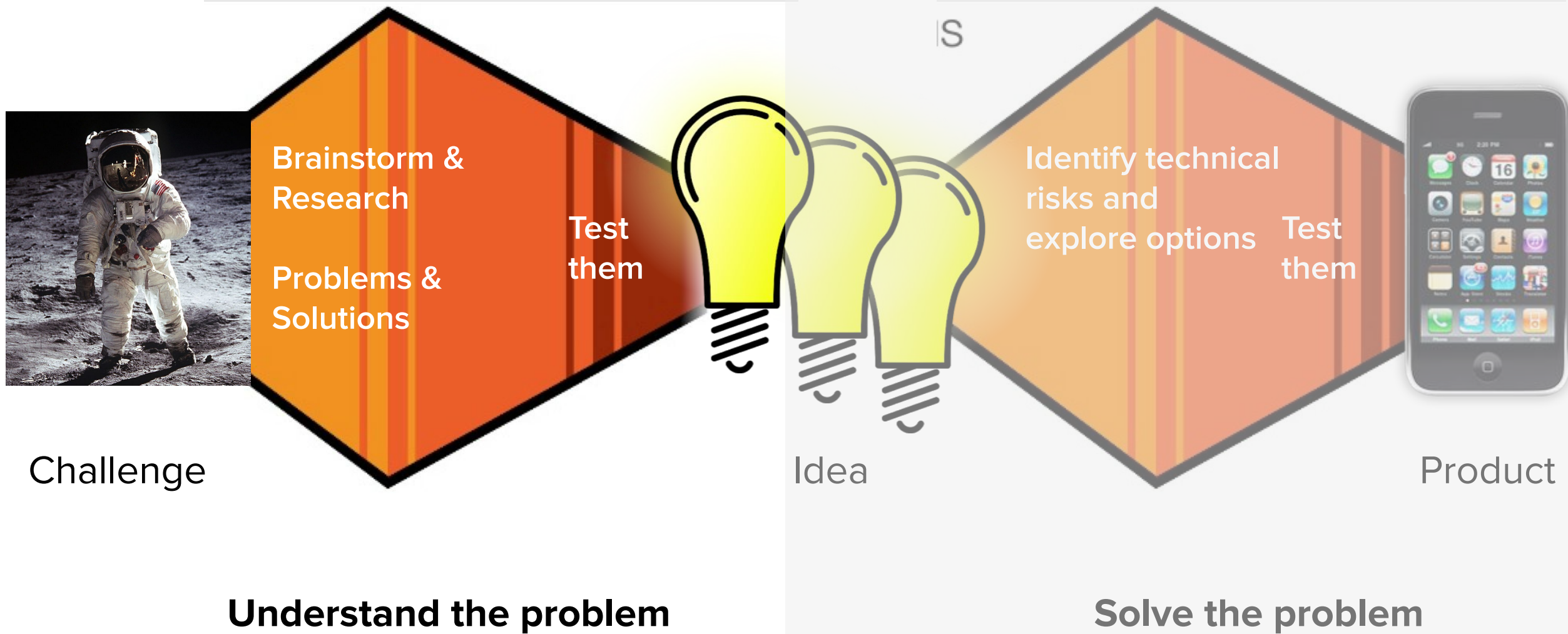
Jamie – a student in User Interface Design who is shy, but is forced to participate in class and fill out a form after class to record her participation

Tell me about the last time you **spoke up in class?**

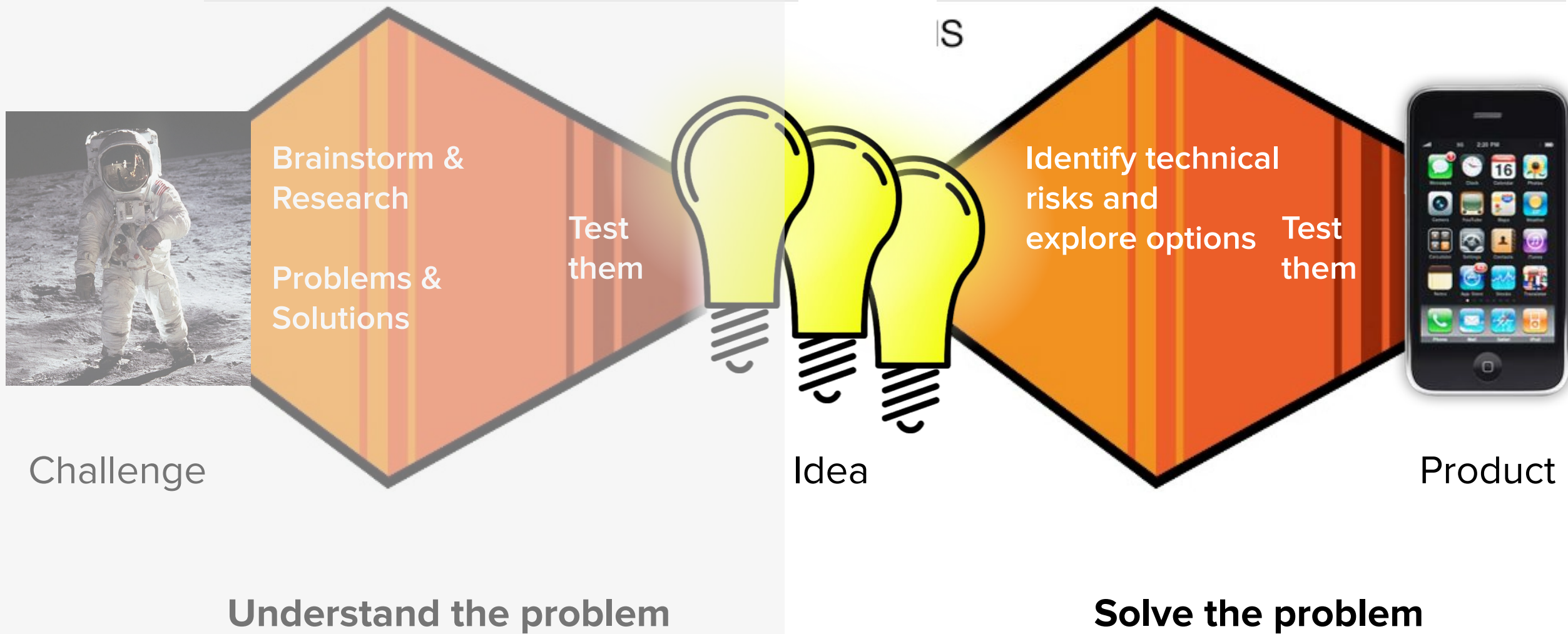
- *I'm worried my accent won't be understood*
- *I only saw something if I'm 100% sure of the answer. I don't like to guess*
- *I'm so nervous about participating that I don't pay attention*
- *I always forget to fill out the participation form*



# Homework 7



# Homework 8-12

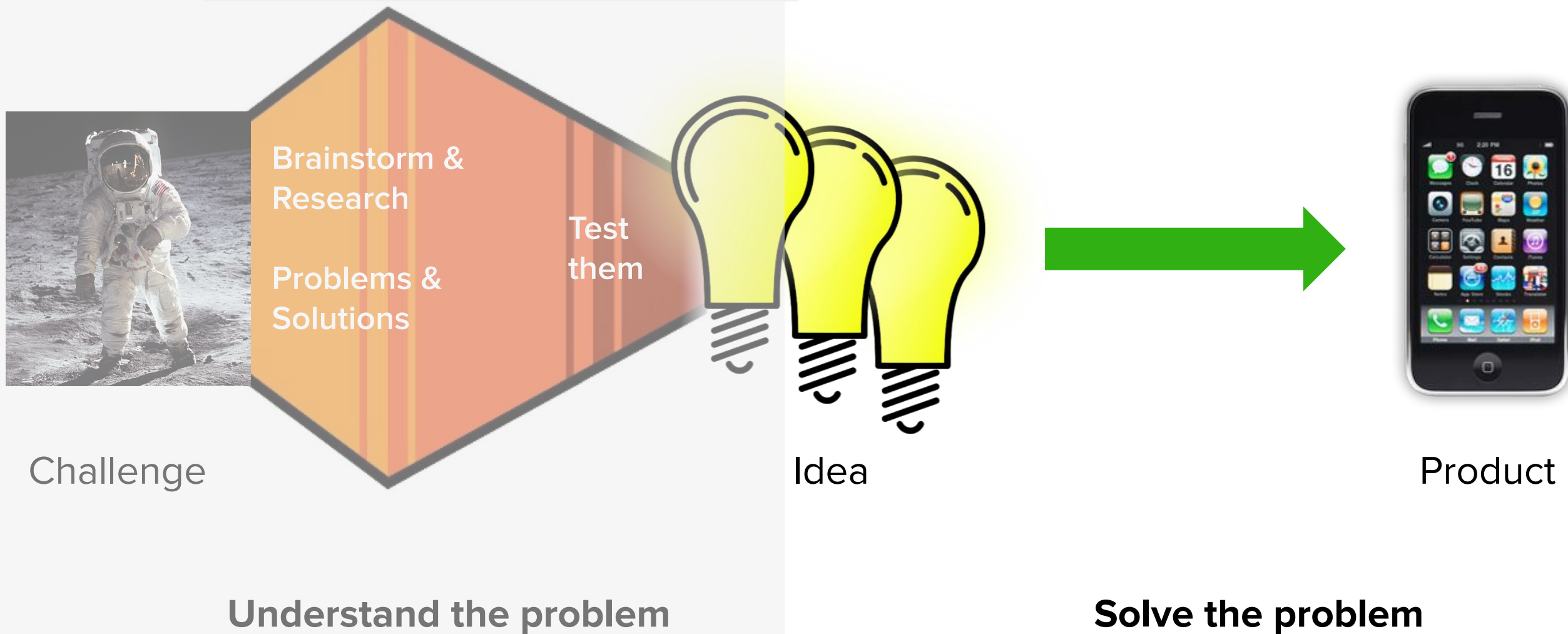


# Phase 2: Solve the Problem



**For many years, software engineers solved problems and made the same mistakes over and over again.**

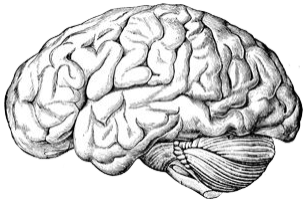
# Common Mistake: Solving problems linearly



# Two Design Processes

The **waterfall model** and **iterative design**

# The Waterfall Model:

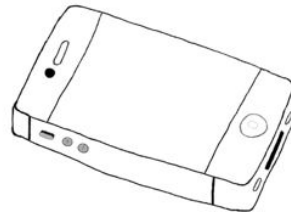


**Idea**

**Requirements**

- One button
- Touch screen
- Soft keyboard

**Design**



**Implement**



**Fix bugs**

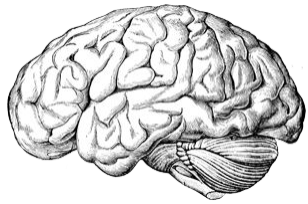


**Ship it**



**Product**

# The Waterfall Model: What's good about it?

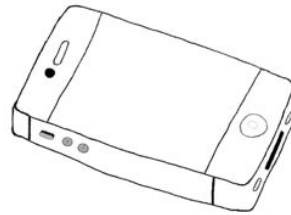


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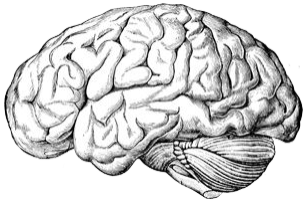
Ship it



Product

It's simple, linear and the steps are certain

# The Waterfall Model: What could go wrong?

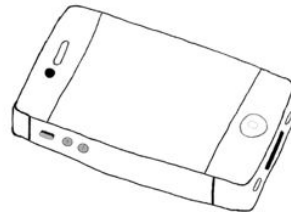


Idea

Requirements

- One button
- Touch screen
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Design



Implement



Fix bugs



Ship it



Product

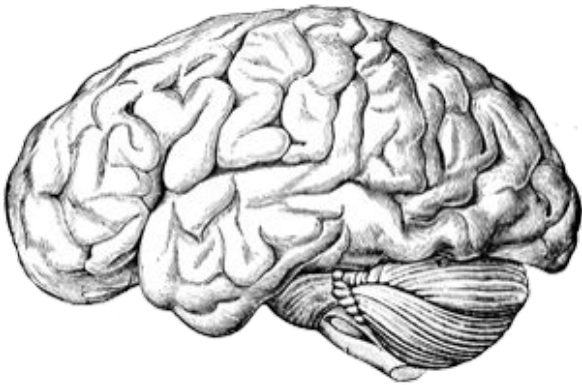
*What if a touch screen can't be implemented?*

*What if the hardware weighs 30 pounds?*

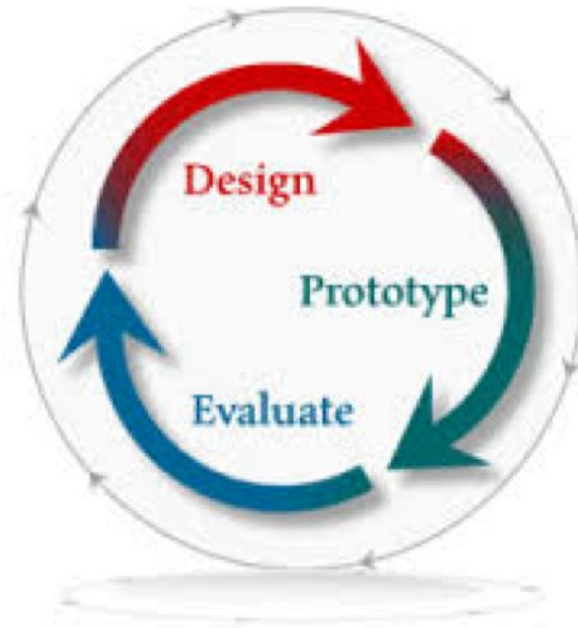
*What if the device is so slow, it's unusable?*

*How can we keep up with the competition?*

# Iterative Design

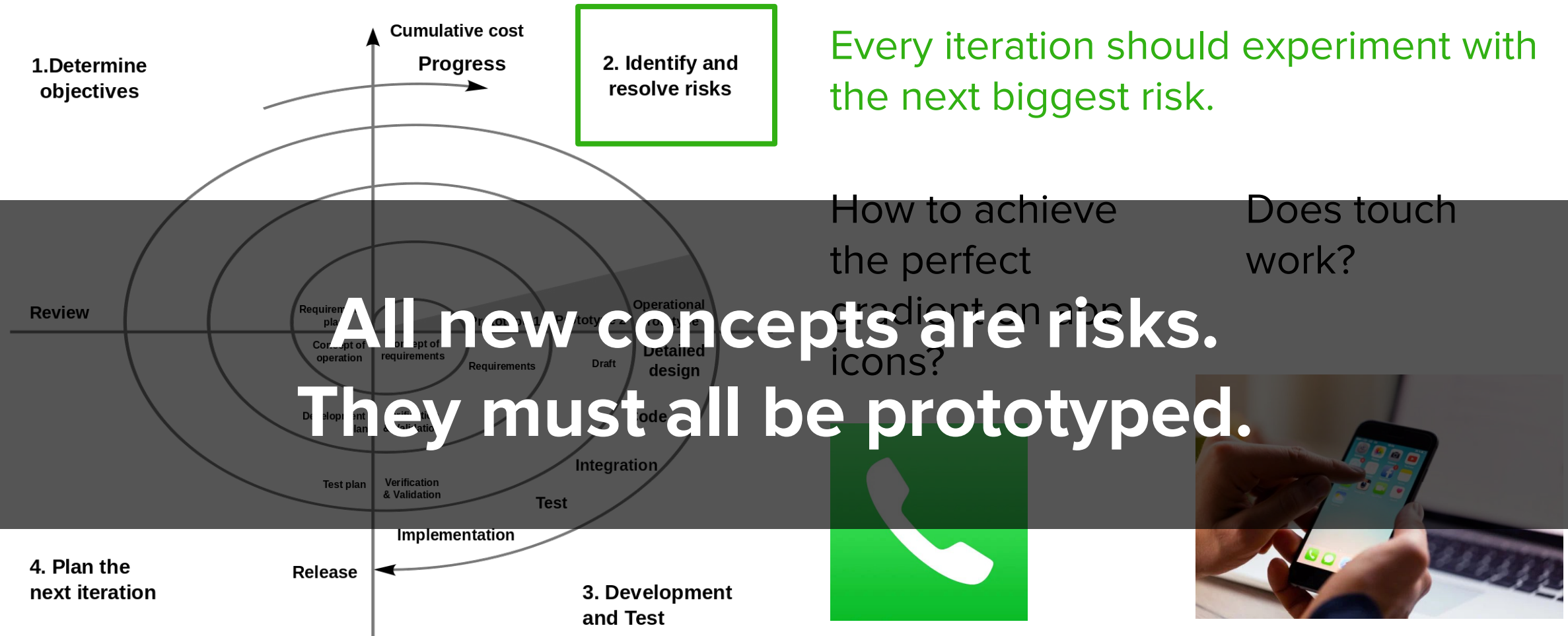


**Idea**

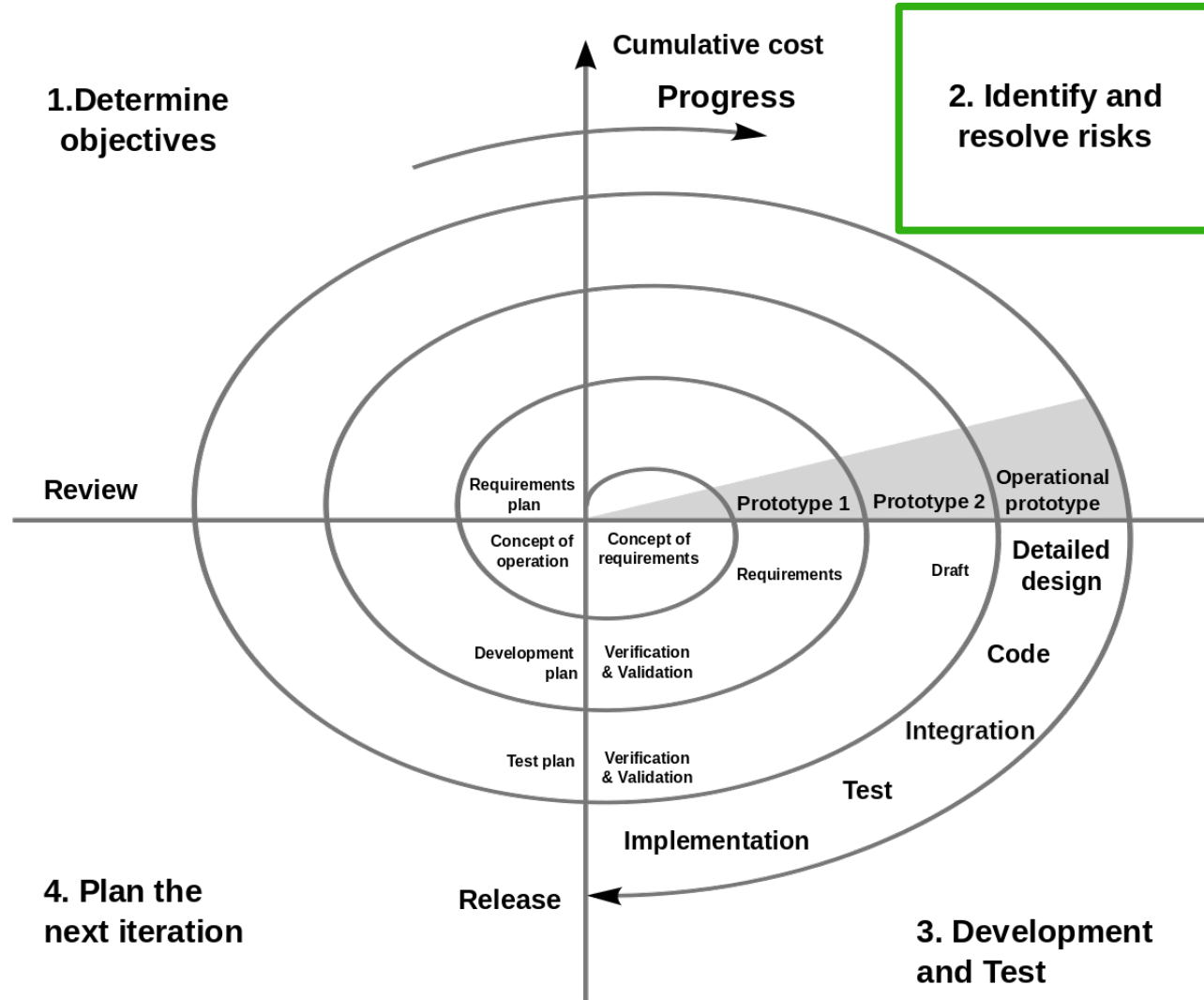


**Product**

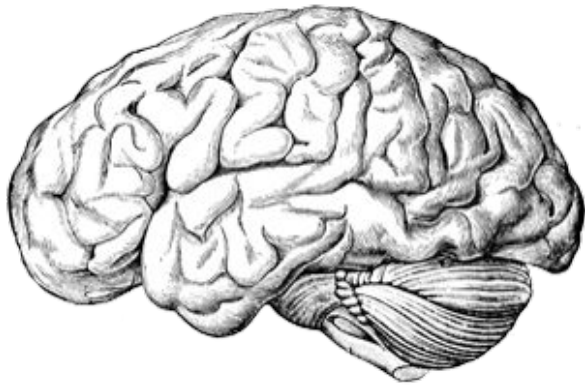
# Iterative Design origins: Spiral Model of software engineering (Barry Boehm, 1988)



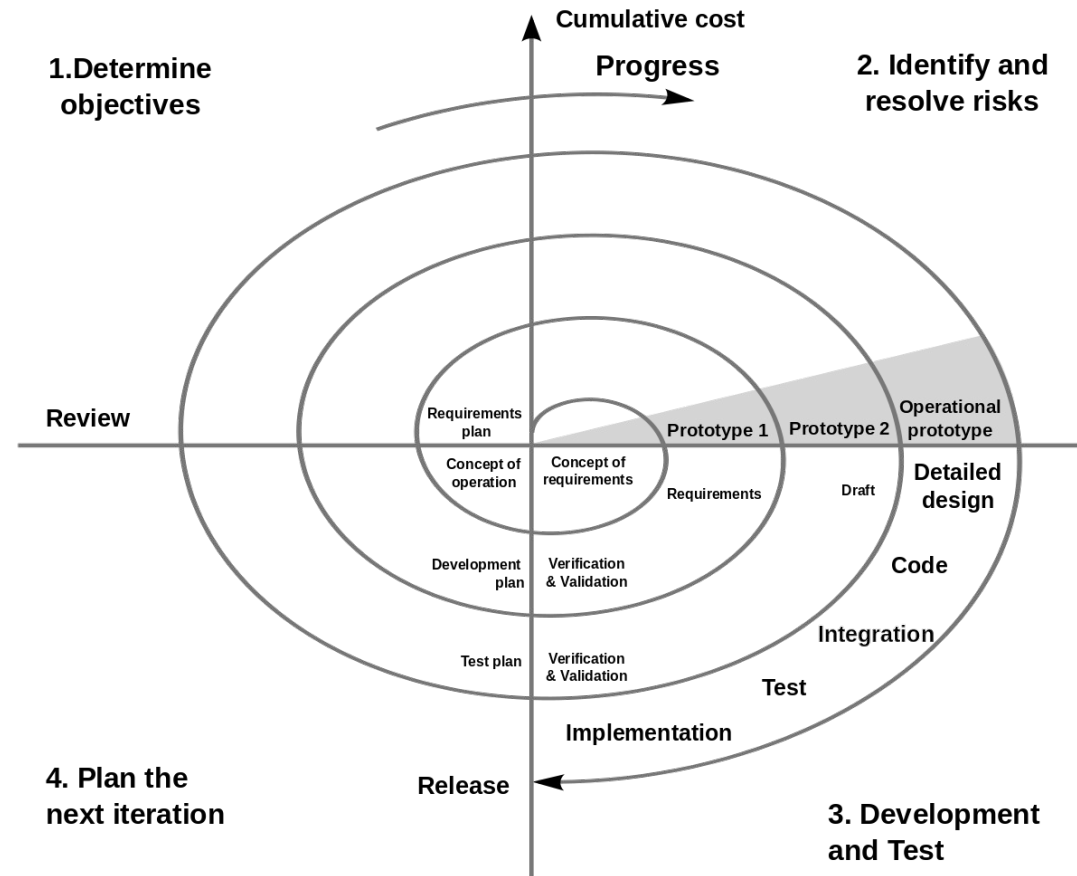
# Iterative Design is good because it minimizes risk



# Iterative Design: What's difficult about it?



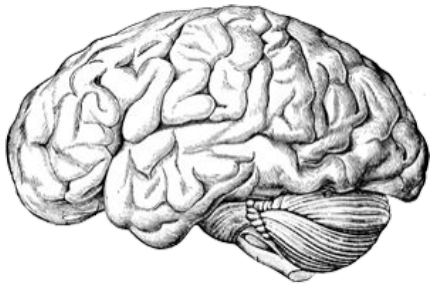
Idea



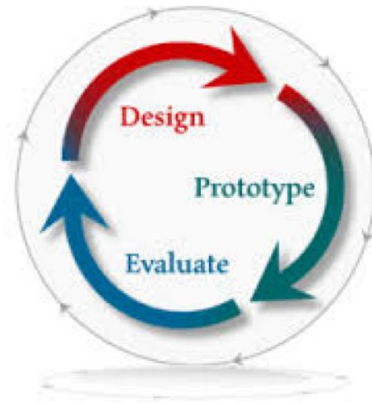
Product

The steps are not certain from the start

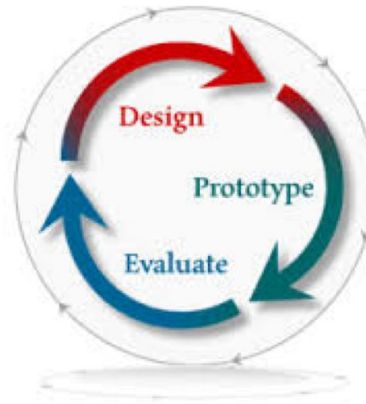
# To minimize risk on novel designs, user iteration on each risky aspect of the design



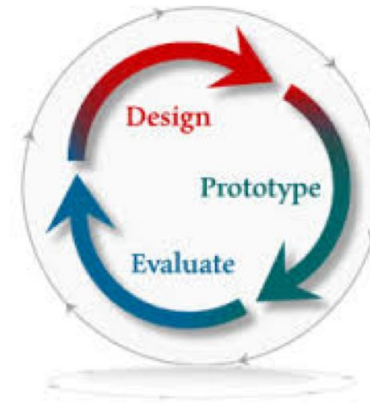
**Idea**



**Touch screen**



**Soft keyboard**

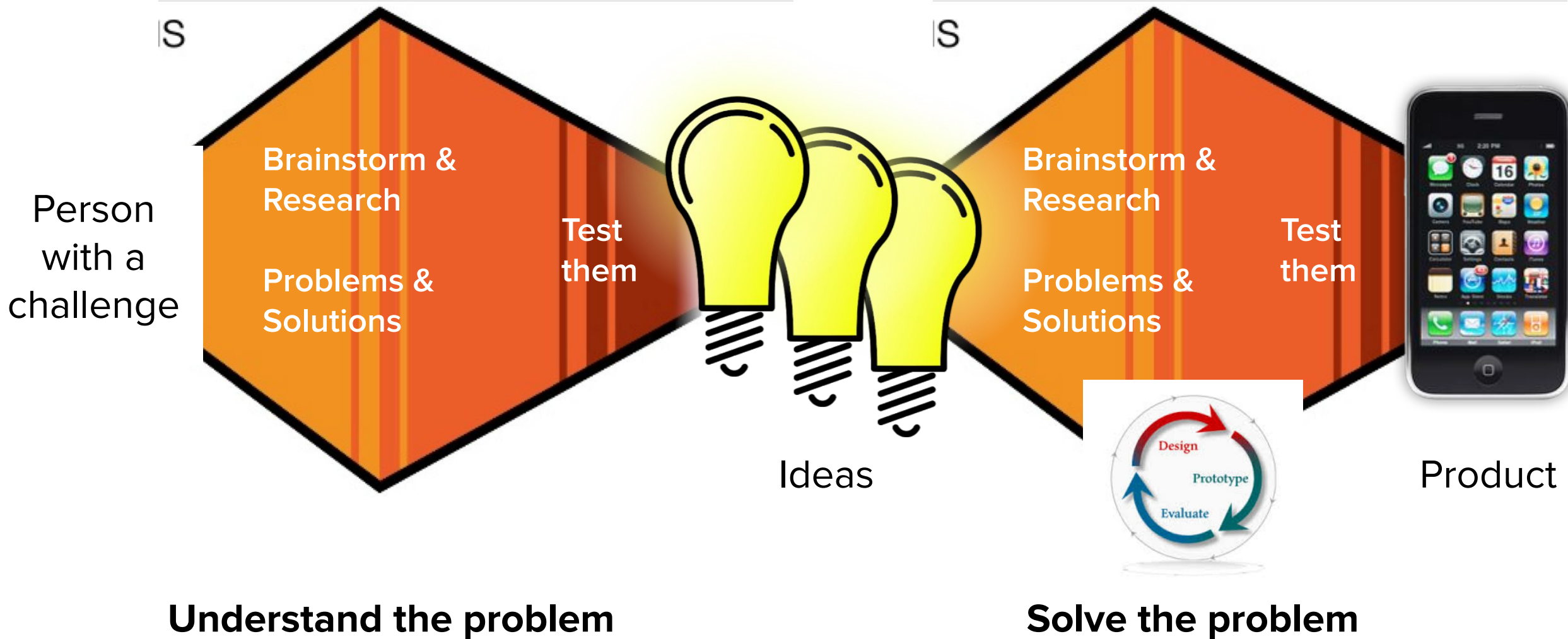


**One button**

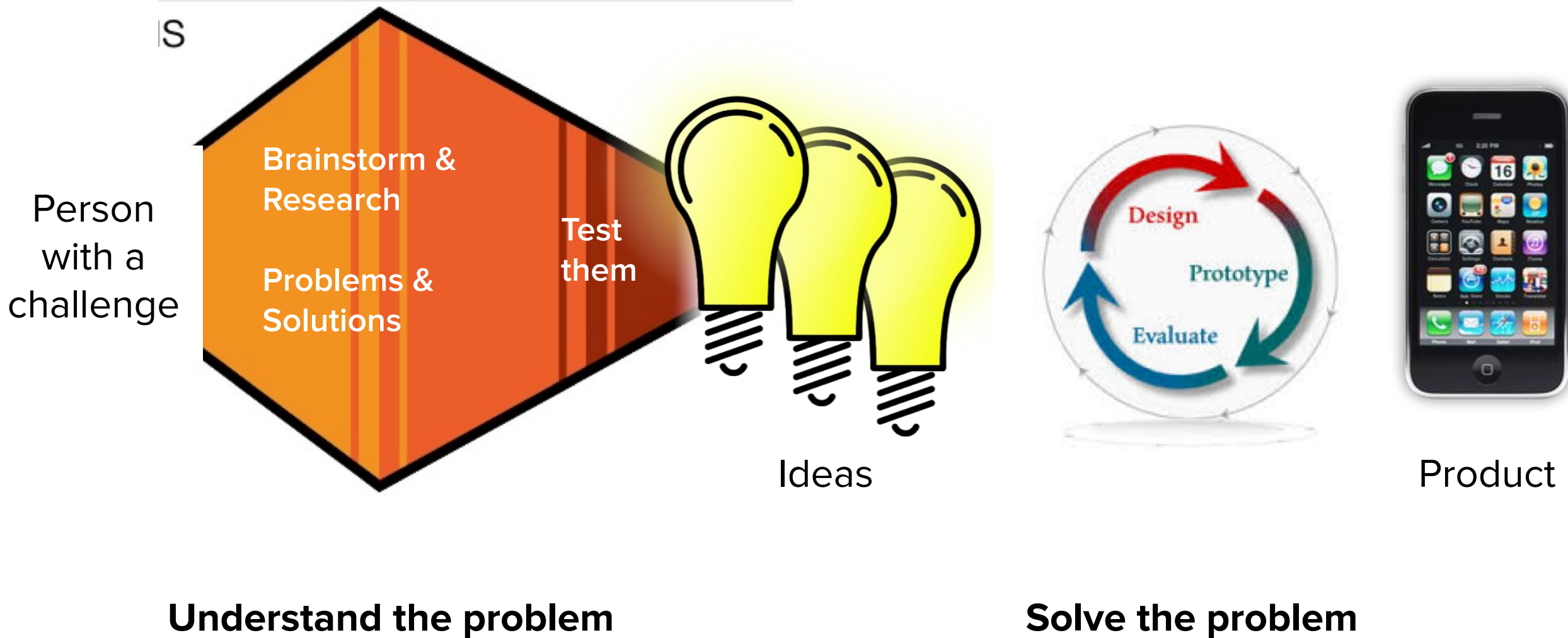


**Product**

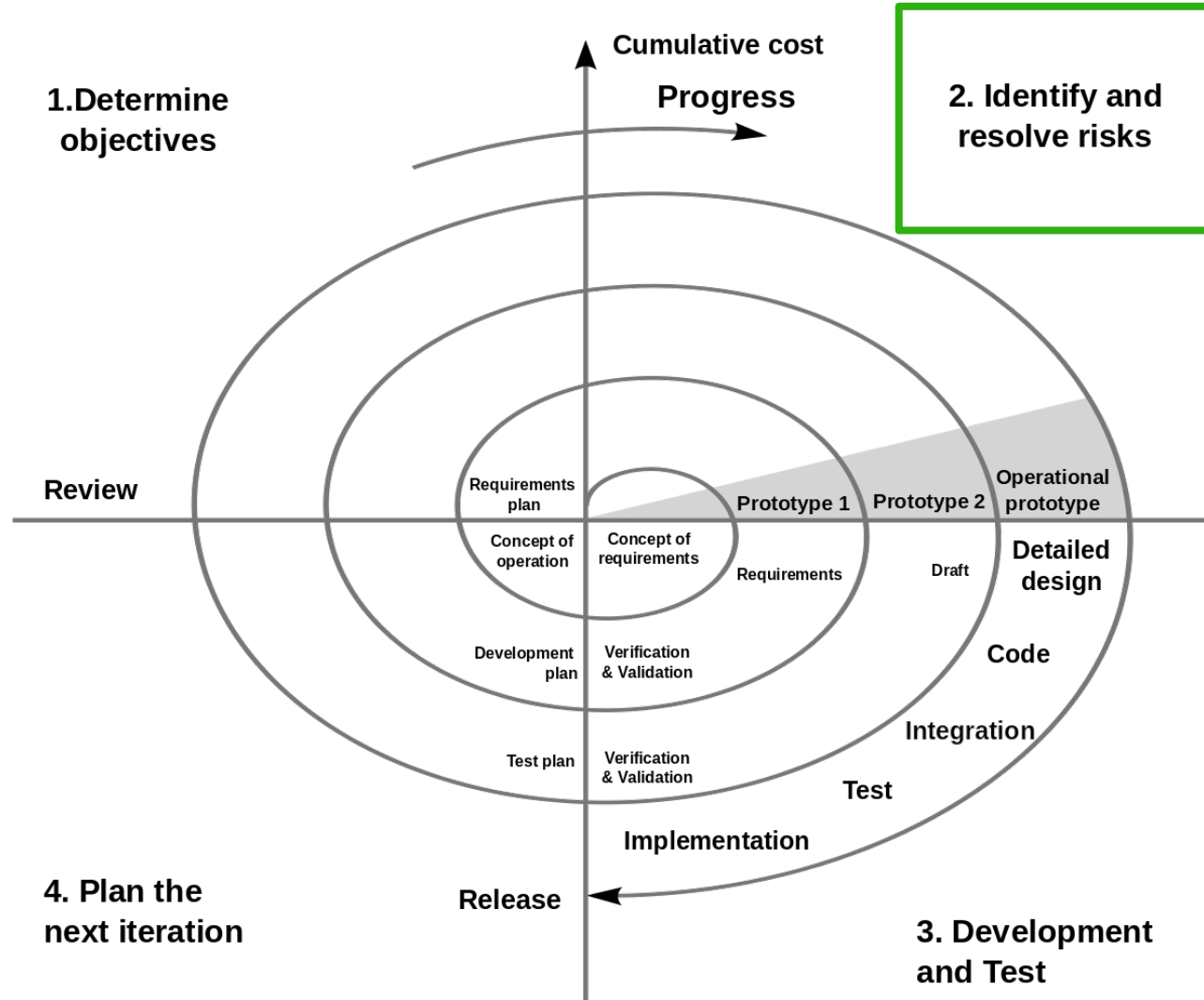
# The Double Diamond Flare and Focus



# Implementation is highly iterative



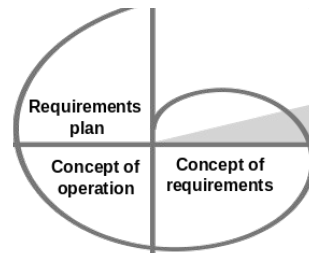
# Iterative Design is good because it minimizes risk



# The first iteration should be as low-fidelity as possible – addressing core concepts

1. Determine objectives

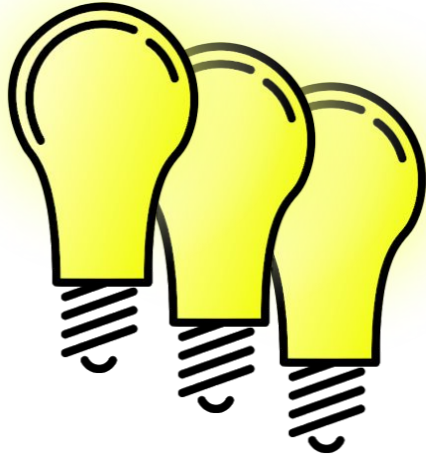
2. Identify and resolve risks



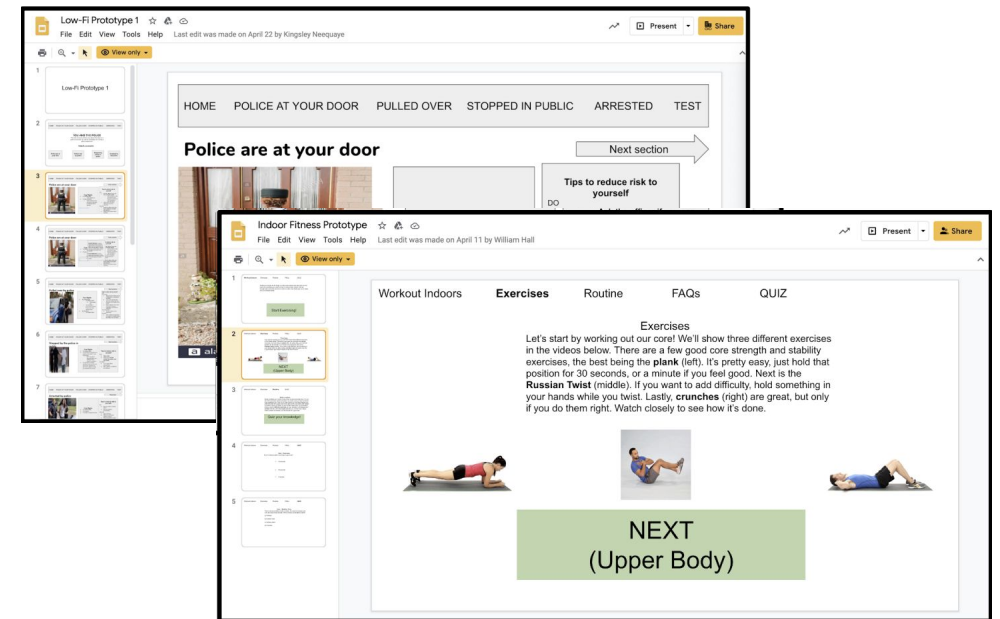
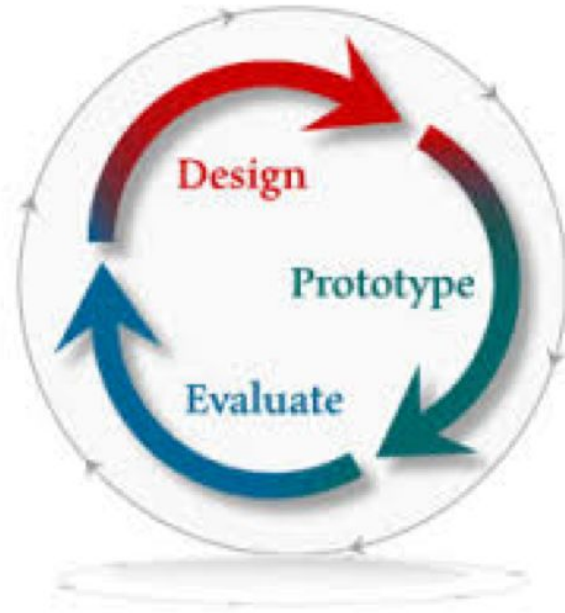
4. Plan the next iteration

3. Development and Test

# Homework 8: Low-Fidelity Prototypes (first design iteration)



Ideas



Low-Fidelity Prototypes

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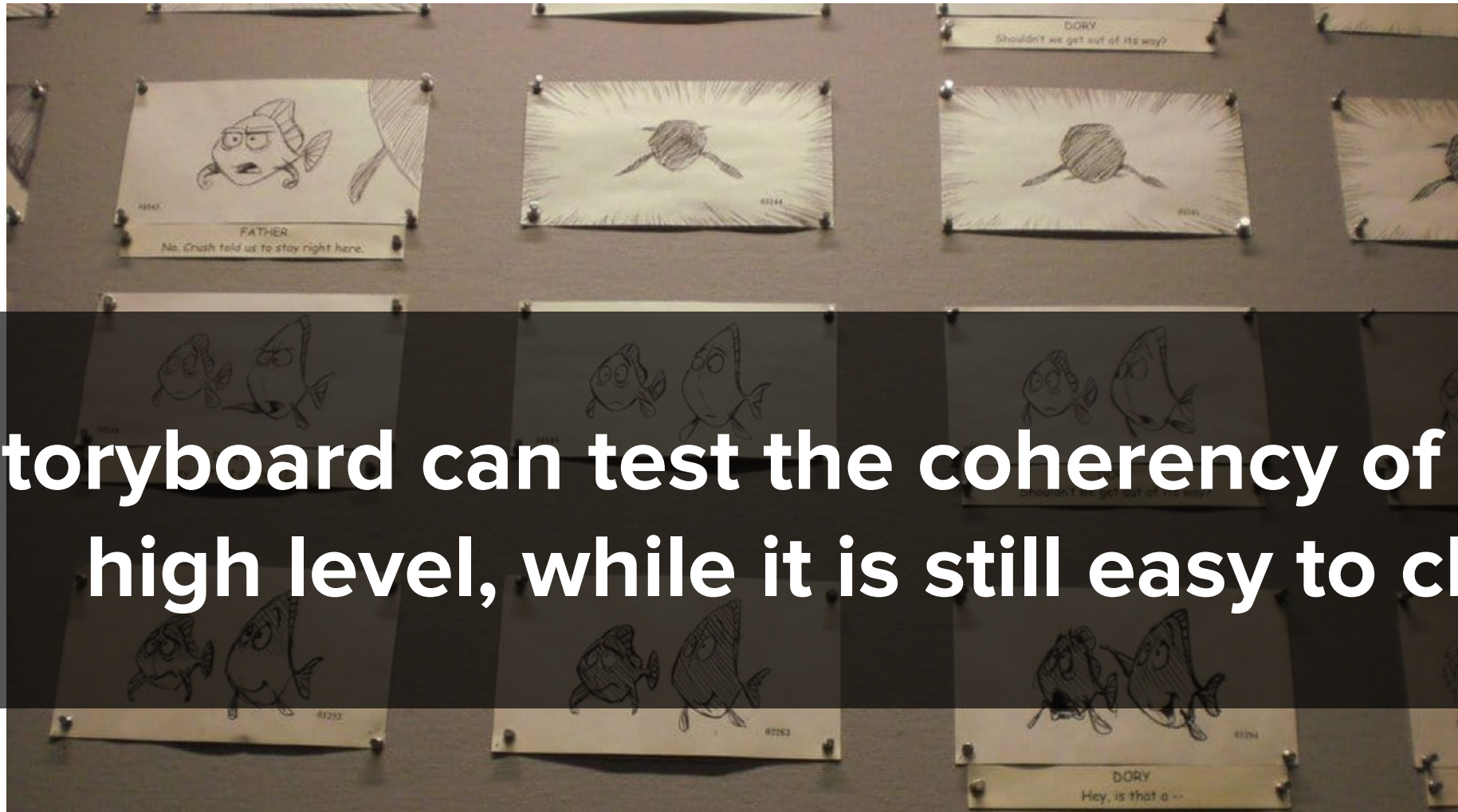
# Low-Fidelity Prototypes



# Pixar makes detailed and beautiful films

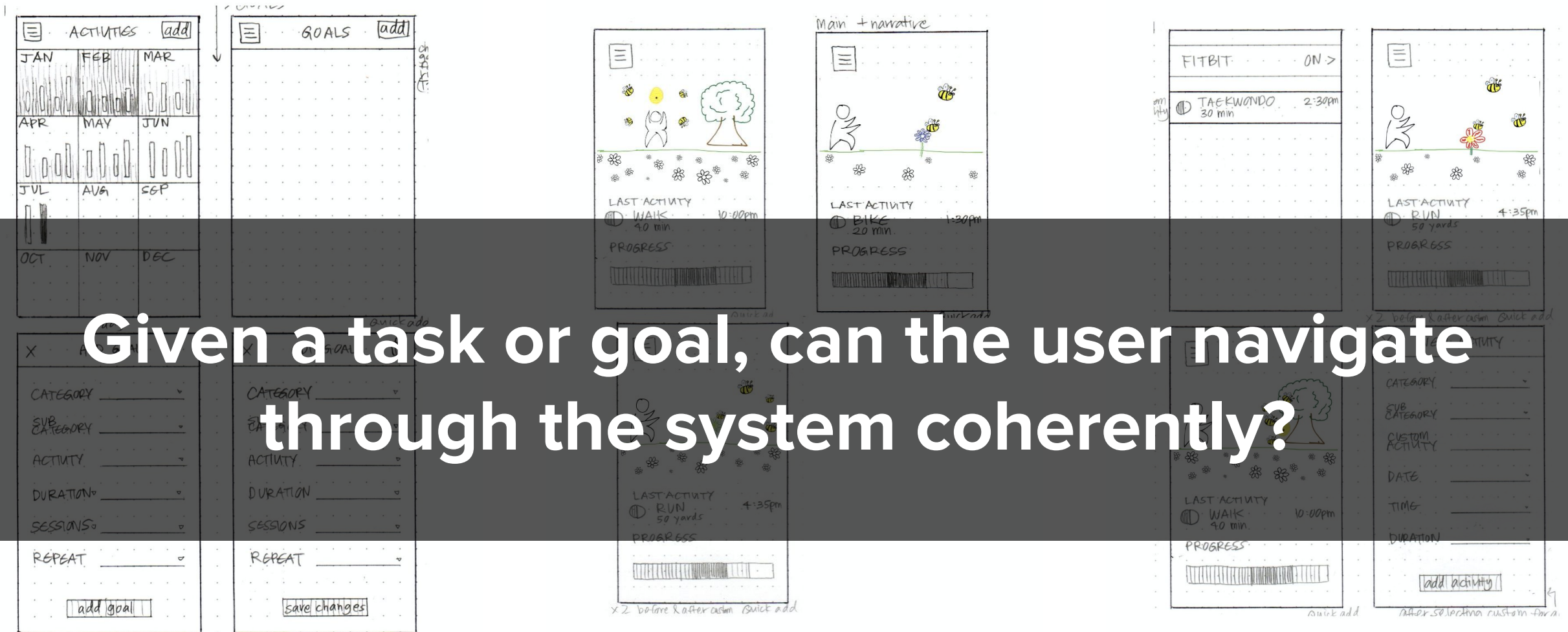


# They **always** start with a storyboard. Why?

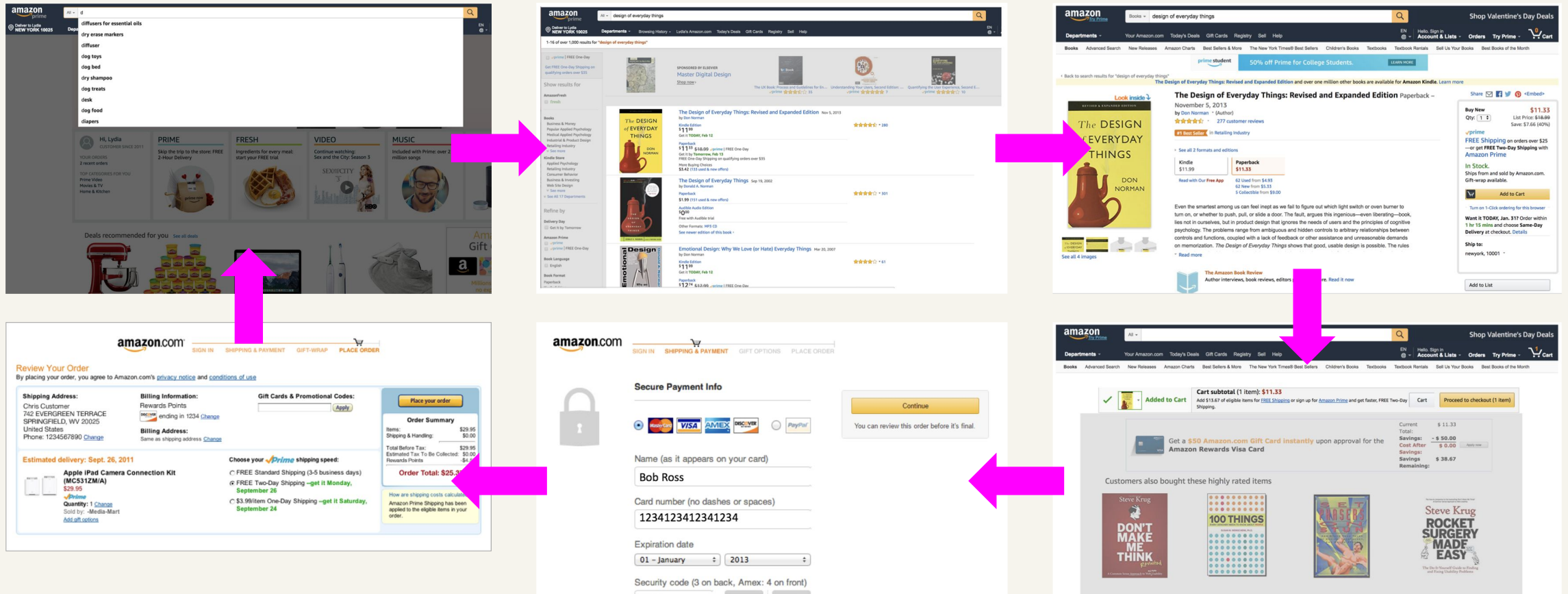


**Storyboard can test the coherency of a story at a high level, while it is still easy to change it**

# Storyboards are also good for **prototyping software interactions**



# For complex goals, guide users' attention by breaking tasks into states, options and transitions to new states



**Main prototype goal:**

“

# Coherence

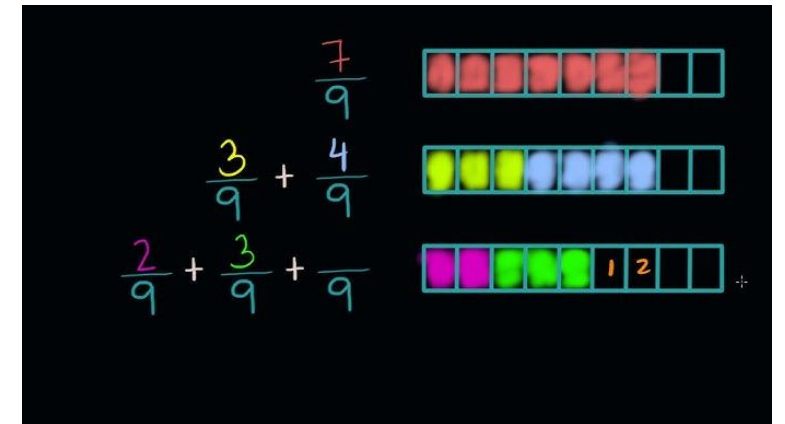
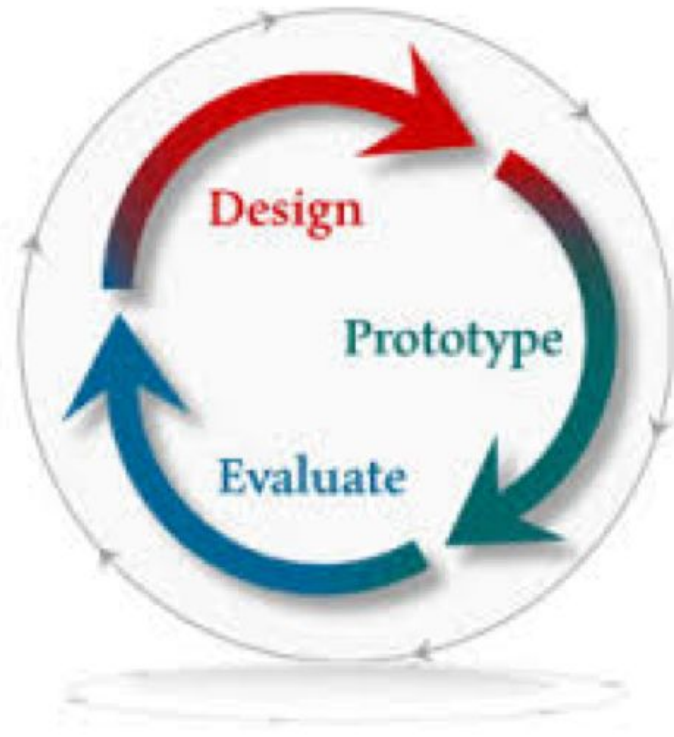
Is there a path through the system by  
which the user can accomplish their goal?



# Translating an idea into a prototype is HARD



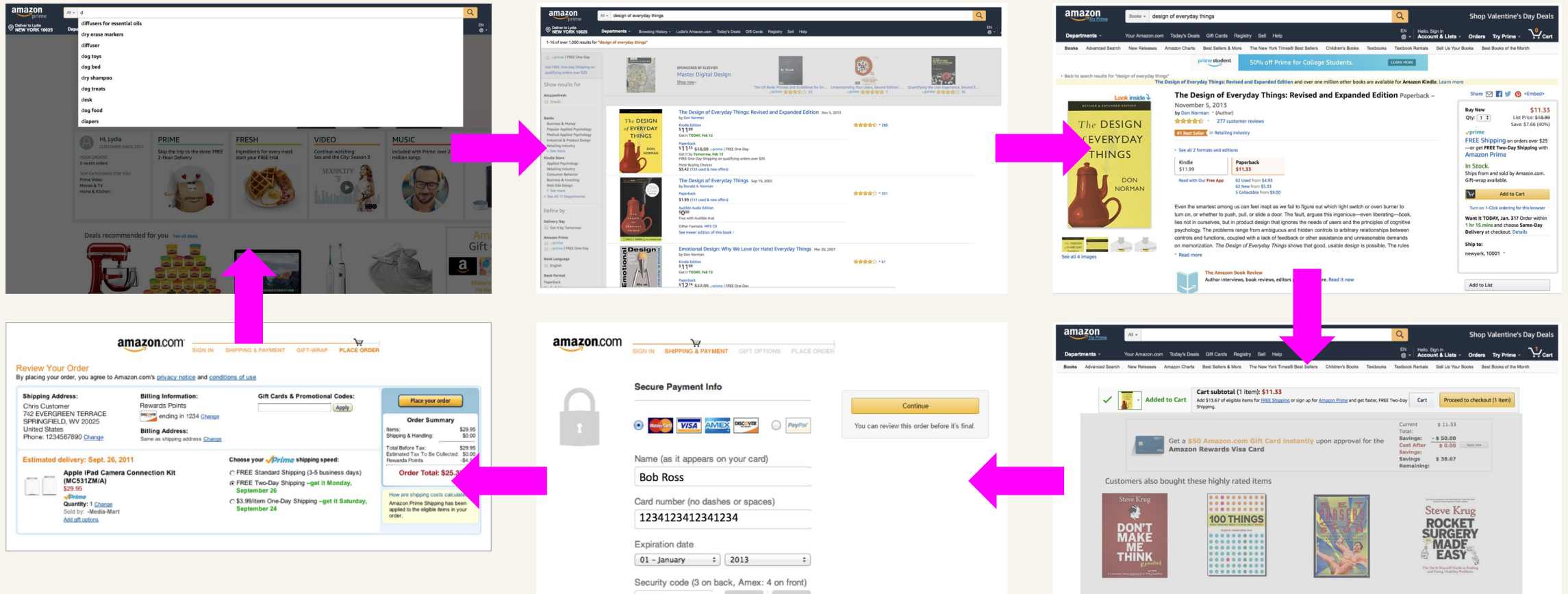
Teach fractions by  
working through  
problems slowly



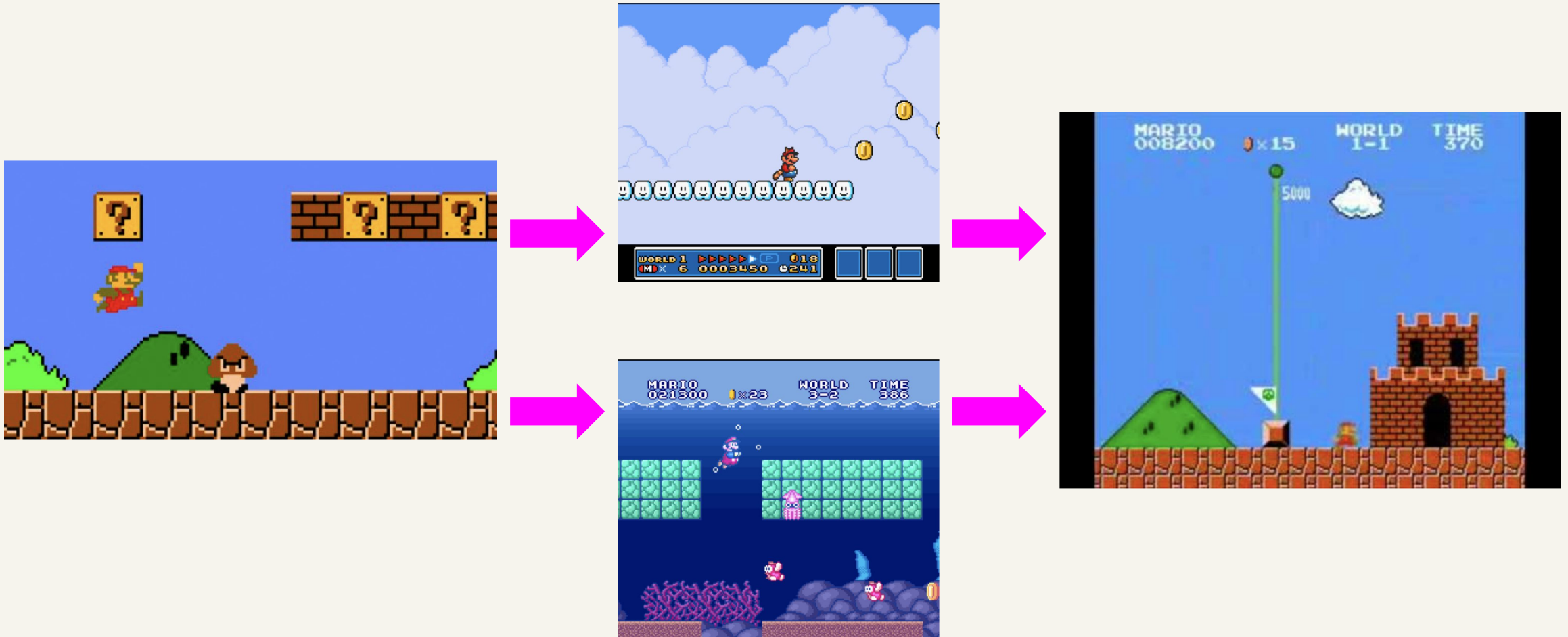
Low-Fidelity Prototype

**You will probably learn as much from making the prototype, as you will from running it.**

# Focus on breaking the task into states, options and transitions



# Think of it like a video game and you are designing the experience of your user



# Prevent errors in options: provide good defaults

StreetEasy

RENT BUY SELL BUILDINGS

Monthly payment calculator

HOME PRICE

\$740,000

DOWN PAYMENT

\$148,000 20%

LOAN TERM

30 years

INTEREST RATE\*

7.375%

MORTGAGE AMOUNT\* \$592,000

MORTGAGE PAYMENT \$4,089

MAINTENANCE FEES \$2,176

Fees co-op owners pay to cover building operating costs and NYC property taxes.

\*All calculations are estimates provided by StreetEasy for informational purposes only. The Mortgage Amount includes principal and interest. Home insurance is not included. Actual amounts or financing terms may vary. Please contact your mortgage provider for specifics.

Estimated monthly payment \$6,265

1 of 10

MAINTENANCE FEES \$2,176/mo

TAXES Included in maintenance fees

About

SELLER'S AGENT

Advertise Sign up / Log in

e.g. address, building, agent

OR SALE ↓ \$35,000

3 rooms 1 bed 1 bath

Resale

REQUEST A TOUR as early as tomorrow at 9:00 am

ASK A QUESTION

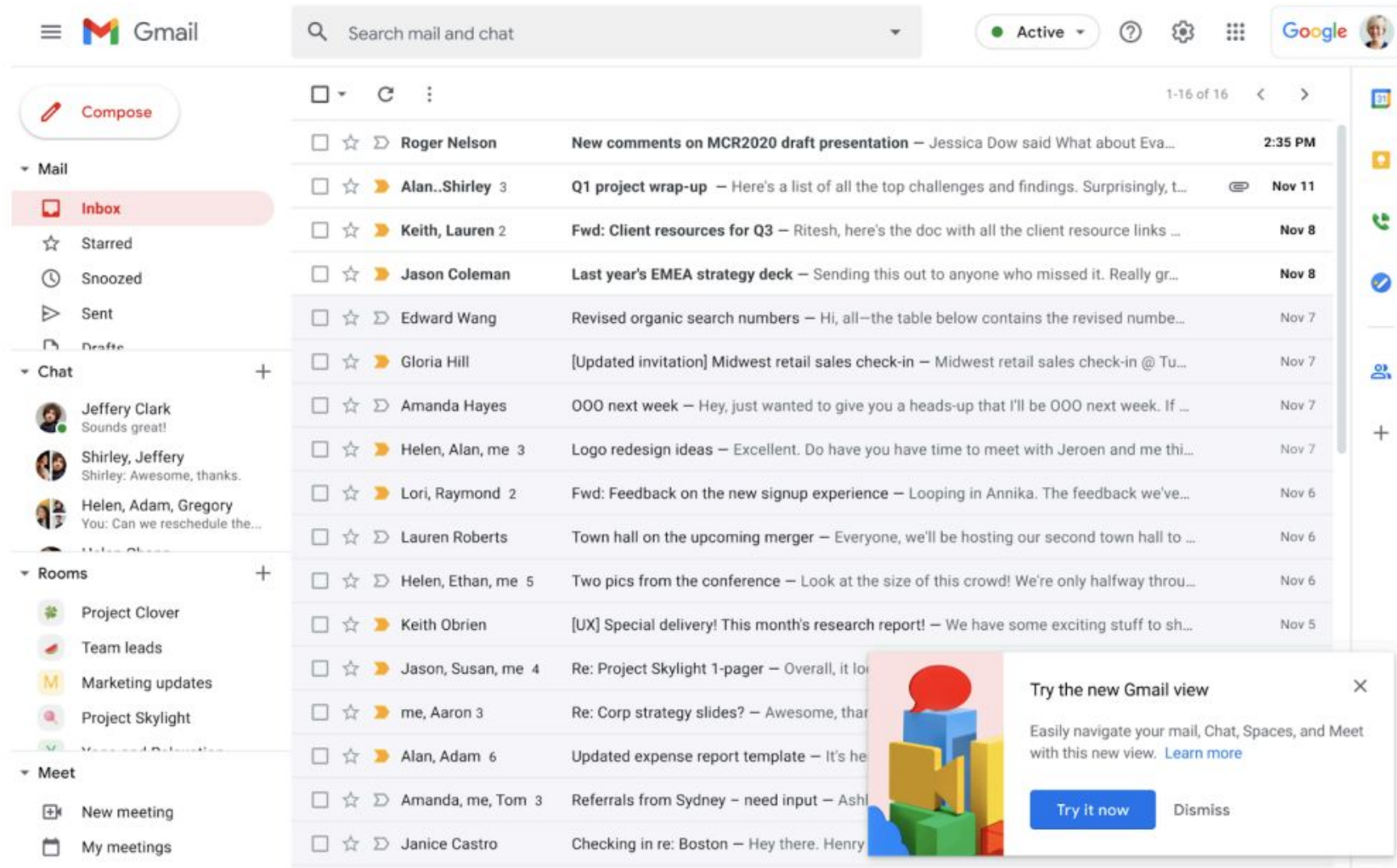
SHARE HIDE

ved by 45 users.

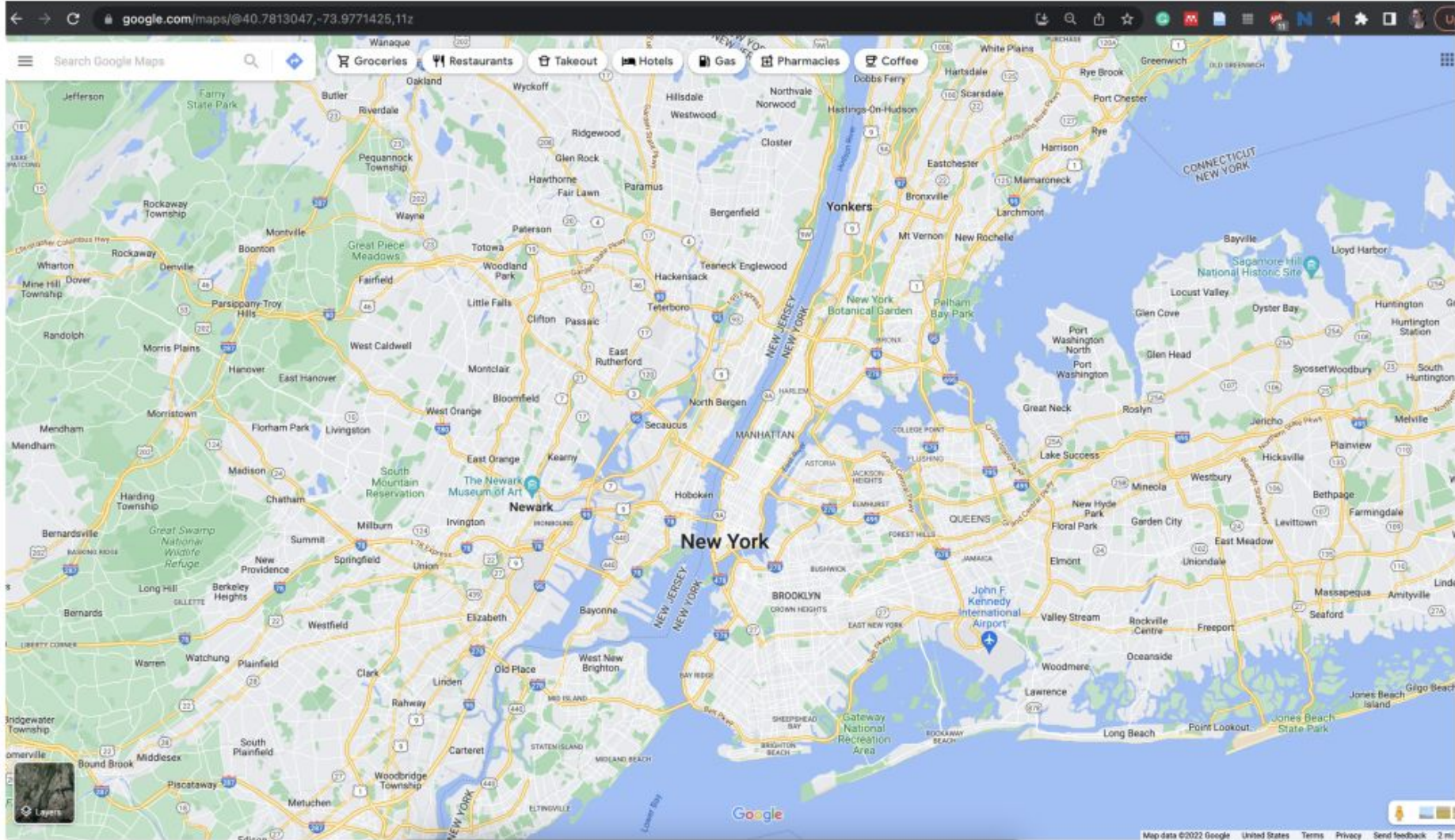
s Stevens, Real Estate Principal Office, 445 Park 10022

Good defaults will help guide the user away from making a mistake because they will be given a context to work within.

# What's the default way Gmail sorts emails?



# What's the default location on Google Maps?



# What's the default number of LABUBU to buy?

LABUBU

POP MART



NEW & FEATURED

CATEGORIES

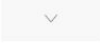
CHARACTERS

COLLABS

ACCESSORIES

MEGA

STORE PICKUP



NEW



## THE MONSTERS x HELLO KITTY - Vinyl Plush Doll

\$149.99

Quantity

- 1 +

Max 1 set per person

World Trade Center



SIGN IN TO BUY (MEMBERS ONLY)

Details



Brand: POP MART

Online Release [Official Website]: March 12, 2026 at 7:00 PM [...]

SIZE:

Product Dimensions: 19\*16\*38cm / 7.48\*6.3\*14.96inches

Package Dimensions: 25\*18\*40.5cm / 9.84\*7.09\*15.94inches

MATERIAL: Polyester / PVC / ABS / Cotton; Stuffing: Polyeste...

Not suitable for persons under 15.

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# Paper Prototype Example



# Paper Prototype Example

**Write down a persona: person, high level goal, 4 or 5 subgoals**

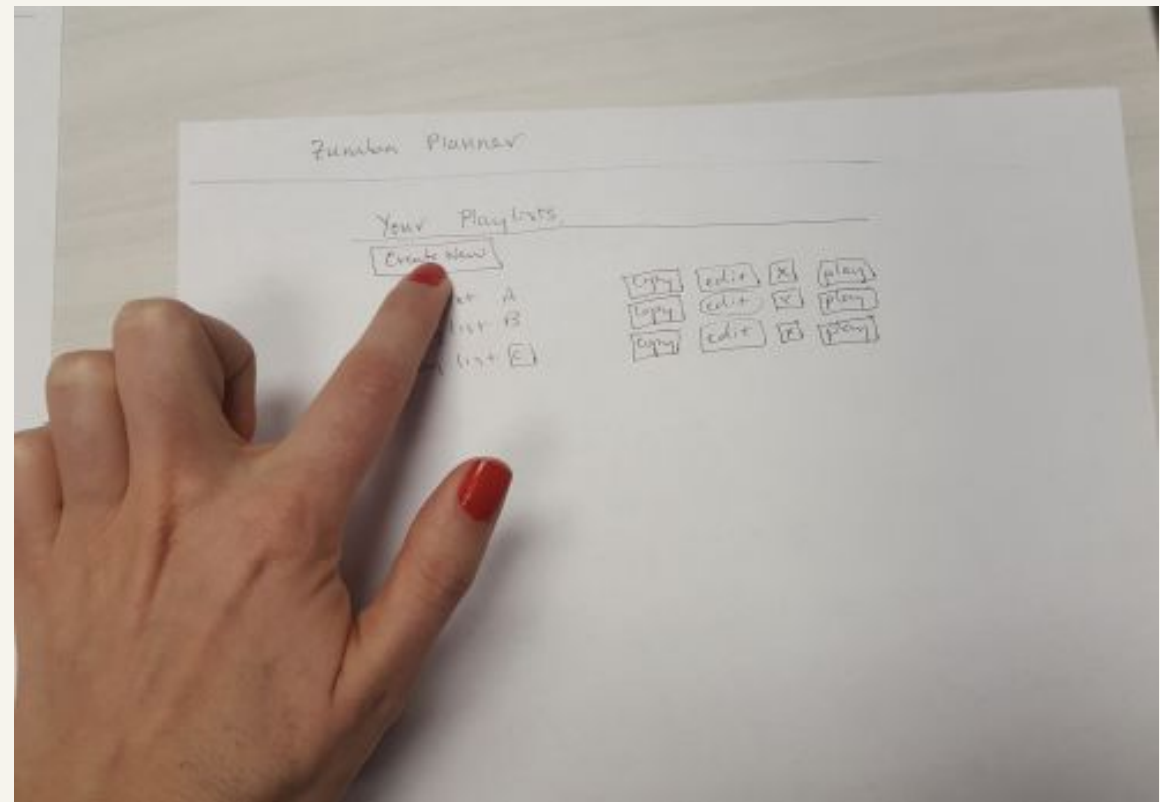
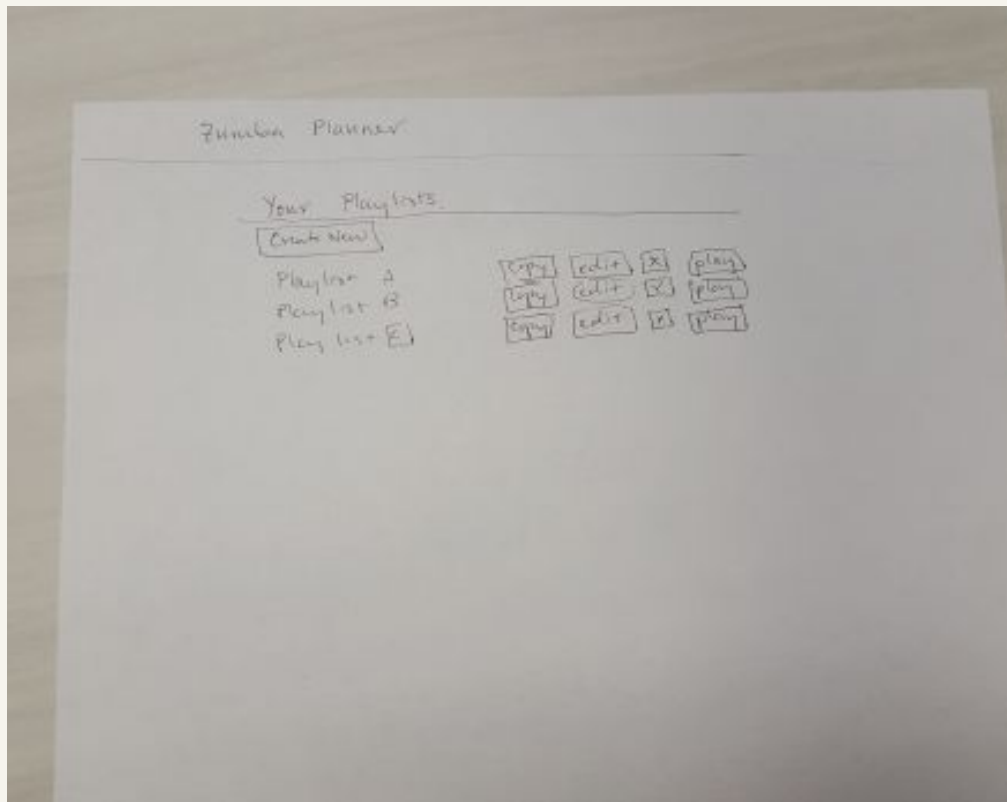
**Idea:** Zumba playlist maker

- **Person**
  - You are Catherine -- a spinning instructor at SoulCycle in New York City
- **Goal**
  - Your goal is to make a playlist of dance songs that last 20 minutes
- **Subgoals**
  - Create a new playlist
  - Add a song to your playlist
  - Add songs to the playlist until the list is 20 minutes long
  - If the playlist is too long, remove a song
  - Play the playlist



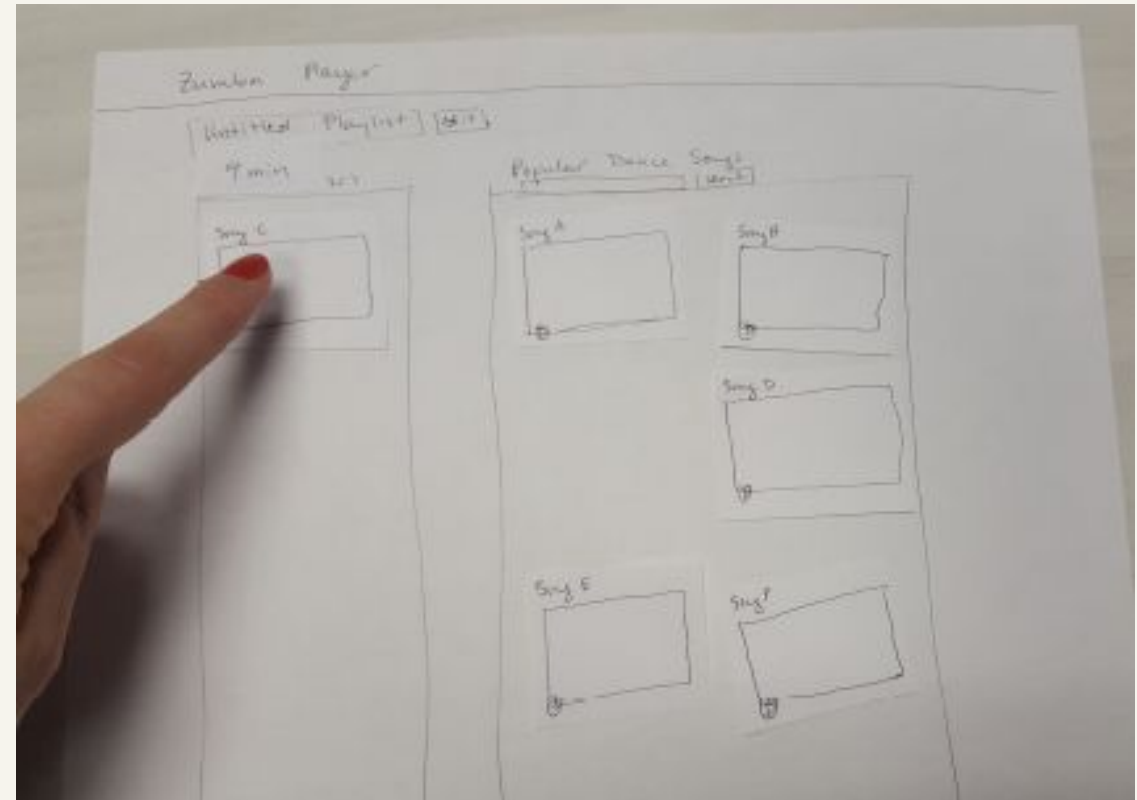
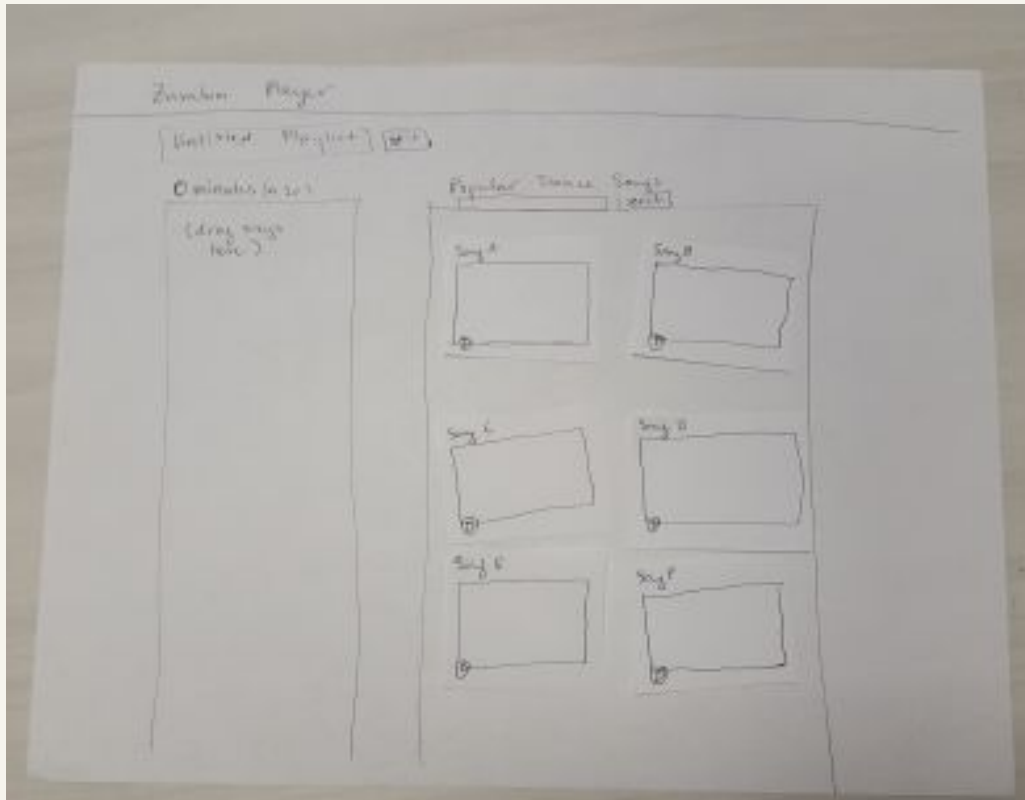
# Paper Prototype Example

## Subgoal 1: Create a new playlist



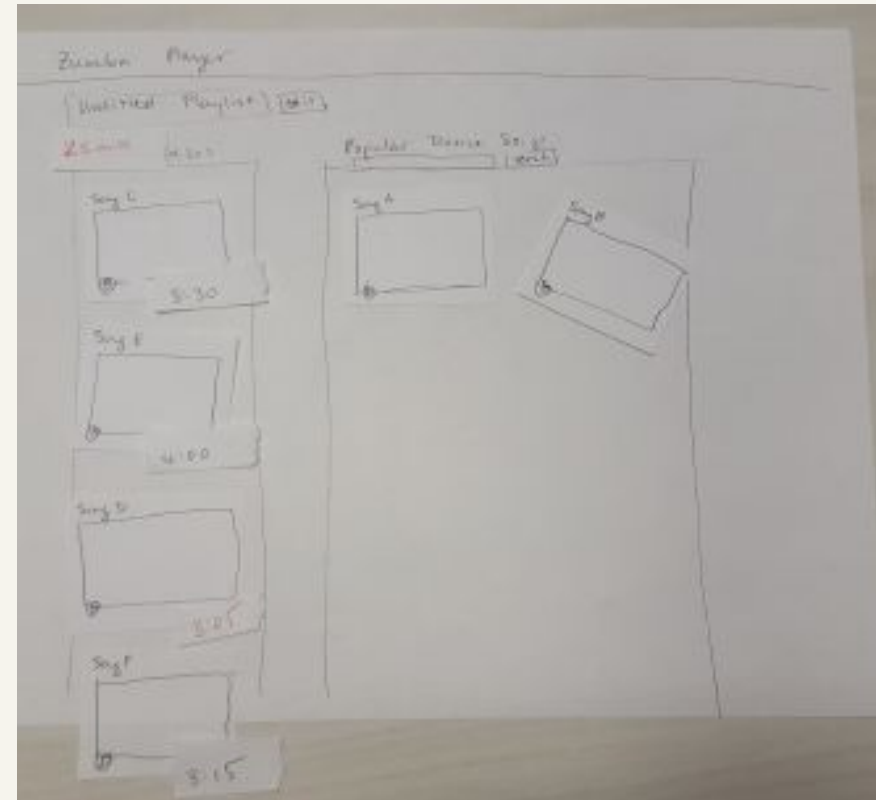
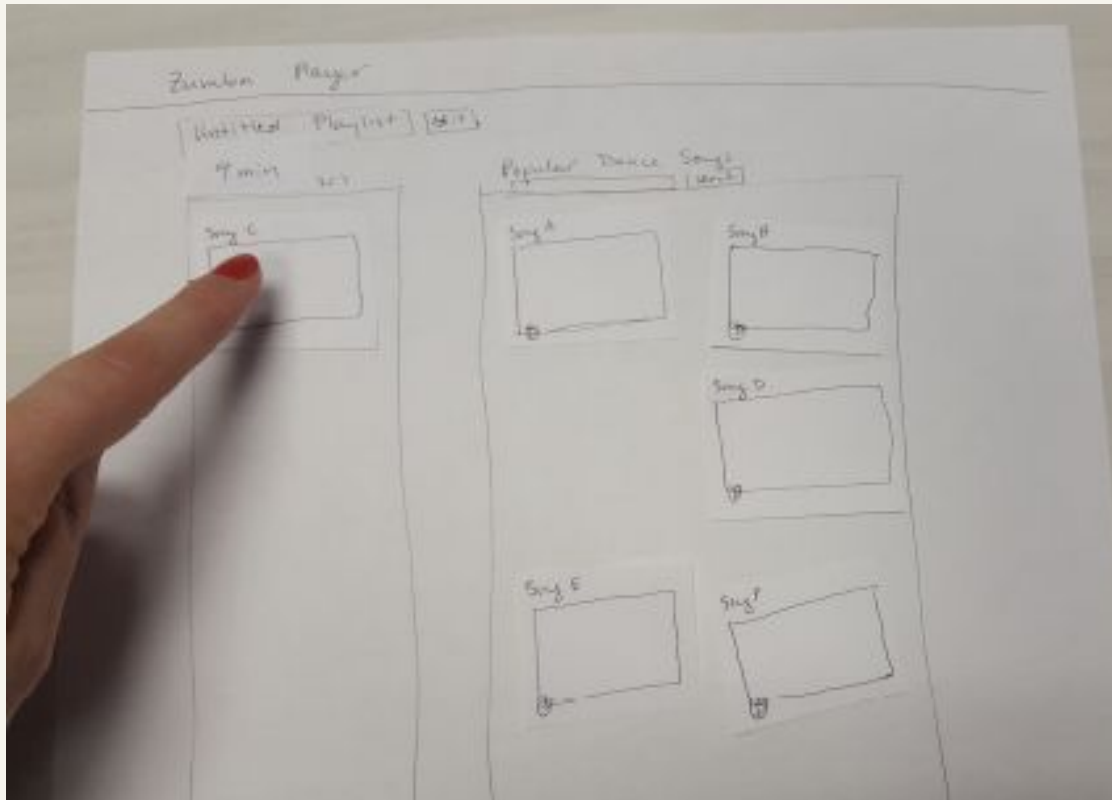
# Paper Prototype Example

## Subgoal 2: Add the first song to the playlist



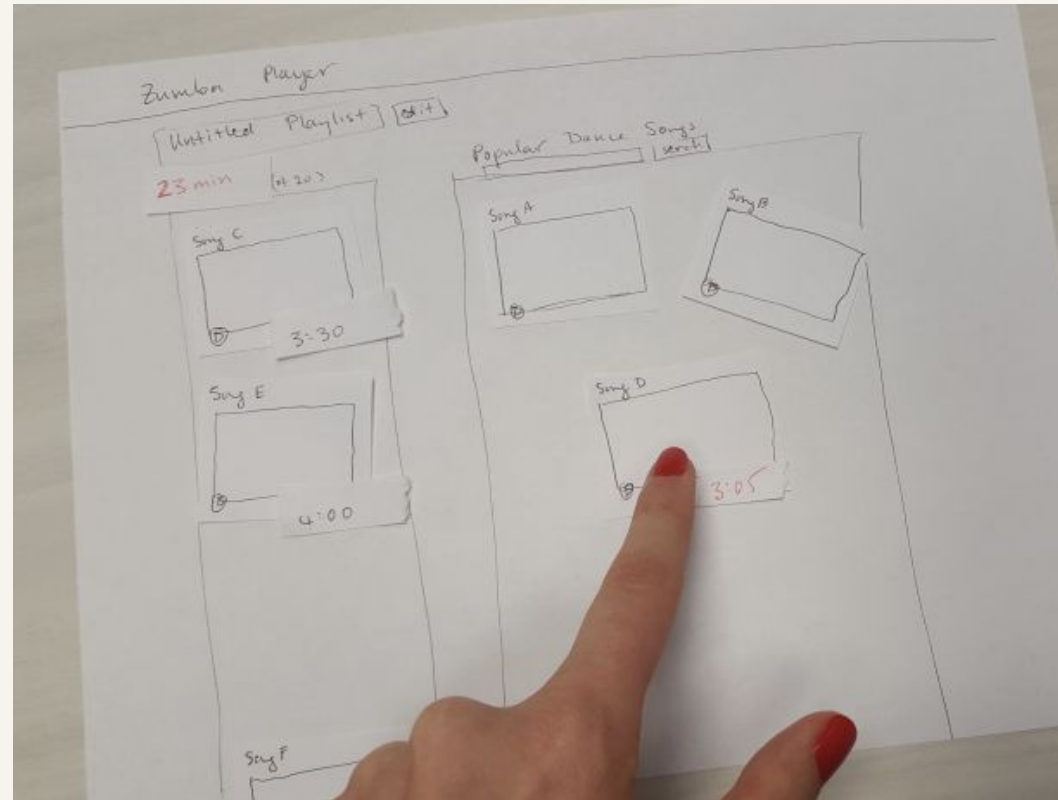
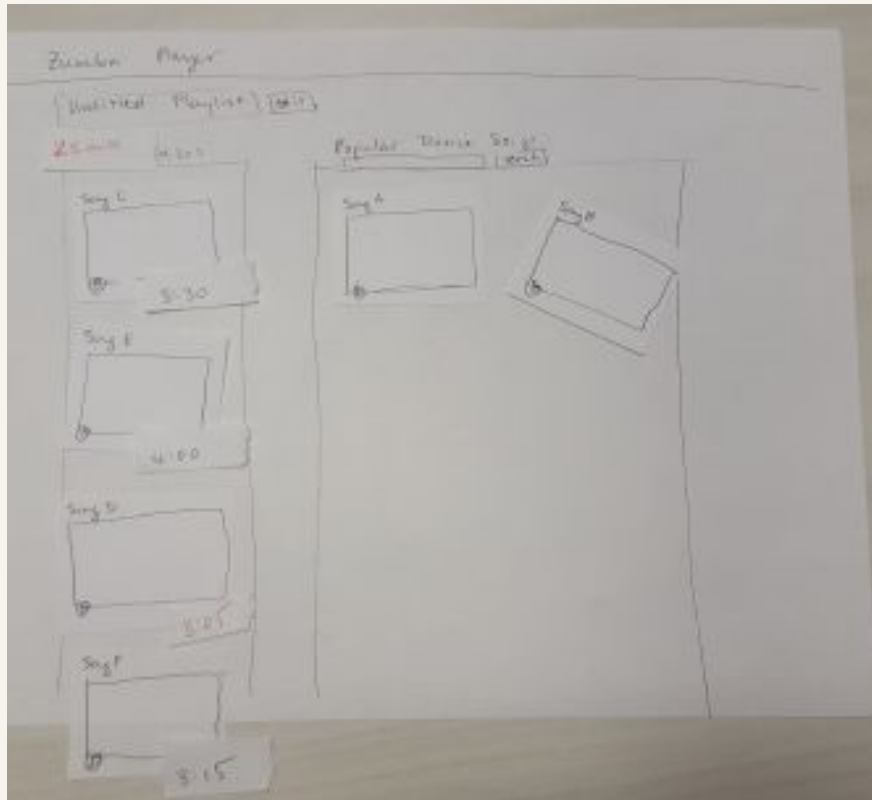
# Paper Prototype Example

## Subgoal 3: Add songs until the playlist is 20 minutes



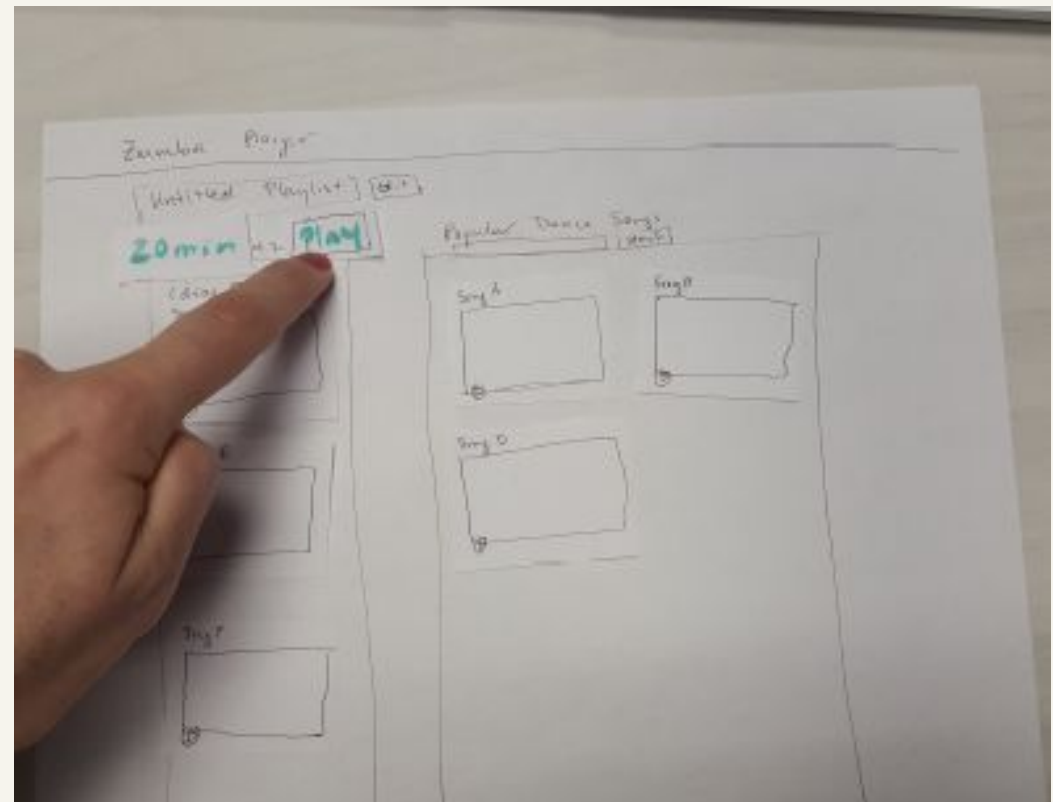
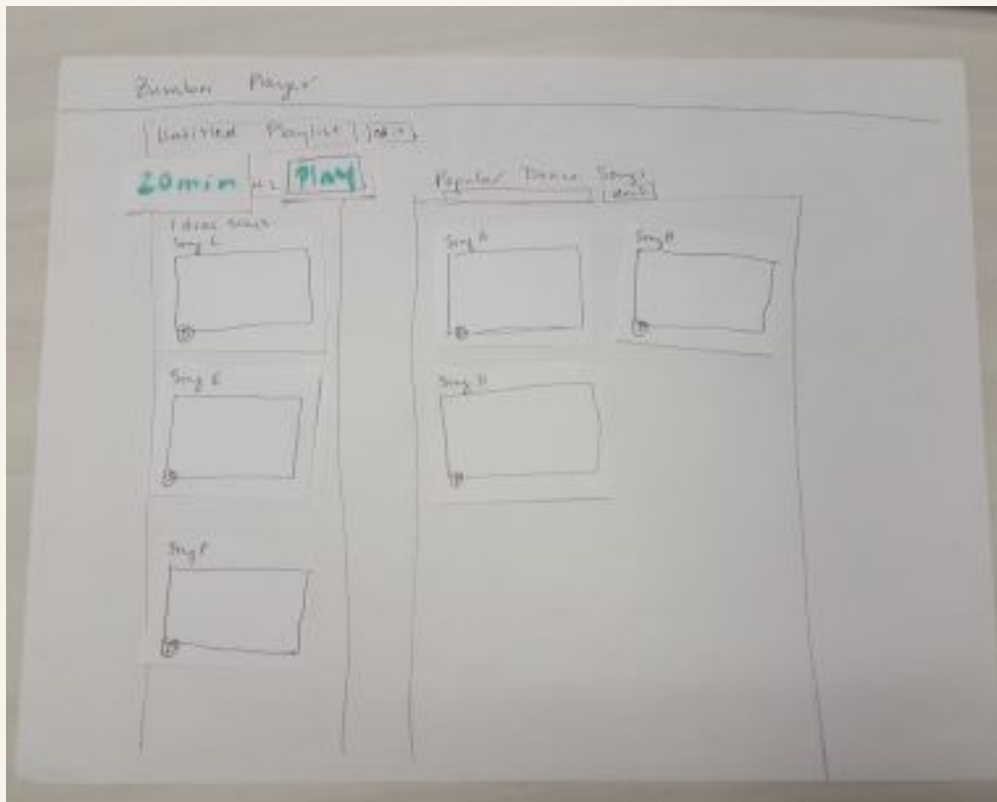
# Paper Prototype Example

## Subgoal 4: Remove songs until the playlist is 18 minutes



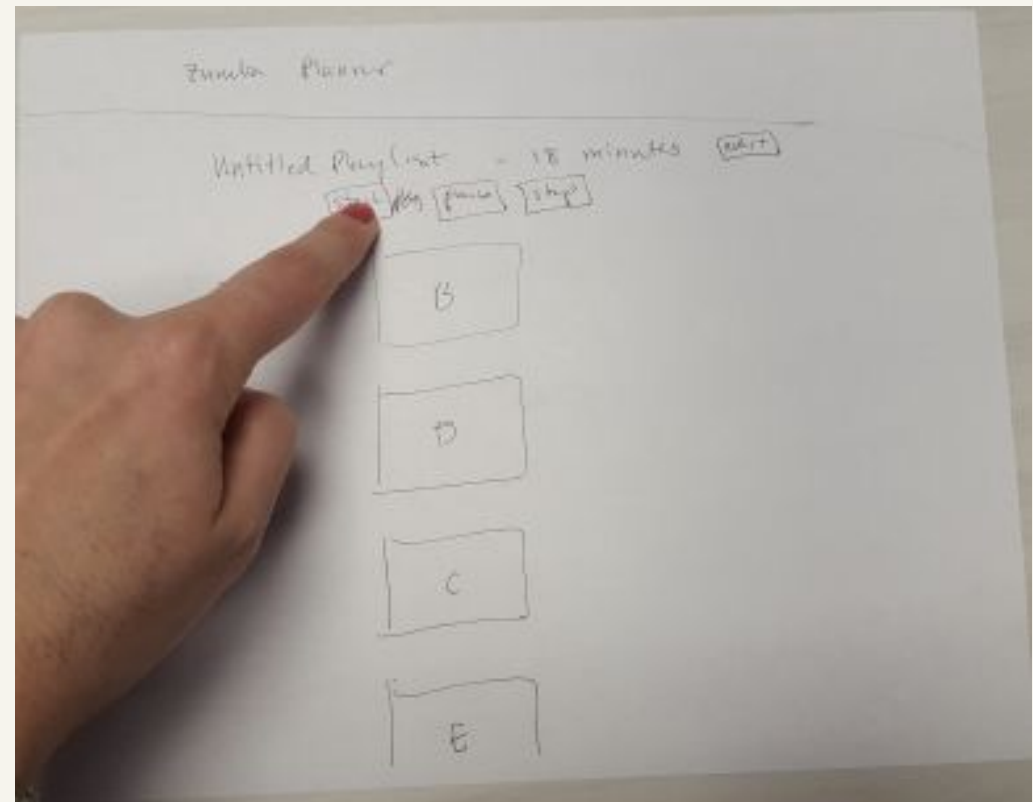
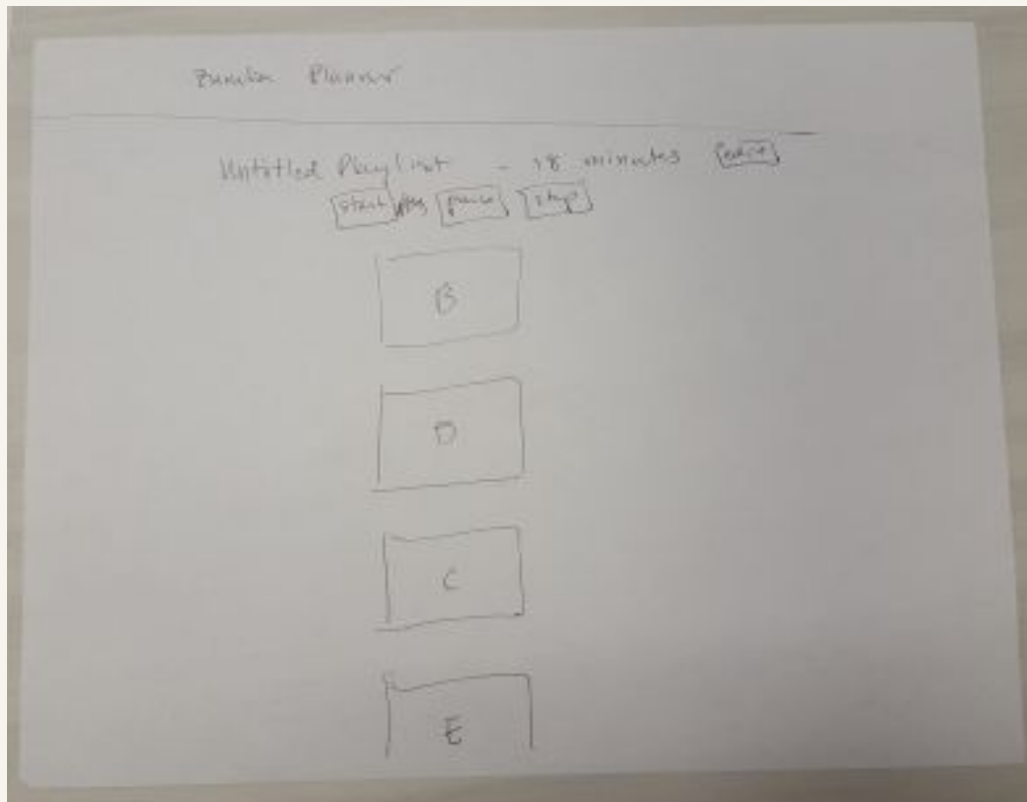
# Paper Prototype Example

## Subgoal 5: Play the playlist (part 1)



# Paper Prototype Example

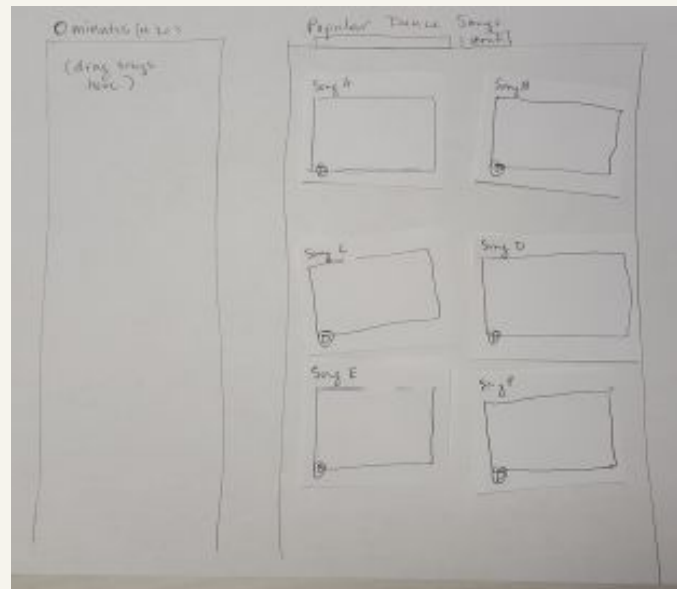
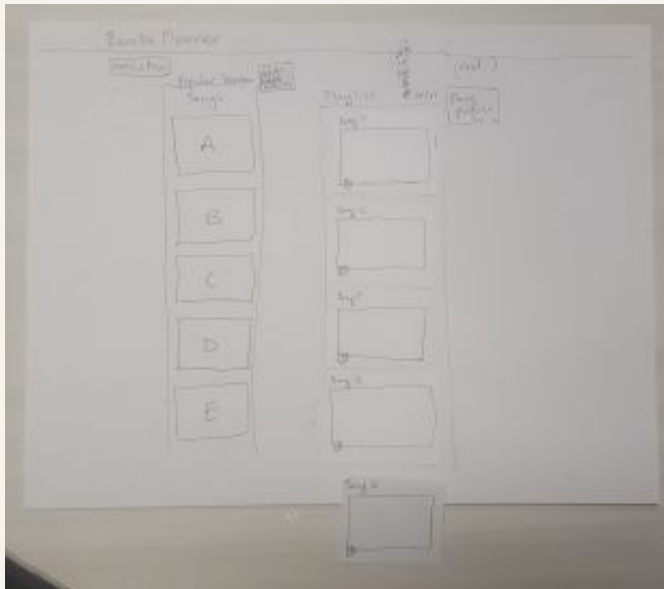
## Subgoal 5: Play the playlist (part 2)



**You will probably learn as much from making the prototype as you will from running it.**

# Paper Prototype Example

We started with the playlist drag interface

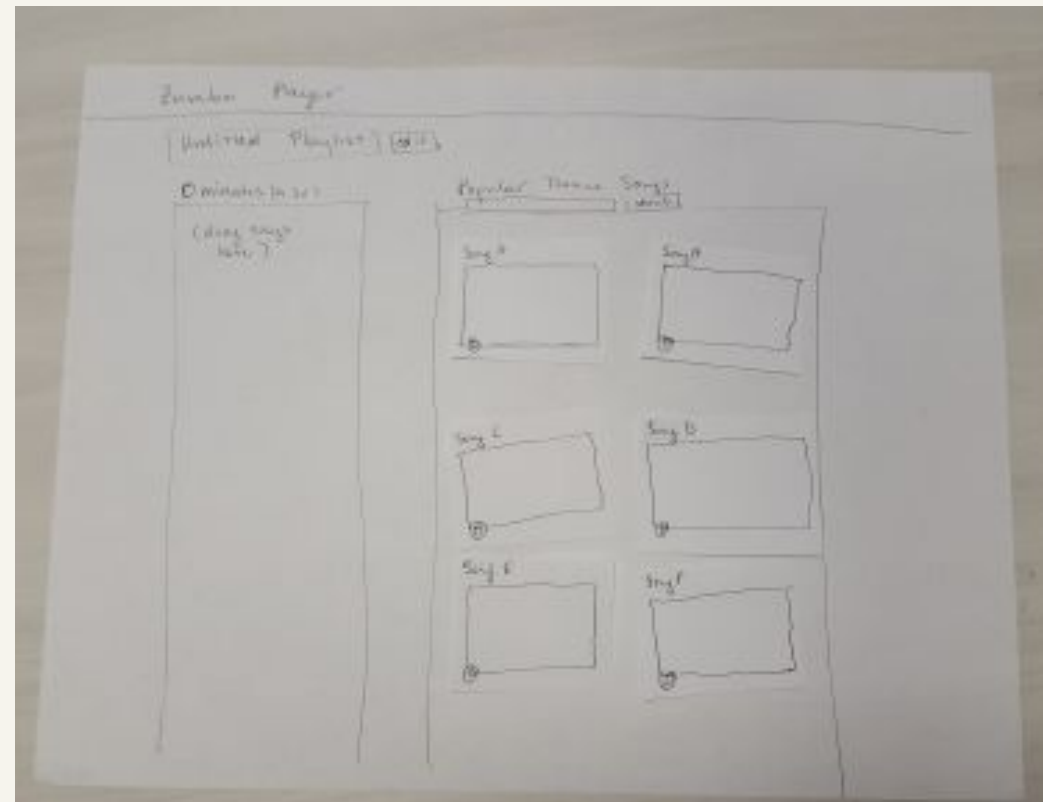
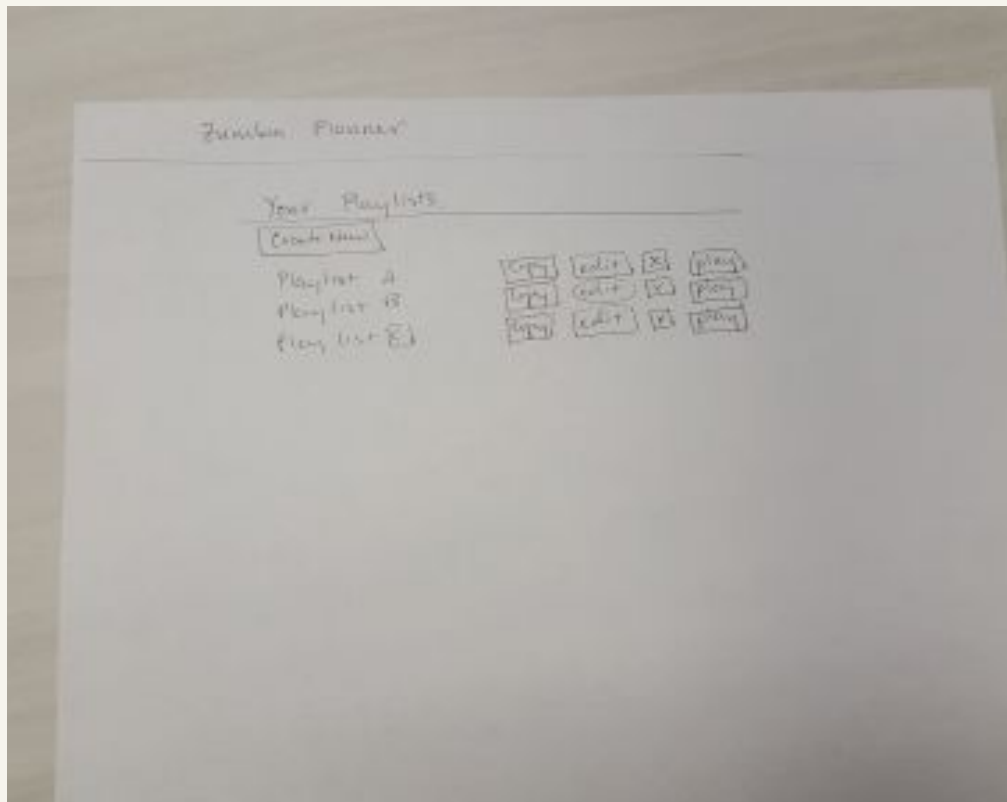


Then realized we needed more songs to pick from, so made the songs to pick from two columns



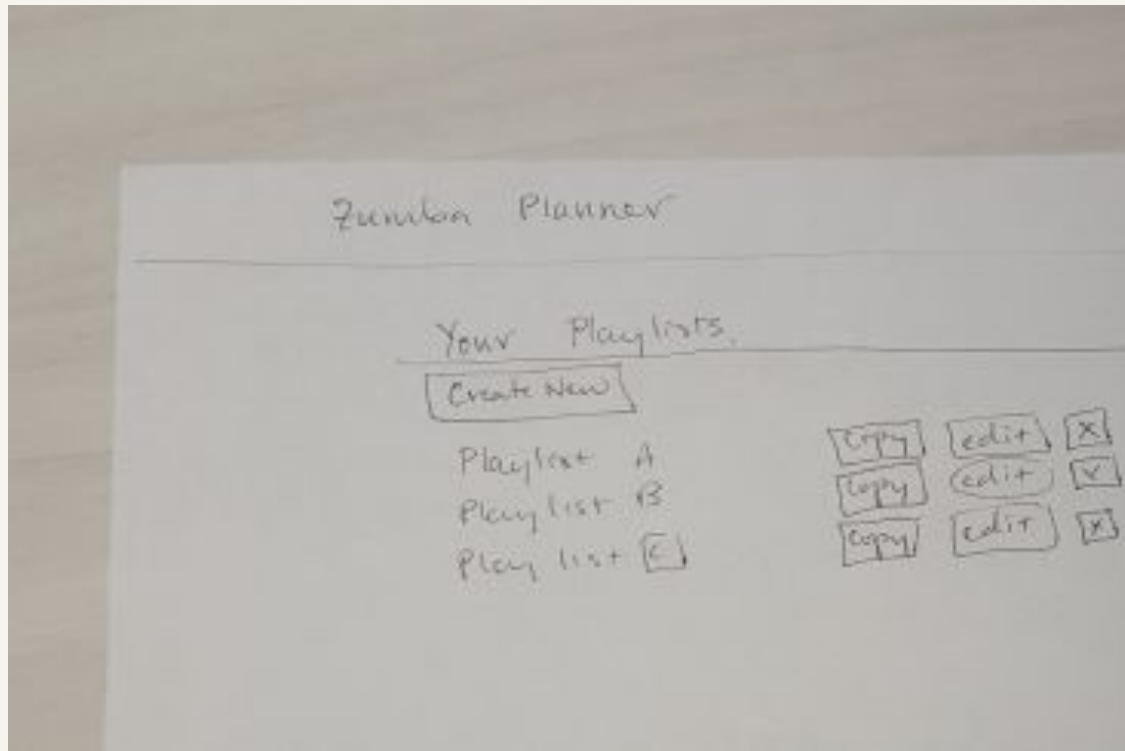
# Paper Prototype Example

We learned that playlists need names



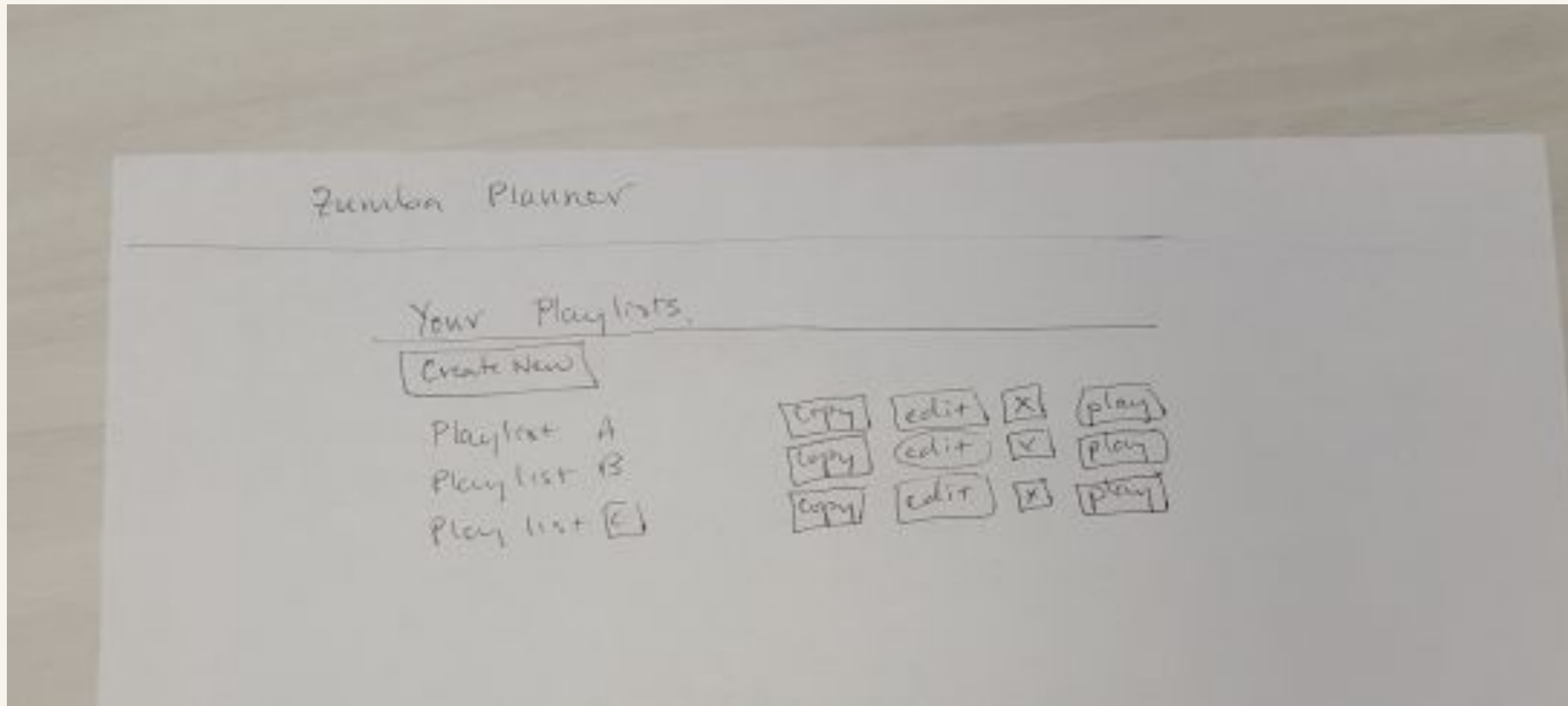
# Paper Prototype Example

We also learned that we need a homepage to create playlists from; and possibly list previously made playlists



# Paper Prototype Example

We learned that we need CRUD operations on the lists of playlists



# **Sam's Design Process for COMS 4170 UID**



# Write down a persona: person, high level goal, subgoals

**Idea:** Learn to tell the difference between impressionism and post-impressionism

- **Person**

- You are David – a Columbia Student new to NYC who loves museums. He likes art but would like to know more about the modern movements which seems confusingly similar to him

- **Goal**

- Be able to determine whether a painting is impressionism or post-impressionism and say why.

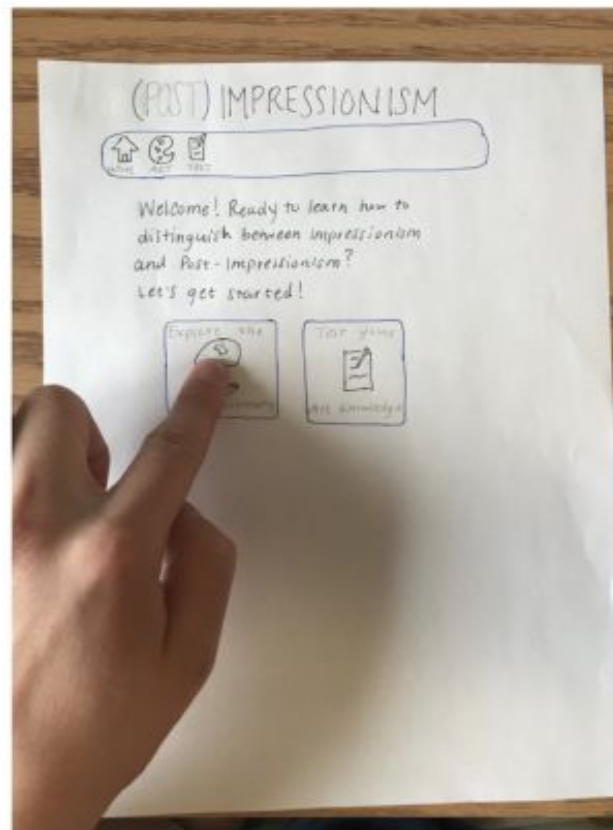
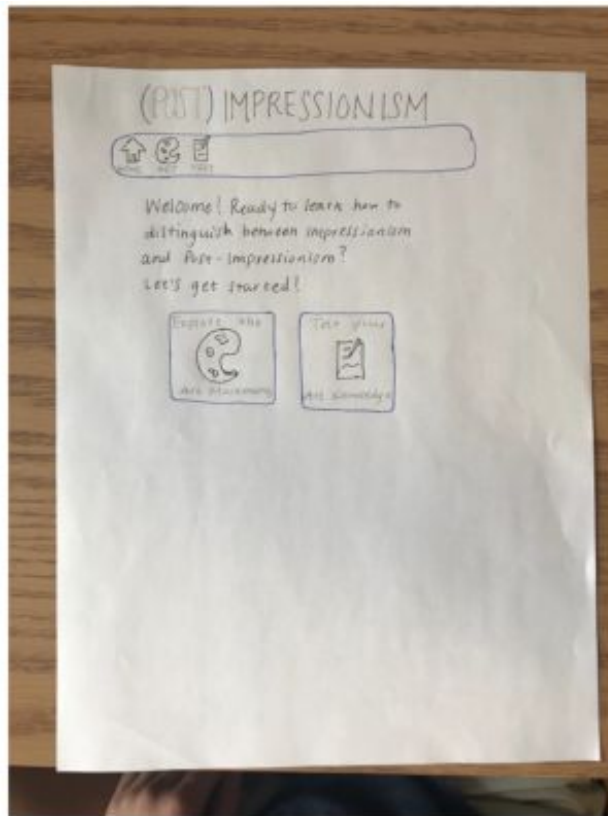
- **Subgoals**

1. Learn the major hallmarks of impressionism
2. Learn the major hallmarks of post-impressionism
3. Take a quiz within 10 minutes



# Sam's Paper Prototype

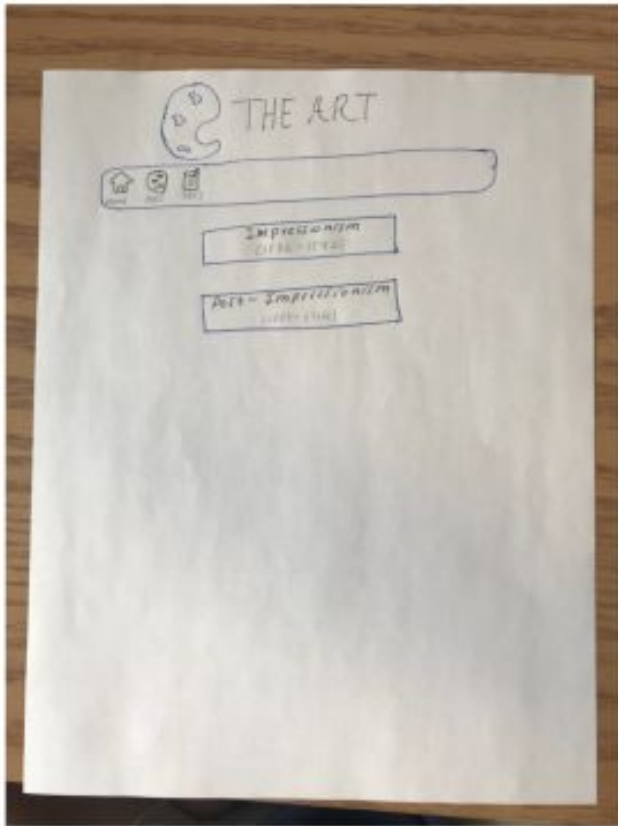
## 1. Home Screen



# Sam's Paper Prototype

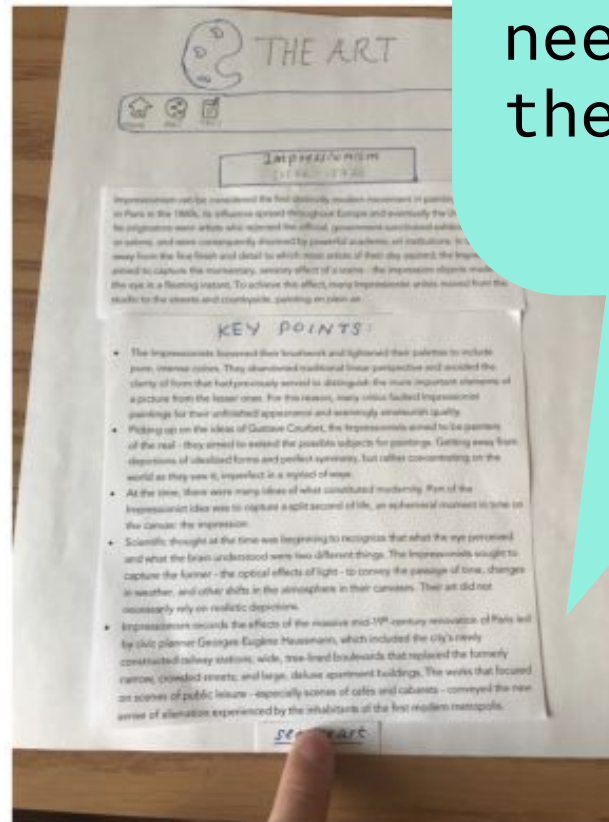
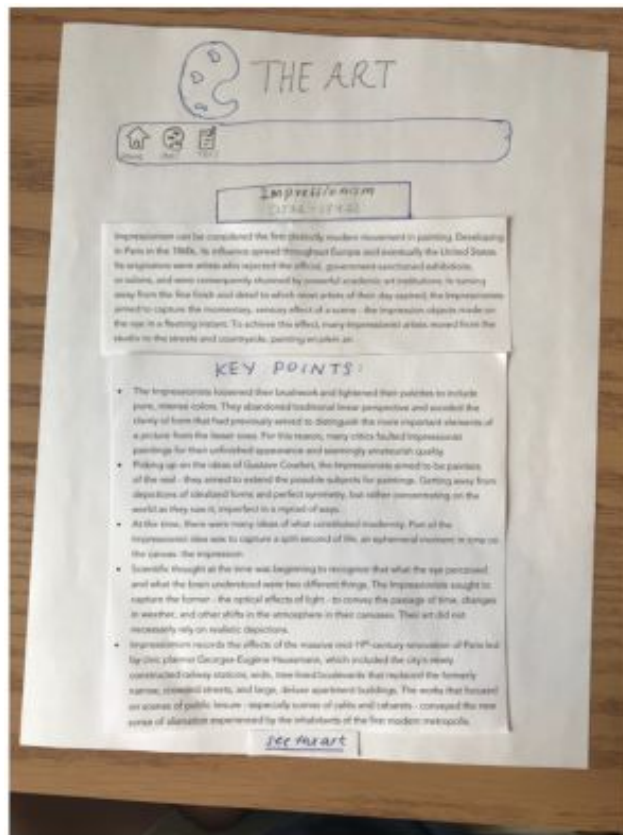
## 2. Select a lesson

This page is weirdly sparse



# Sam's Paper Prototype

## 3. Learn Impressionism

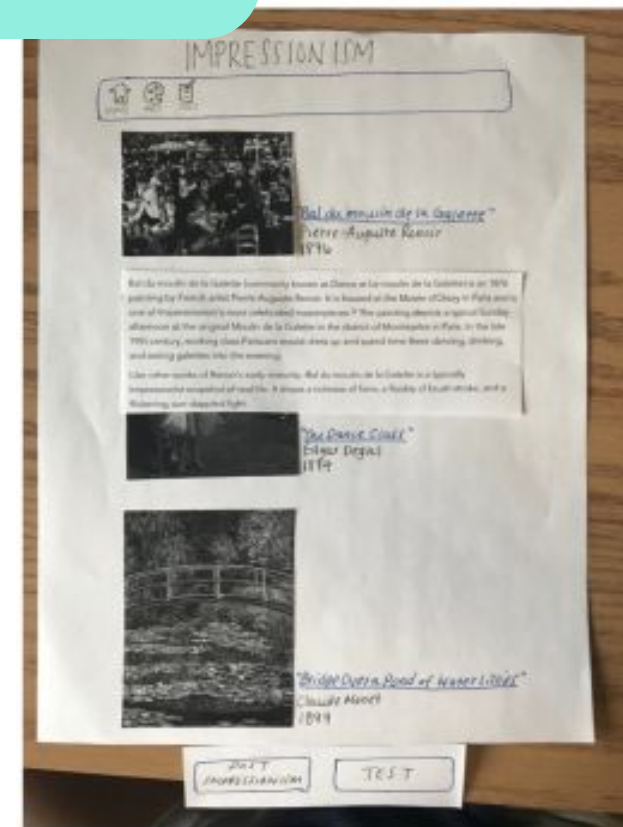
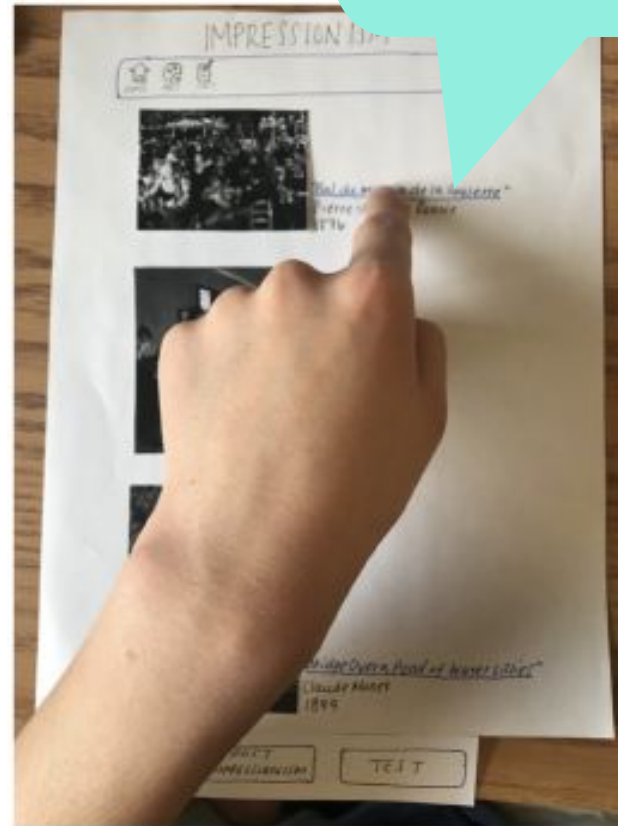
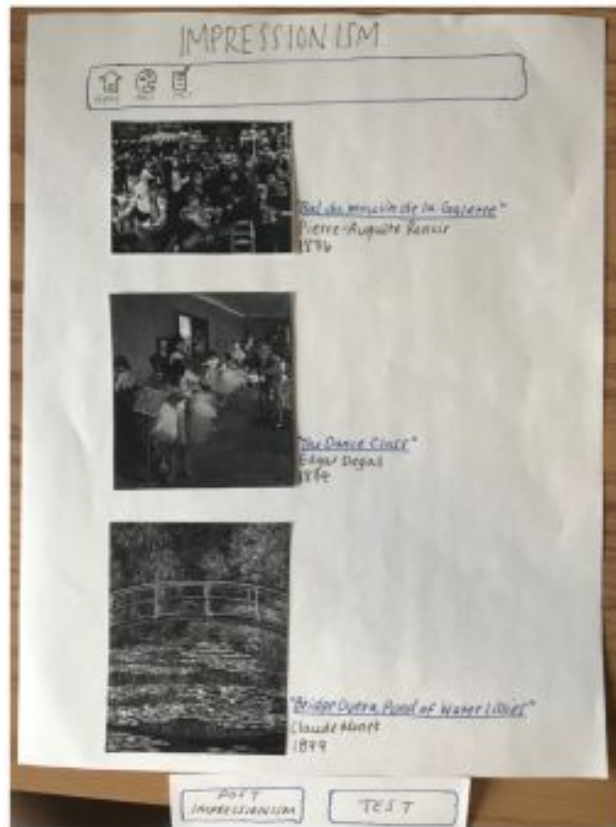


Hmmm...people are not really reading this text...maybe people need to see art as they read about it.

# Sam's Paper Prototype

## 4. See art examples

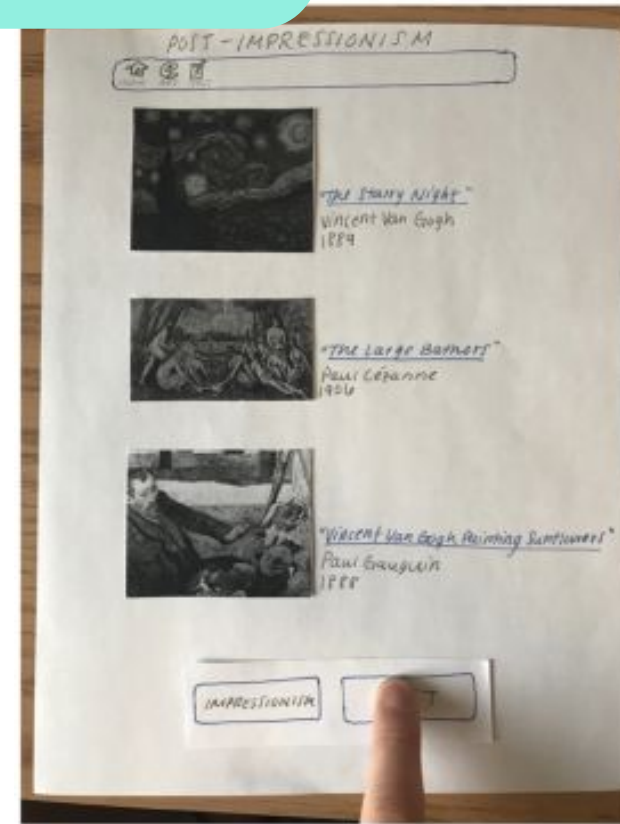
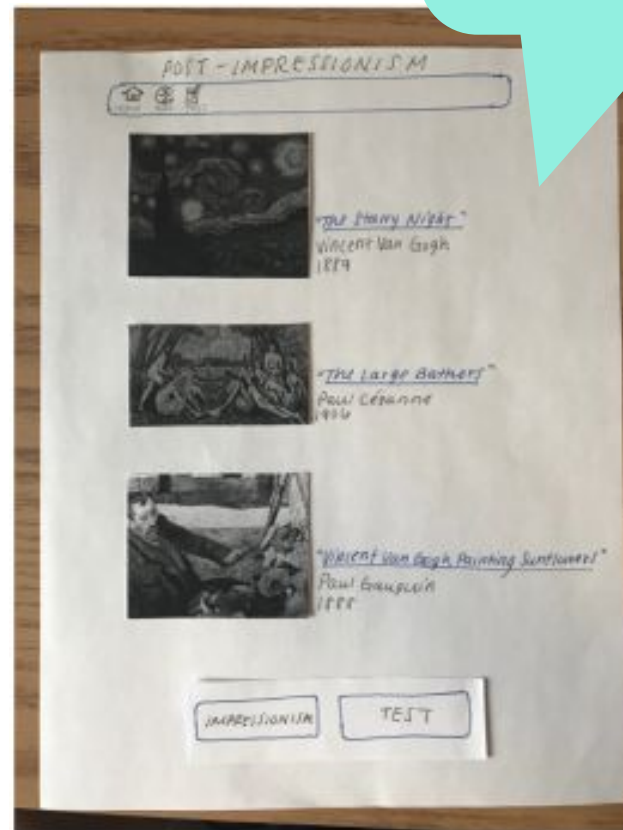
I didn't know you could click on that.



# Sam's Paper Prototype

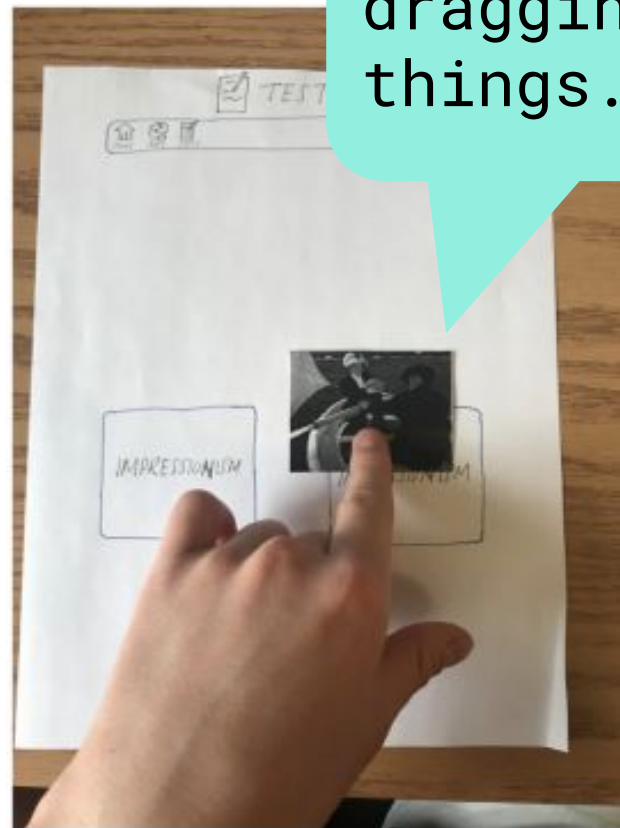
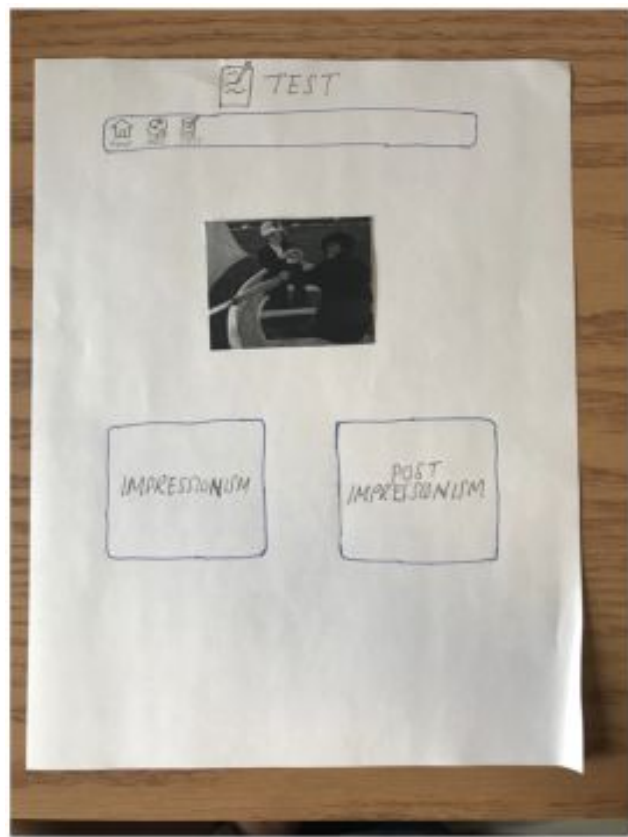
## 5. Learn Post-Impressionism

oh crap! I forgot the overview.



# Sam's Paper Prototype

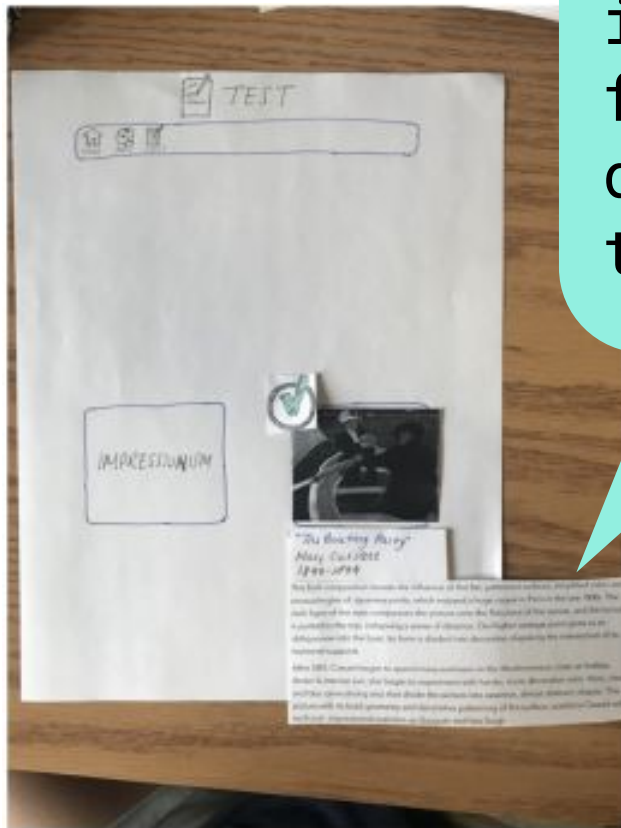
## First Quiz Question



People like dragging things.

# Sam's Paper Prototype

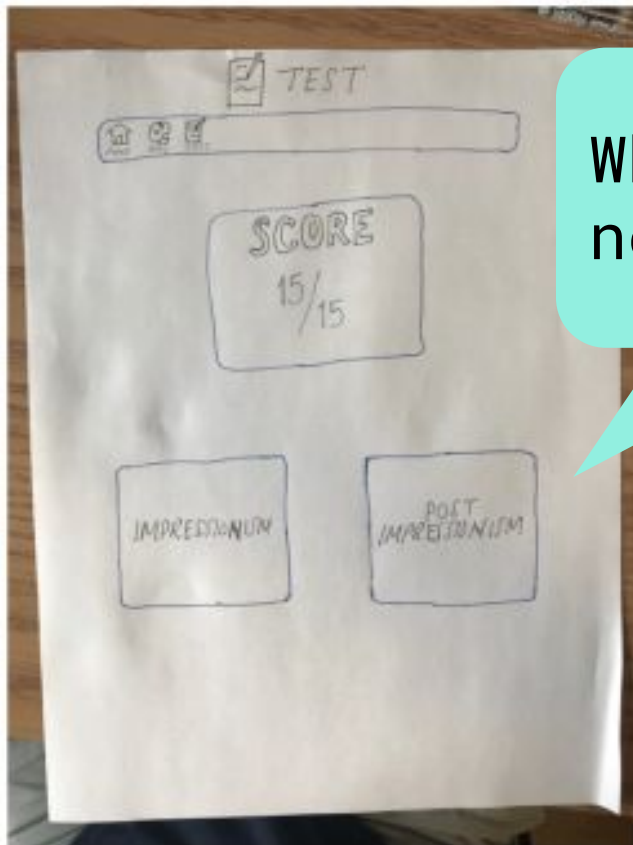
## Feedback



People like immediate feedback; but don't read the text.

# Sam's Paper Prototype

## Final Feedback



Where do we go next?

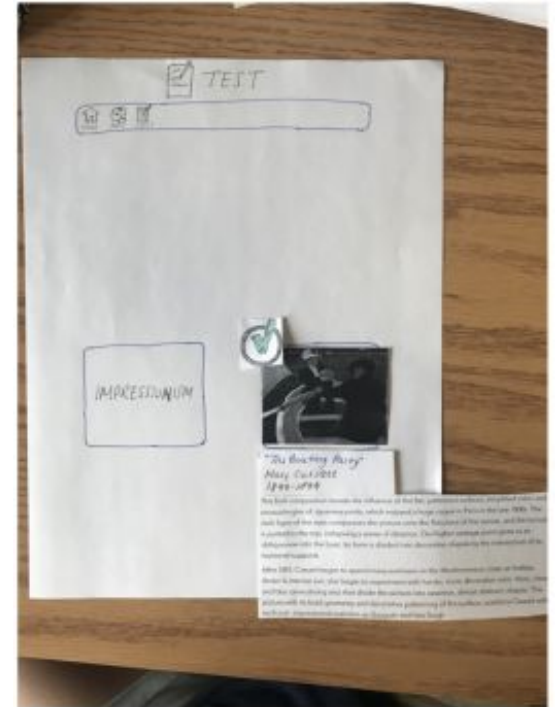
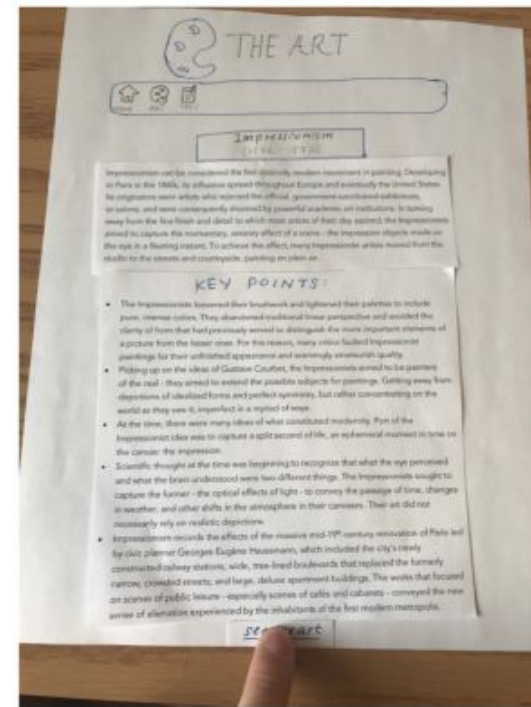
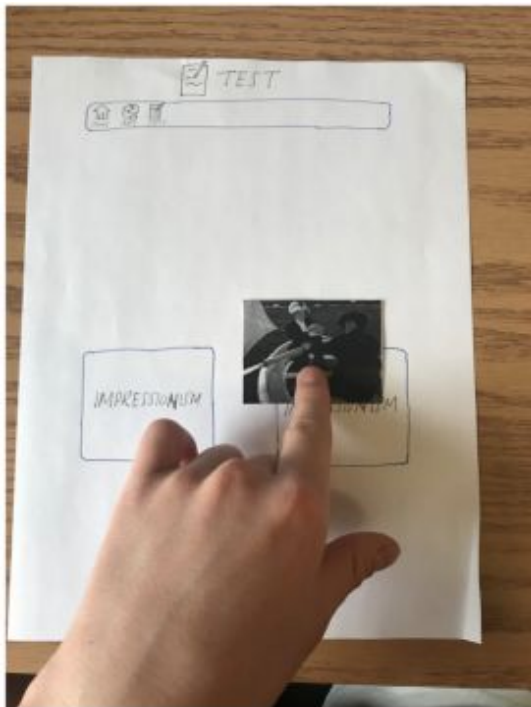
# Feedback Summary: What did we learn?

Good:

People like the quiz. Drag and drop was easy and fun.

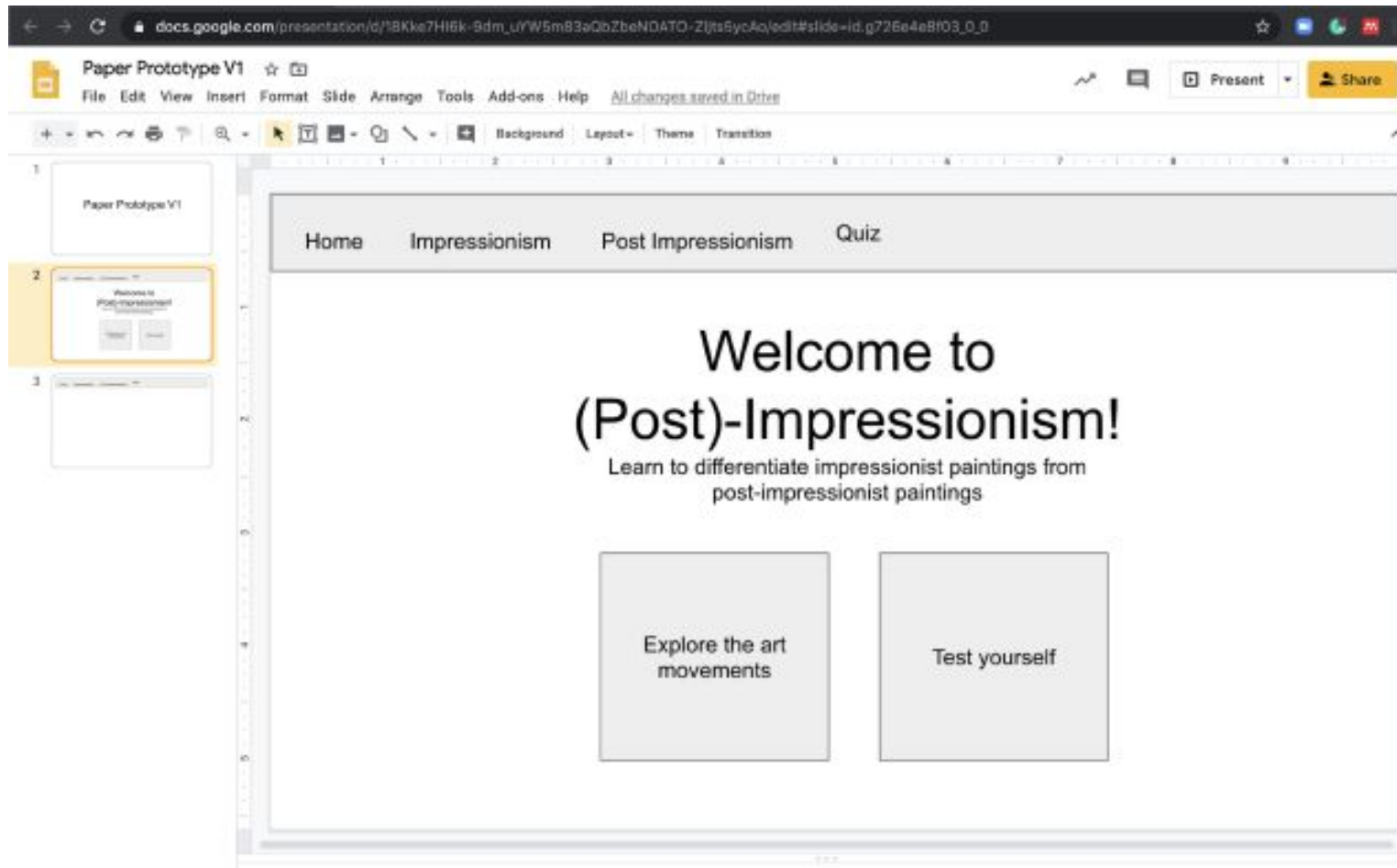
Bad:

People do not read long blocks of text.



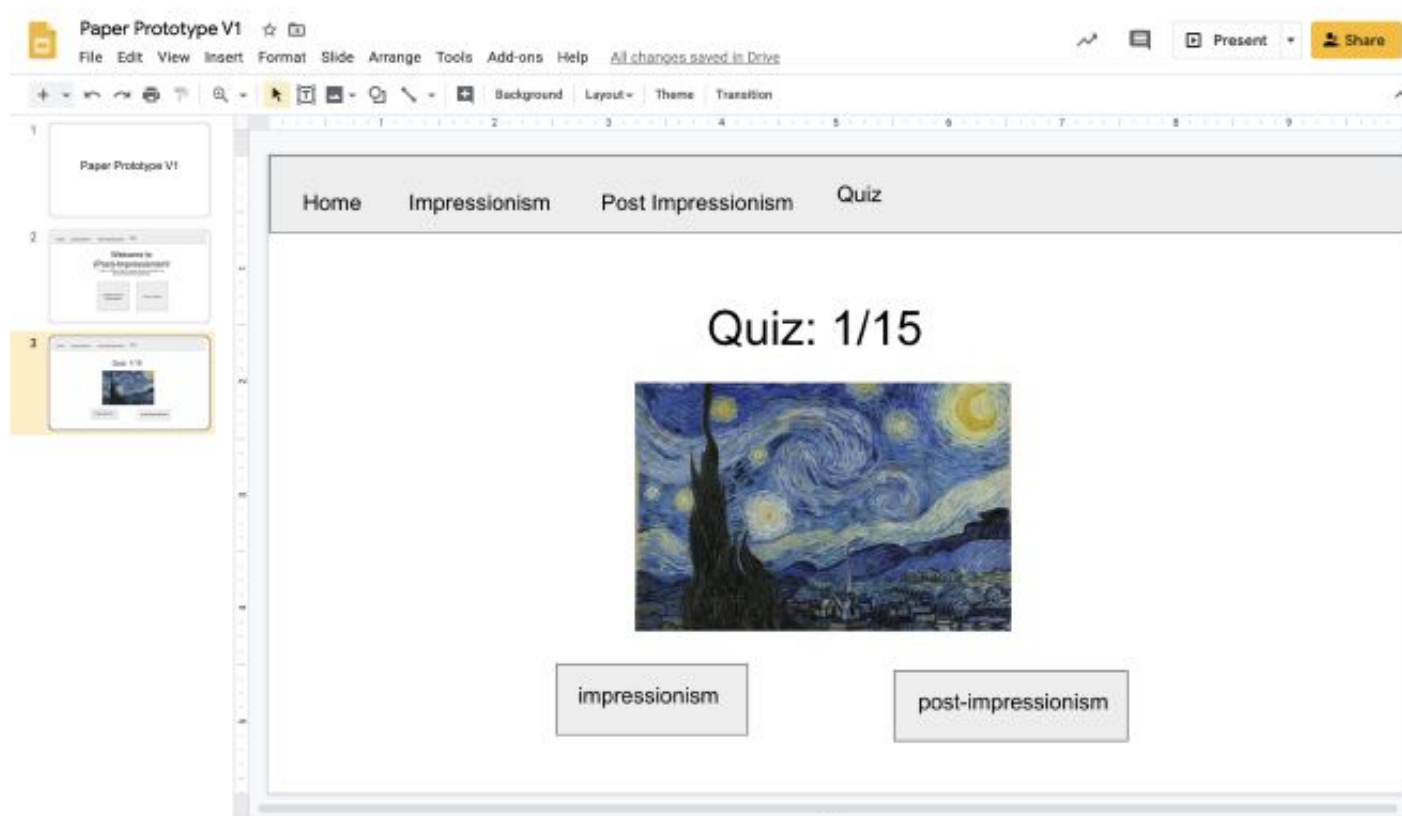
# Google Slides (not paper)

In this class, we'll use Google Slides to make our prototypes



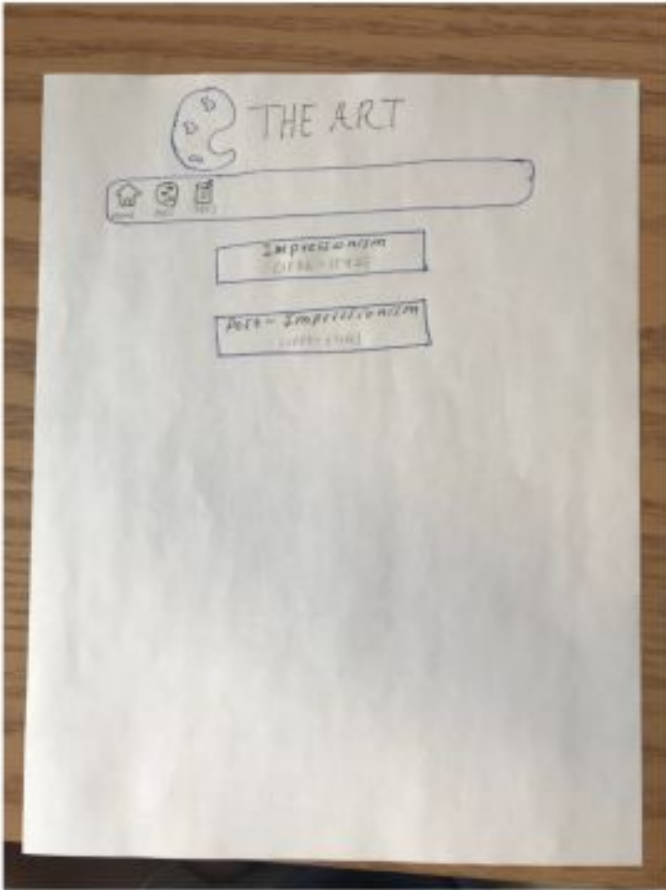
# Google Slides

Use **real examples of media** in your Google Slide prototype



# What should be the default art to learn first?

## Impressionism or Post-Impressionism?



**Bad:**

*I don't know, so ... let the user choose*

**Good:**

*Impressionism. For most users, I think they're going to learn the most from contrasting post-impressionism with impressionism; therefore, it's best if they need to introduce impressionism first.*

# Prevent errors in options: provide good defaults

StreetEasy

RENT BUY SELL BUILDINGS

Monthly payment calculator

HOME PRICE

\$740,000

DOWN PAYMENT

\$148,000 20%

LOAN TERM

30 years

INTEREST RATE\*

7.375%

MORTGAGE AMOUNT\* \$592,000

MORTGAGE PAYMENT \$4,089

MAINTENANCE FEES \$2,176

Fees co-op owners pay to cover building operating costs and NYC property taxes.

\*All calculations are estimates provided by StreetEasy for informational purposes only. The Mortgage Amount includes principal and interest. Home insurance is not included. Actual amounts or financing terms may vary. Please contact your mortgage provider for specifics.

Estimated monthly payment \$6,265

1 of 10

MAINTENANCE FEES

\$2,176/mo

TAXES

Included in maintenance fees

About

SELLER'S AGENT

Advertise Sign up / Log in

e.g. address, building, agent

OR SALE ↓ \$35,000

3 rooms 1 bed 1 bath

Resale

REQUEST A TOUR

as early as tomorrow at 9:00 am

ASK A QUESTION

SHARE HIDE

ved by 45 users.

s Stevens, Real Estate Principal Office, 445 Park 10022

Good defaults will help guide the user away from making a mistake because they will be given a context to work within.

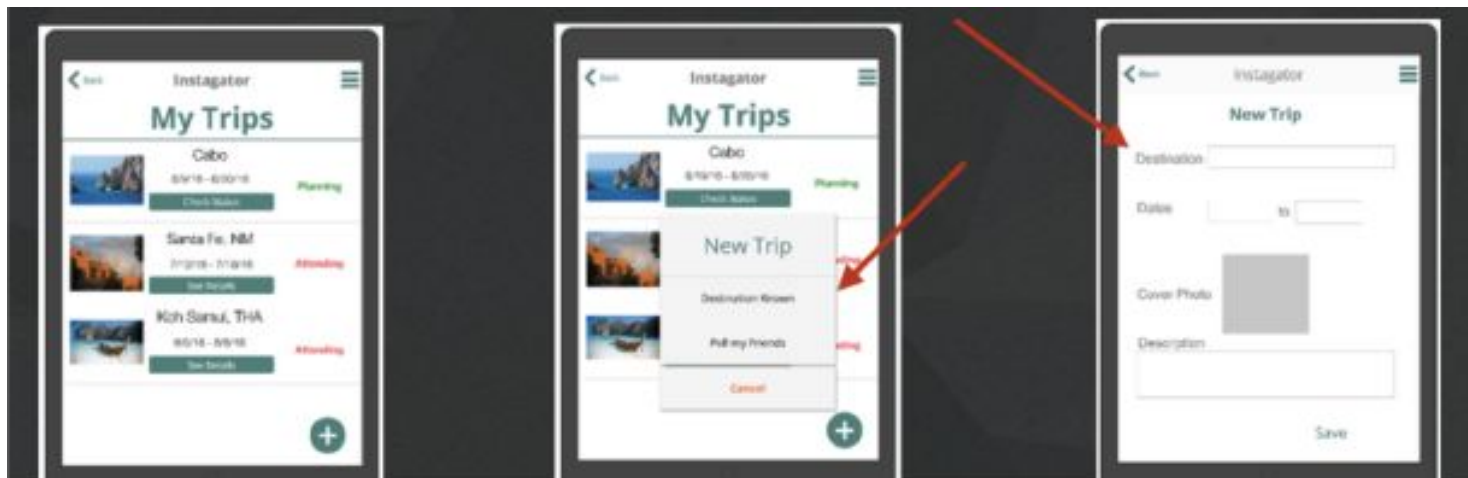
**Why is low fidelity better than high fidelity at early stages?**



# What if the prototype is too polished?



1. It takes a long time to make
2. Designers become attached to designs they spend the most time on
3. You get feedback on the wrong thing:
  - color
  - images
  - fonts
  - wording



COMS 4170 · SPRING 2026

# Running a prototype



# Running Prototypes



1. Put your low fidelity prototype in front of users
2. Read them the goal
3. Ask them to think out loud as they do the task
4. **Do not** interrupt them
5. **Do not** lead them
6. Observe **“critical incidents”**:
  - Times they are unsure
  - Times they did something you did not intend
7. Write it down, take screenshots

# You get the best feedback when you are observing and listening. Not instructing:

- Give the user a **goal**:
  - Create a playlist
  - Learn about the difference between impressionism and post-impressionism
- **Observe** what they do
  - Encourage people to think-aloud
  - Look for critical incidents
    - Times were users are unsure what to do, do the wrong thing
  - Resist the temptation to “rescue” the interviewee or tell them what to do



Your role when running a prototype:

“

**Listen**

...and take notes.



## Your role when running a prototype:

- **Do not** explain your prototype
- **Do not** guide the user
- **Do not** help the user (unless they are truly stuck)
  
- **Do learn** all the things you did right and wrong



**Pretend you are the TA using  
a prototype...**

What's your experience and what's your feedback?

Prototype 1:

# Learn the Arabic Alphabet



# Pronounce the Arabic Alphabet

With **28 letters** each letter has **3 Pronunciations**

Arabic is read  
from **right** to **left**



**Start Learning Now**

## Home Alphabet Quiz

خ

ح

ج

ث

ت

ب

أ

Click a  
letter to  
start

ص

ش

س

ز

ر

ذ

د

ق

ف

غ

ع

ظ

ط

ض

ي

و

هـ

ن

م

ل

ك

# Home Alphabet Quiz

Alif



The letter is pronounced  
"ah"



Back to Alphabet

Next Letter

## Home Alphabet Quiz

Alif



The letter is pronounced  
"oh"



Back to Alphabet

Next Letter

# Home Alphabet Quiz

Alif



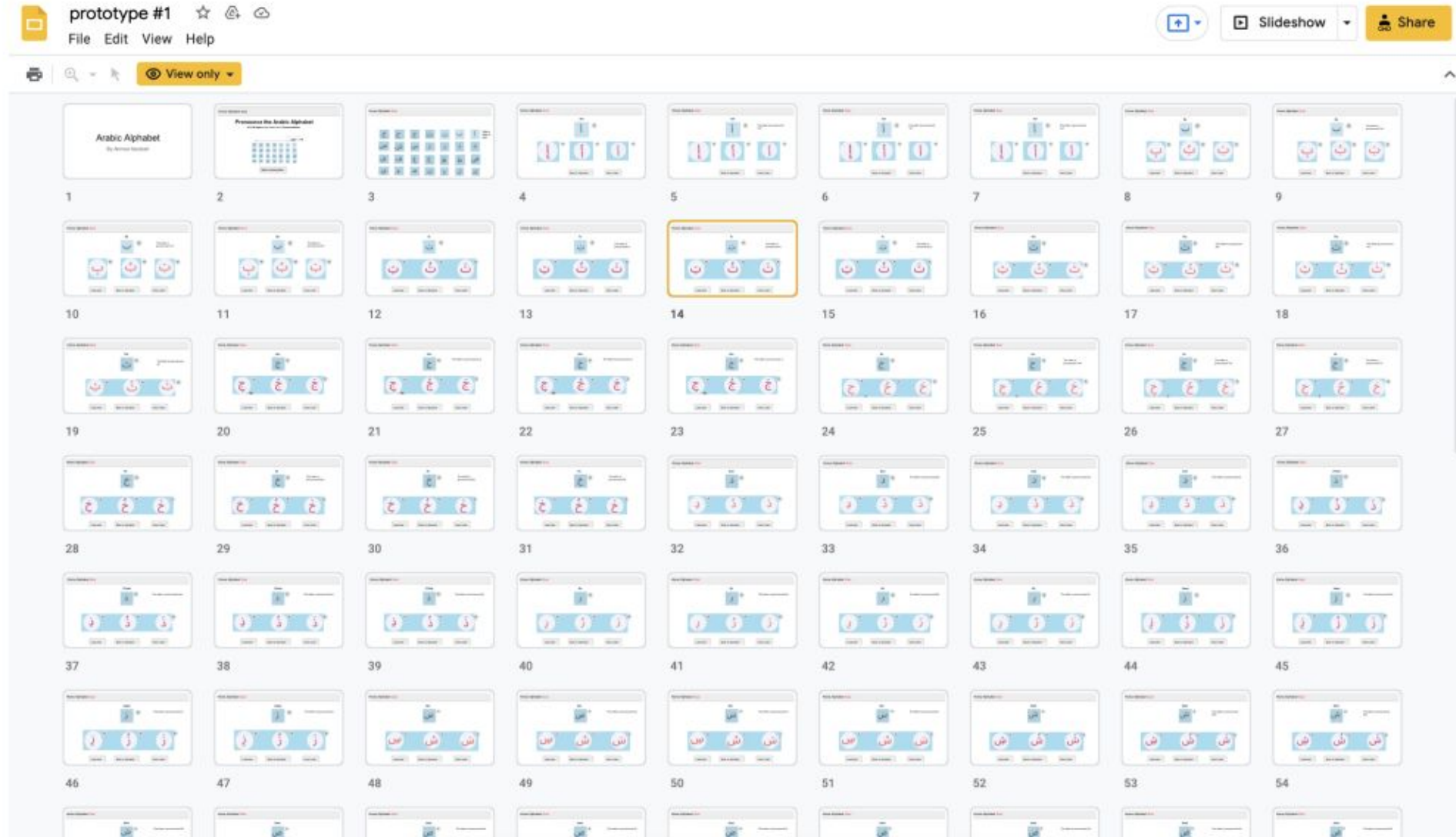
The letter is pronounced  
"ee"



Back to Alphabet

Next Letter

# How many of these are there?





## Drag and Drop Letters

Match the letter with the correct location

Submit

'Ayn

Kh

Nuun

Haa'

Saad

Faa'

Waaw

Raa'

Alif

ر

ن

و

ف

ه

أ

ص

خ

ع

Congrats, You scored 9/9

ع

خ

ن

هـ

ص

ف

و

ر

أ

Return to Alphabet Main Screen

# Lessons from Prototype 1: Learn Arabic Alphabet

- Good topic
  - Interesting, introductory topic for this audience
  - Media is good
- **Too long**
- Needs stronger default for first letter to select
- Perhaps next button needs to be on the left, not the right
- **Not entirely satisfying to just memorize a bunch of letter. I'd like to be able to make words – things I could say in conversation. Like “Hello” or “My name is \_\_\_\_”**



# TA Suggestion

1. Pick fewer letters (maybe 5?) for the user to memorize
2. Then show how they're combined into words
3. Topic pivot: How to read Arabic words



Prototype 2:

# How to build a computer



Welcome to Build a Computer, where we teach you how to build a computer. First, we'll show you through the different parts of a computer and what to look for. Next, we'll show you walk you through the process of building it, step by step. After that, we'll explain some common problems that you could face while building. Finally, we'll do a quick quiz on you to make sure you paid attention.



**START BUILDING!**



## Parts

One of the most daunting tasks to building a computer is learning what all the different components are and what they do. We'll display an image of each one, along with what to look out for when buying one. We'll link info for a budget, regular, and high-power option for each part, so you have the ability to build as we go along.

**NEXT**  
**(Graphics Card)**

Graphics Card

If you're gaming or video editing, the graphics card is probably the most important piece of hardware in your PC. The important specs to look out for are **Video Memory** and **Clock Speed**, higher numbers are better.

|        | Name                               | Link                                                                                                                   | Price  | Memory    | Clock Speed |
|--------|------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------|-----------|-------------|
| Low    |                                    |                                                                                                                        |        |           |             |
| Medium | Zotac GeForce GTX 1660 Ti Twin Fan | <a href="#">ZOTAC Gaming GeForce GTX 1660 Ti 6GB GDDR6 192-bit Gaming Graphics Card Super Compact - ZT-T16610F-10L</a> | 279.99 | 6 GB DDR6 | 1770MHz     |
| High   |                                    |                                                                                                                        |        |           |             |



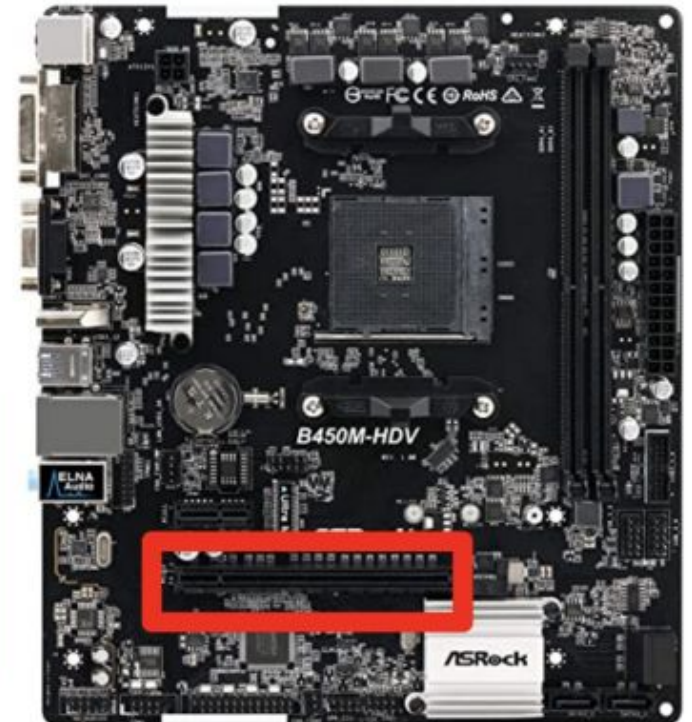
NEXT  
(CPU)

### Building Time - Graphics Card

Alright, now it's time to put in the card! Just push it into the slot in the motherboard, and wait for the click. After it's in, screw it into the case using the screws from the mounting.



NEXT  
(CPU)



### Troubleshooting

Screen won't turn on

- Reseat your graphics card

Computer turning on and off

- Check RAM

Fans really loud

- Check power connections, thermal paste on CPU



Quiz your knowledge!



## Quiz - Parts

If you're trying to build a PC for gaming, what is the most important part?



☐ RAM



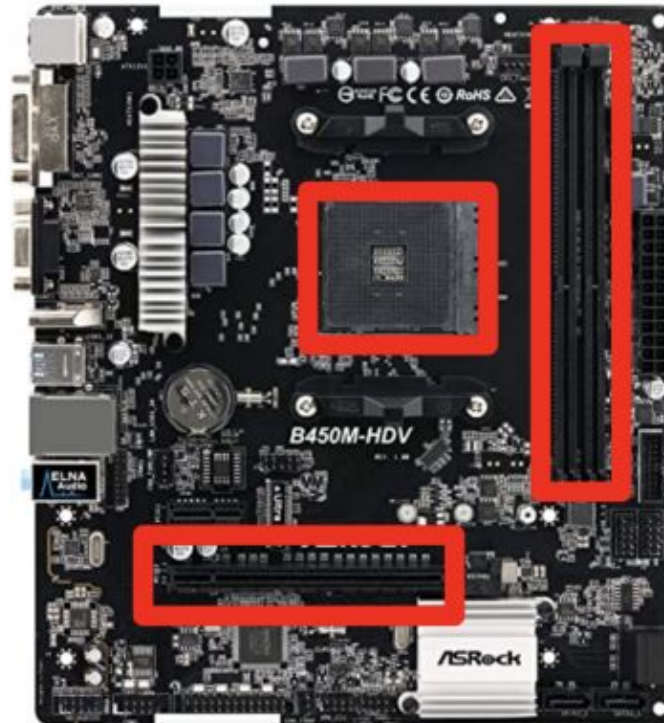
☐ CPU



☐ Graphics Card

### Quiz - Building Time

Time to test your building knowledge. Drag the component to the correct slot and click submit when you think you have it.



## Lessons from Prototype 2: How to build a computer

- Interesting topic, but the material is too easy
- The quiz can be passed without using the learning material
- From the quiz, it's not clear you're learning how to build a computer. Perhaps what you're actually learning is *how to identify the parts of the computer?*



Prototype 3:

**The police are at  
your door**



[HOME](#)   [POLICE AT YOUR DOOR](#)   [PULLED OVER](#)   [STOPPED IN PUBLIC](#)   [ARRESTED](#)   [TEST](#)

# YOU AND THE POLICE

This site serves to inform you about the laws in a police encounter as well as strategies for having a good experience.

## Select a scenario

**Police are at  
your door**

**Pulled over  
by police**

**Stopped by  
police in  
public**

**Arrested by  
the police**

## Police are at your door

Next section



### Your Rights

- To remain silent
  - This means xyz
- To see a warrant
  - This means you do not have to let them in until you see a warrant

What are warrants?

### Tips to reduce risk to yourself

#### DO

- Ask the officer if you can see a warrant and ID
  - Ask the officer to slip the warrant under the door or hold it up to the window so you can read it
- Write down everything officers do if you are searched

#### DO NOT

- Invite the officer into your house without seeing ID and warrant
- Speak to the officers about anything

## Police are at your door

Next section



### Your

- To remain
- To see a

**Search Warrant:** A search warrant allows police to enter the address listed on the warrant, but officers can only search the areas and for the items listed. It must be signed by a judicial officer and lists your address as a place to be searched

**Arrest Warrant:** An arrest warrant allows the police to make lawful arrest. It must have your name on it as the subject of an arrest warrant .

What are warrants?

**to reduce risk to yourself**

**Ask the officer if you can see a warrant and ID**

Ask the officer to slip the warrant under the door or hold it up to the window so you can read it

Put down everything officers do if you are searched

**Do not invite the officer into your house without seeing ID and warrant**

- Speak to the officers about anything

## Pulled over by police



### Your Rights

- As a passenger
  - To remain silent
    - This means xyz
  - You can ask if you're free to leave. If yes, you may silently leave.
- As the driver
  - To remain silent

### Next section

#### Tips to reduce risk to yourself

##### DO:

##### As a driver

- **Stop and turn off the car in a safe place as quickly as possible.**
- Turn on the internal light, open the window part way, and place your hands on the wheel.
- **Upon request**, show police your driver's license, registration, and proof of insurance.

##### As a passenger

- Put your hands on the dashboard.

##### DO NOT

- Make **sudden movements**, and **keep your hands where the officer can see them.**

## Stopped by the police in public

[Next section](#)

### Your Rights

- To remain silent
- To not have to consent to a search of yourself or your belongings
  - BUT, the police may pat down your clothing if they suspect a weapon.
- To not have to answer questions about where you were born, whether you are a U.S. citizen, or how you entered the country.

### Tips to reduce risk to yourself

#### DO

- **Stay calm.**
- **Keep your hands where the police can see them.**

#### DO NOT

- Lie or give false documents.
- Run, resist, or obstruct the officers.

Take test!

## Arrested by police



### Your Rights

- To remain silent
- To a lawyer
- To make a local phone call

### Tips to reduce risk to yourself

#### DO

- Say you wish to remain silent
- Ask for a lawyer immediately.

#### DO NOT

- Sign anything
- Make any decisions without a lawyer
- Allow the police to listen to phone call only if you call a lawyer.

## Test-Part ¼ “Police are at your door”

Stuck? Click to review!



1. What is a right that you do not have?

- a. Right to stay silent
- b. Right to call a lawyer
- c. Right to read the search warrant
- d. Right to close the door

2. What should you do if the police ask to come in?

- a. Ask the officers to show identification
- b. Ask the officers to show you the warrant
- c. Allow the police to come in
- d. Ask the officers to say your name

Next

## Test-Part ¼ “Police are at your door”

Stuck? Click to review!



1. What is a right that you do not have? ✗
- a. **Right to stay silent**
  - b. Right to call a lawyer
  - c. Right to read the search warrant
  - d. **Right to close the door**

2. What should you do if the police ask to come in? ✓
- a. **Ask the officers to show identification**
  - b. Ask the officers to show you the warrant
  - c. Allow the police to come in
  - d. Ask the officers to say your name

Next

## Test-Part ¾ “Stopped in public”

Stuck? Click to review!



5. What can the police legally do in this situation?

- a. Force you to answer about your birthplace
- b. Force you to answer about your citizenship status
- c. Force you to answer how you entered the US
- d. Pat down your clothing to check you

6. The police has a gun on you what should you do?

- a. Confront the police officer about the gun
- b. Run away fast so that you don't get hurt
- c. Keep your hands in the air
- d. Keep your hands hidden

Next

## Test-Part 4/4 “Arrested by police”



Stuck? Click to review!

7. If you call your \_\_\_\_\_ the police can listen to your conversation.

8. You should not \_\_\_\_\_ any documents that the police give you

Next

## Test-Review

Your score: X/8

Want to learn more?

See

<https://www.aclu.org/know-your-rights/stopped-by-police/>

## Lessons from Prototype 3: The police are at your door

- Great topic, just the right amount of information
- Good states and transitions
- Information on each page is hard to read
  - Organize information on the page better
  - Split up into two states if there is too much information



Prototype 3:

**The police are at your door**  
***Second iteration***



[Home](#)[At Your Door](#)[Pulled Over](#)[Stopped in Public](#)[Arrested](#)[Test](#)

## You and the police

This site serves inform you about the laws in a police encounter as well as strategies for having a good experience.

Police Are at Your Door



Police Pull You Over



Police Stop You in Public



Police Arrest You



Home

At Your Door

Pulled Over

Stopped in Public

Arrested

Test

## Police are at Your Door



### Your Rights

To remain silent: This means you can say you want to exercise the 5th amendment if you want to

To see a warrant: This means you do not have to let them in until you see a warrant

#### DO

#### **Ask the officer if you can see a warrant and ID**

Ask the officer to slip the warrant under the door or hold it up to the window so you can read it

Write down everything officers do if you are searched

#### DO NOT

#### **Do not Invite the officer into your house without seeing ID and warrant**

Do not Speak to the officers about anything

Next

[Home](#)[At Your Door](#)[Pulled Over](#)[Stopped in Public](#)[Arrested](#)[Test](#)

## Test-Part 1/4 "Police at Door"



### 1. What is a right that you do not have?

- (a) Right to stay silent
- (b) Right to call a lawyer
- (c) Right to read the search warrant
- (d) Right to close the door

[Stuck? Click to Review!](#)[Next](#)

Home

At Your Door

Pulled Over

Stopped in Public

Arrested

Test

## Test-Part 1/4 "Police at Door"



### 1. What is a right that you do not have?

- (a) Right to stay silent
- (b) Right to call a lawyer
- (c) Right to read the search warrant
- (d) Right to close the door

Stuck? Click to Review!

Next

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## Test-Part 4/4 "Arrested"



7. If you call your \_\_\_\_\_ the police cannot listen to your conversation.

Stuck? Click to Review!

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## Test-Part 4/4 "Arrested"



7. If you call your **lawyer** the police cannot listen to your conversation.

Stuck? Click to Review!

Next

Home

At Your Door

Pulled Over

Stopped in Public

Arrested

Test



# The End. Thank you!

Your Score: X/8

Want to learn more?

See:

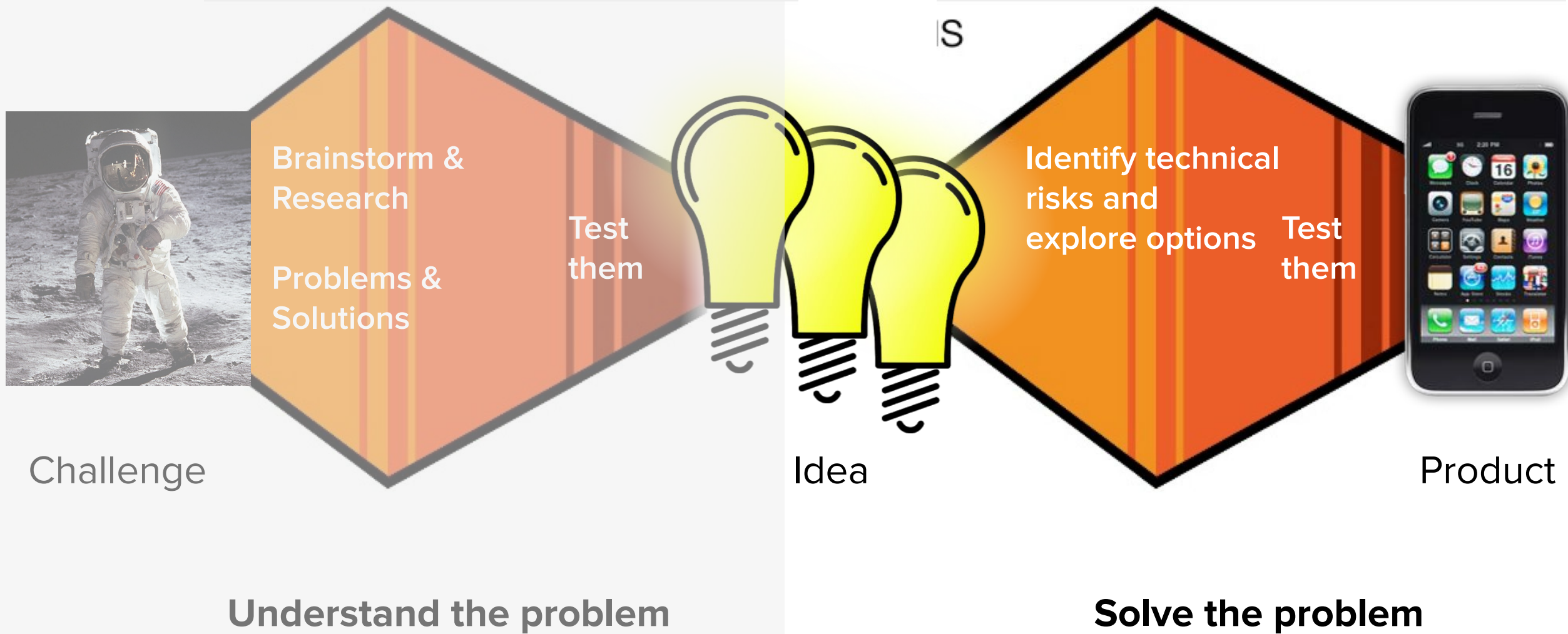
<https://www.aclu.org/know-your-rights/stopped-by-police>

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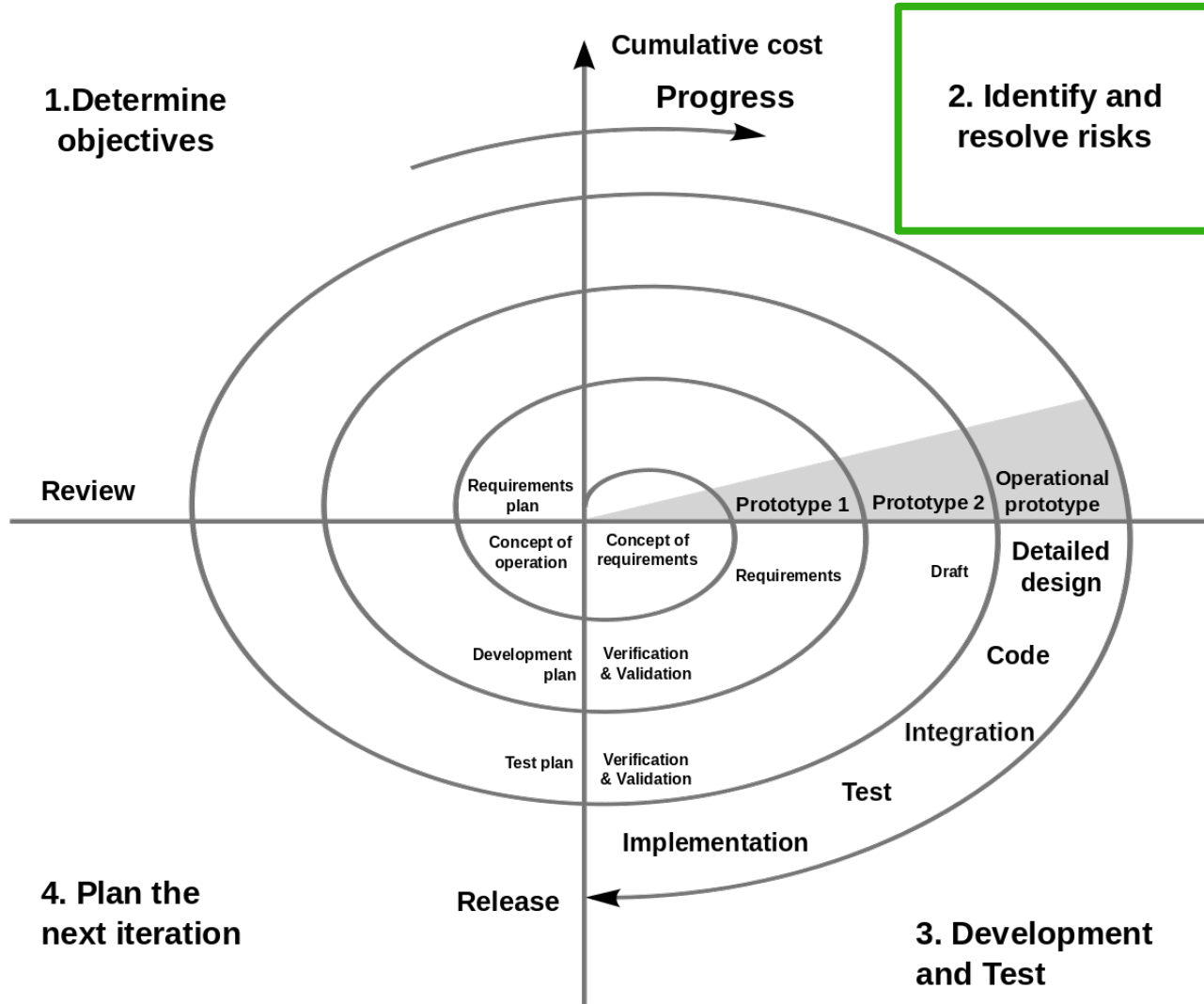
# Summary



# Homework 8-12



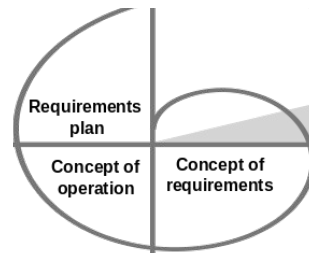
# Iterative Design is good because it minimizes risk



# The first iteration should be as low-fidelity as possible – addressing core concepts

1. Determine objectives

2. Identify and resolve risks



4. Plan the next iteration

3. Development and Test

**Main prototype goal:**

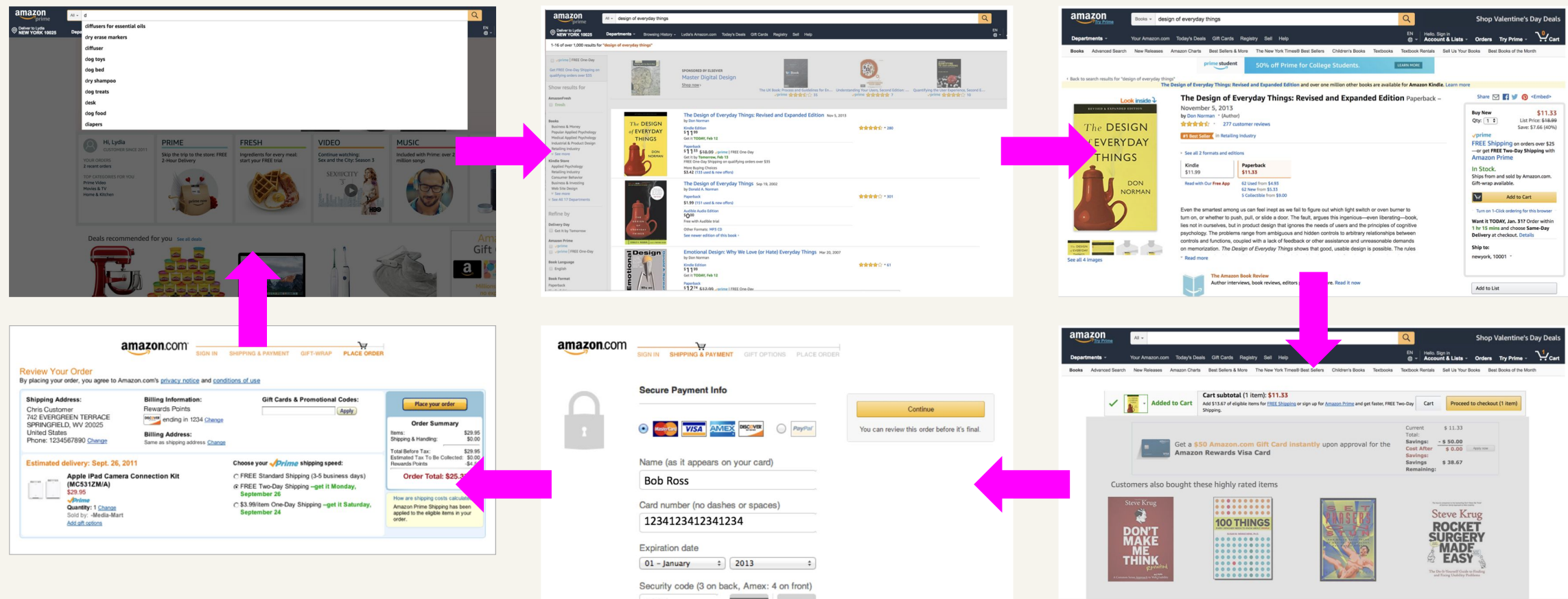
“

# Coherence

Is there a path through the system by  
which the user can accomplish their goal?



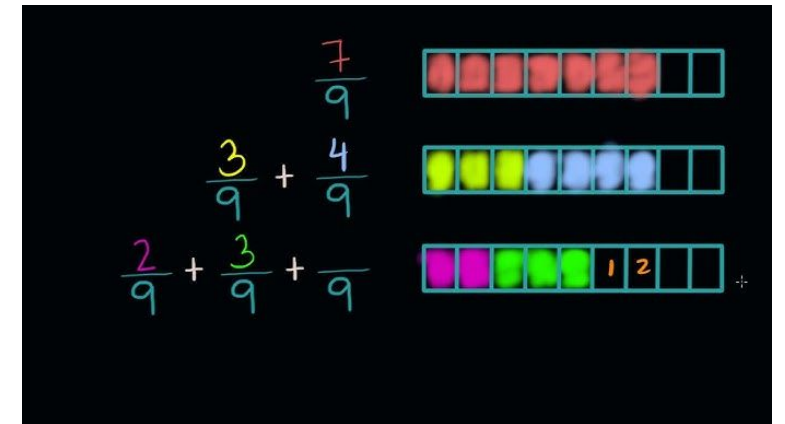
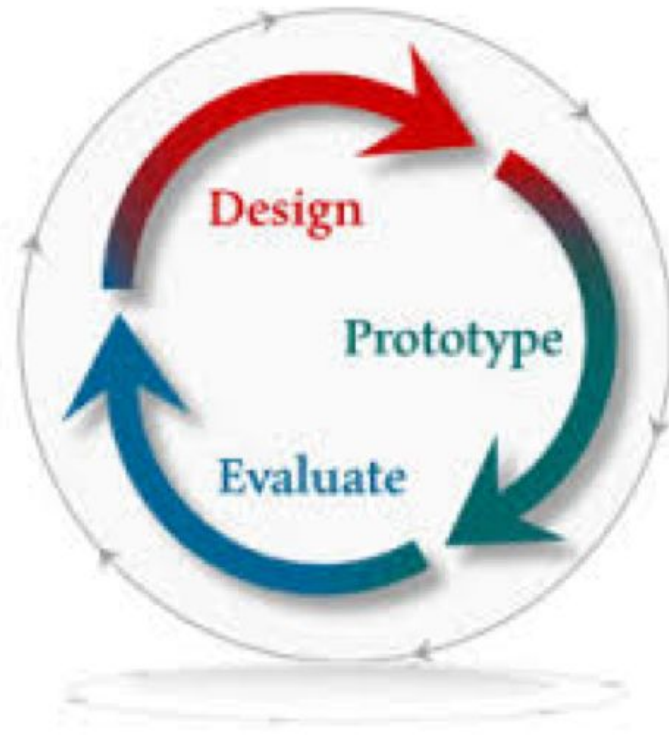
# Focus on breaking the task into states, options and transitions



# Translating an idea into a prototype is HARD



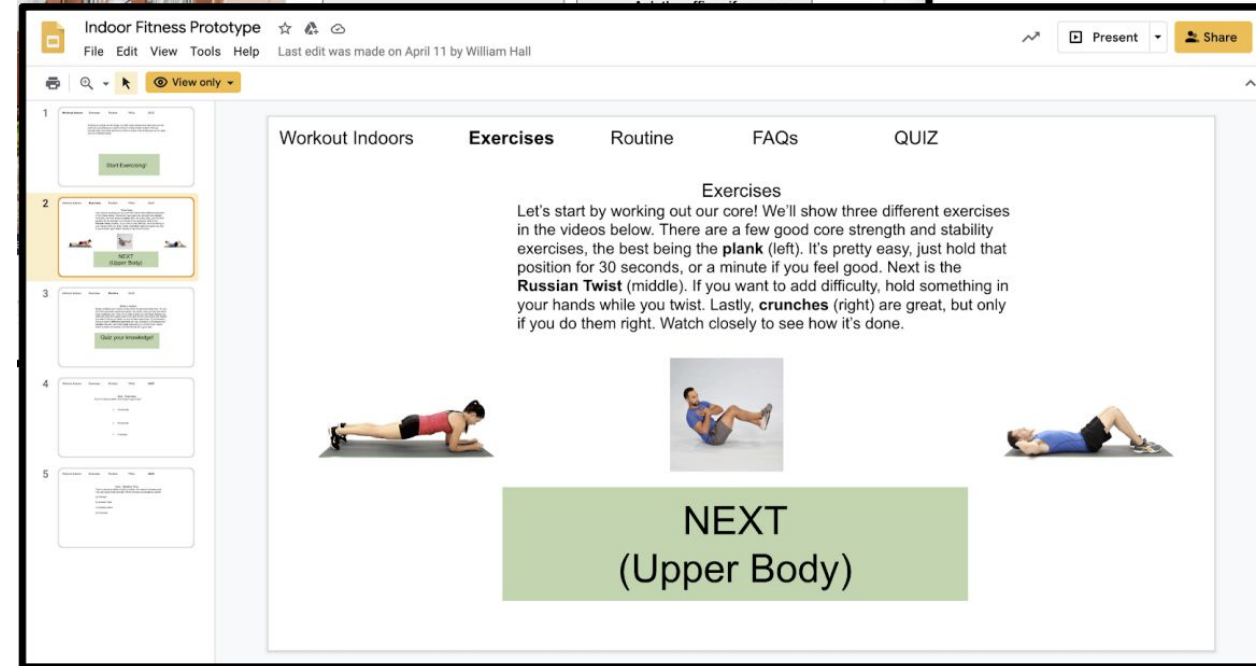
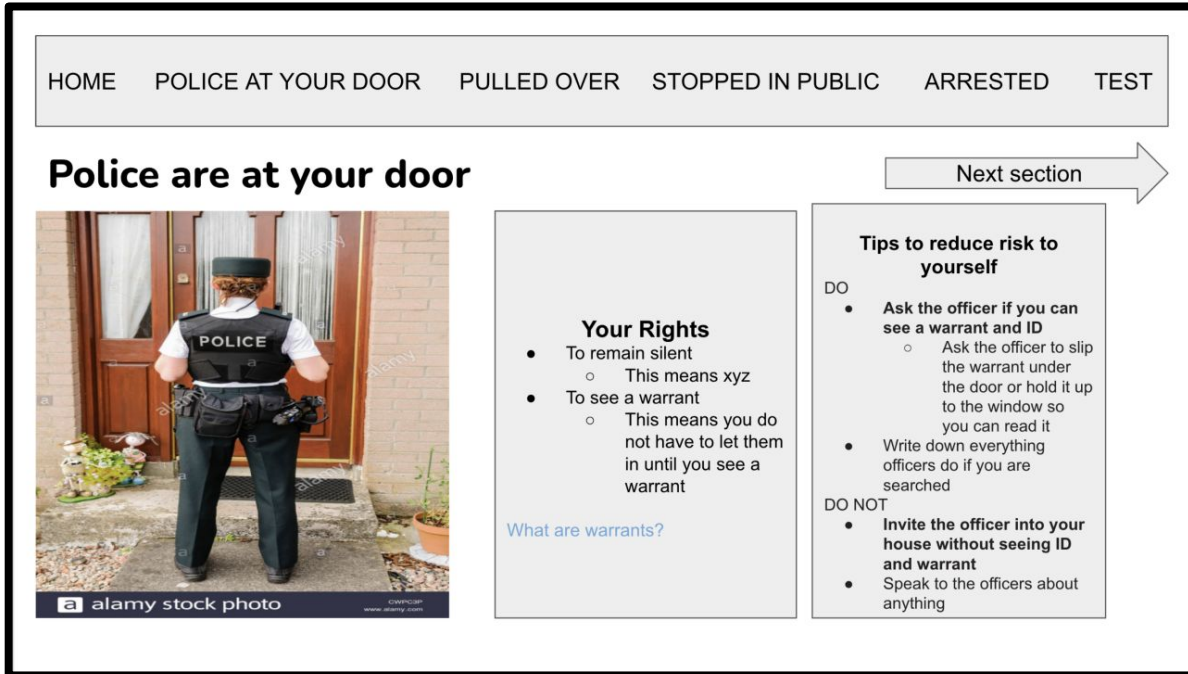
Teach fractions by  
working through  
problems slowly



Low-Fidelity Prototype

**You will probably learn as much from making the prototype, as you will from running it.**

# Homework 8: Low-Fidelity Prototypes in Google Slides



Every team member will make their own prototype, and run it on someone in the class. You will get TA feedback during the next group session.

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# Weekly Group Session



# Group Session Guidelines – Presenter

1. Provide a **one sentence description** of your user goal
2. Provide a **one minute demo** of your progress,
3. Provide a **one question** you have that you would like feedback on

# Group Session Guidelines – Giving Feedback

You all should have started on your **user journeys**, **competitive analysis** and the **low-fidelity prototype**.

1. Start with clarifying questions. Remember, the design should be derived from user **insights** and **opportunities**.
2. Lead into explorations of alternatives. e.g. “Based on what you’ve presented here today, have you also thought about \_\_\_\_\_?”
3. Avoid statements that refer to absolutes.
  - a. Bad example: “*This sucks and it’s ugly*”
  - b. Better example: “*I think there’s something missing between step 3 and 4. It’s not clear to me what the sequence of operations is. How do you envision people knowing where to click?*”

# Group Session Guidelines – Receiving Feedback

You all should have started on your **user journeys**, **competitive analysis** and the **low-fidelity prototype**.

1. Try to talk as little as possible, and let the time be used for critique, not for defense. If you don't agree with what's being said, just say "Thank you" and move on.
2. Ask clarifying questions. For example:
  - a. *"When you say the style in my design is sloppy, do you mean that the lines aren't sharp, or that the composition isn't balanced quite right? Can you show me exactly what you mean?"*

# Group Session Guidelines – Receiving Feedback

3. Refer back to the user **insights** and **opportunities**. If you're not getting what you want from the critique, ask the to get feedback in the context of the stated goals.
4. Ask for what changes you can make to satisfy criticism. For example:
  - a. If you agree with a criticism, but don't see a path to improvement, ask the person providing the feedback for one.
  - b. Turn it around to the person who made the comment: *“Good point. So do you see anything I can do to improve on that?”*

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# Homework 08



# Homework 8: Low-Fidelity Prototypes

Each group member will create their own **Google Slide Prototype** of how you would **teach a topic interactively through media**. Remember, the goal of a low-fidelity prototype is to determine if the design concept has end to-end coherence.

## Prototype.

- Create slides as described in this lecture.
- Run the prototype on a target user as described in class.
- Be prepared to present your prototype and findings during TA session.

**Prototype Assessment.** Determine where do you think you are and what ideas you might want to keep working on for the next iteration.



# 10% of your grade is participation in your TA section

- Show up on time
- Be prepared to present
- Present your materials clearly and concisely.
- Actively, listen to other presentations.
- If you're virtual, keep your camera on at all times. No exceptions.
- Take feedback from your TA.
- If you can't attend a feedback session, email your TA in advance to schedule a makeup session before **Friday 11:59PM**



